

Violent behavior can also occur with phencyclidine use, as intoxicated individuals may believe that they are being attacked.

Prevalence

Data on the prevalence of phencyclidine use disorder are not available, but rates appear to be low (based on rates of the overall category of hallucinogen use disorder, which includes phencyclidine, of about 0.1% among individuals age 12 and older in the United States). Furthermore, among U.S. substance use treatment facility admissions, only 0.3% of the admitted individuals endorsed phencyclidine as their primary drug.

Risk and Prognostic Factors

In a general population study in Australia, ketamine users were more likely to be men and to have consumed more than 11 standard drinks per day.

Sex- and Gender-Related Diagnostic Issues

The gender ratio for phencyclidine use disorder is not known, but among U.S. substance use treatment facility admissions endorsing phencyclidine as the primary drug, 62% were men.

Diagnostic Markers

Laboratory testing may be useful, as phencyclidine is present in the urine in intoxicated individuals up to 8 days after ingestion. The individual's history along with certain physical signs (e.g., nystagmus, analgesia, prominent hypertension) may aid in distinguishing the phencyclidine clinical picture from that of other hallucinogens.

Functional Consequences of Phencyclidine Use Disorder

In individuals with phencyclidine use disorder, there may be physical evidence of injuries from accidents, fights, and falls. Chronic use of phencyclidine can lead to acute and persistent cognitive impairment; urinary tract and intestinal symptoms; abdominal pain, chest pain, palpitations, and tachycardia; respiratory depression; sleep disorders; and depression.

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Differential Diagnosis

Other substance use disorders. Distinguishing the effects of phencyclidine from those of other substances may be important, because phencyclidine can be an additive to other substances (e.g., cannabis, cocaine).

Phencyclidine intoxication and phencyclidine-induced mental disorders. Phencyclidine use disorder is differentiated from phencyclidine intoxication and phencyclidine-induced mental disorders (e.g., phencyclidine-induced psychotic disorder) in that phencyclidine use disorder describes a problematic pattern of phencyclidine use that involves impaired control over phencyclidine use, social impairment attributable to phencyclidine use, risky phencyclidine use (e.g., driving while intoxicated), and pharmacological symptoms (the development of tolerance), whereas

phencyclidine intoxication and phencyclidine-induced mental disorders describe psychiatric syndromes that occur in the context of heavy use. Phencyclidine intoxication and phencyclidine-induced mental disorders occur frequently in individuals with phencyclidine use disorder. In such cases, a diagnosis of phencyclidine intoxication or a phencyclidine-induced mental disorder should be given in addition to a diagnosis of phencyclidine use disorder, the presence of which is indicated in the diagnostic code.

Independent mental disorders. Some of the effects of phencyclidine use may resemble symptoms of independent mental disorders, such as psychosis (schizophrenia); low mood (major depressive disorder); and violent, aggressive behaviors (conduct disorder, antisocial personality disorder). Discerning whether these behaviors occurred before the intake of the drug is important in the differentiation of acute drug effects from a preexisting mental disorder.

Comorbidity

Conduct disorder in adolescents and antisocial personality disorder may be associated with phencyclidine use. Other substance use disorders, especially alcohol, cocaine, and amphetamine use disorders, are common among those with phencyclidine use disorder.

Other Hallucinogen Use Disorder

Diagnostic Criteria

- A. A problematic pattern of hallucinogen (other than phencyclidine) use leading to clinically significant impairment or distress, as manifested by at least two of the following, occurring within a 12-month period:
1. The hallucinogen is often taken in larger amounts or over a longer period than was intended.
 2. There is a persistent desire or unsuccessful efforts to cut down or control hallucinogen use.
 3. A great deal of time is spent in activities necessary to obtain the hallucinogen, use the hallucinogen, or recover from its effects.
 4. Craving, or a strong desire or urge to use the hallucinogen.
 5. Recurrent hallucinogen use resulting in a failure to fulfill major role obligations at work, school, or home (e.g., repeated absences from work or poor work performance related to hallucinogen use; hallucinogen-related absences, suspensions, or expulsions from school; neglect of children or household).
 6. Continued hallucinogen use despite having persistent or recurrent social or interpersonal problems caused or exacerbated by the effects of the hallucinogen (e.g., arguments with a spouse about consequences of intoxication; physical fights).

7. Important social, occupational, or recreational activities are given up or reduced because of hallucinogen use.
8. Recurrent hallucinogen use in situations in which it is physically hazardous (e.g., driving an automobile or operating a machine when impaired by the hallucinogen).
9. Hallucinogen use is continued despite knowledge of having a persistent or recurrent physical or psychological problem that is likely to have been caused or exacerbated by the hallucinogen.
10. Tolerance, as defined by either of the following:
 - a. A need for markedly increased amounts of the hallucinogen to achieve intoxication or desired effect.
 - b. A markedly diminished effect with continued use of the same amount of the hallucinogen.

Note: Withdrawal symptoms and signs are not established for hallucinogens, and so this criterion does not apply.

Specify the particular hallucinogen.

Specify if:

In early remission: After full criteria for other hallucinogen use disorder were previously met, none of the criteria for other hallucinogen use disorder have been met for at least 3 months but for less than 12 months (with the exception that Criterion A4, "Craving, or a strong desire or urge to use the hallucinogen," may be met).

In sustained remission: After full criteria for other hallucinogen use disorder were previously met, none of the criteria for other hallucinogen use disorder have been met at any time during a period of 12 months or longer (with the exception that Criterion A4, "Craving, or a strong desire or urge to use the hallucinogen," may be met).

Specify if:

In a controlled environment: This additional specifier is used if the individual is in an environment where access to hallucinogens is restricted.

Code based on current severity/remission: If a hallucinogen intoxication or another hallucinogen-induced mental disorder is also present, do not use the codes below for hallucinogen use disorder. Instead, the comorbid hallucinogen use disorder is indicated in the 4th character of the hallucinogen-induced disorder code (see the coding note for hallucinogen intoxication or specific hallucinogen-induced mental disorder). For example, if there is comorbid hallucinogen-induced psychotic disorder and hallucinogen use disorder, only the hallucinogen-induced psychotic disorder code is given, with the 4th character indicating whether the comorbid hallucinogen use disorder is mild, moderate, or severe: F16.159 for mild hallucinogen use disorder with hallucinogen-induced psychotic disorder or F16.259 for a moderate or severe hallucinogen use disorder with hallucinogen-induced psychotic disorder.

Specify current severity/remission:

F16.10 Mild: Presence of 2–3 symptoms.

F16.11 Mild, In early remission

F16.11 Mild, In sustained remission

F16.20 Moderate: Presence of 4–5 symptoms.

F16.21 Moderate, In early remission

F16.21 Moderate, In sustained remission

F16.20 Severe: Presence of 6 or more symptoms.

F16.21 Severe, In early remission

F16.21 Severe, In sustained remission

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Specifiers

“In a controlled environment” applies as a further specifier of remission if the individual is both in remission and in a controlled environment (i.e., in early remission in a controlled environment or in sustained remission in a controlled environment). Examples of these environments are closely supervised and substance-free jails, therapeutic communities, and locked hospital units.

Diagnostic Features

Hallucinogens comprise a diverse group of substances that despite having different chemical structures and possibly involving different molecular mechanisms, produce similar alterations of perception, mood, and cognition in users. Hallucinogens included are phenylalkylamines (e.g., mescaline, DOM [2,5-dimethoxy-4-methylamphetamine], and MDMA [3,4-methylenedioxymethamphetamine; also called “ecstasy” or “molly”]); the indoleamines, including psilocybin (and its metabolite psilocin, the compound primarily responsible for the psychedelic effects of hallucinogenic mushrooms) and dimethyltryptamine (DMT); and the ergolines, such as LSD (lysergic acid diethylamide) and morning glory seeds. In addition, miscellaneous other ethnobotanical compounds are classified as hallucinogens, of which *Salvia divinorum* and jimsonweed are two examples. Excluded from the hallucinogen group are cannabis and its active compound, delta-9-tetrahydrocannabinol (THC) (see the section “Cannabis-Related Disorders”). These substances can have hallucinogenic effects but are diagnosed separately because of significant differences in their psychological and behavioral effects.

Hallucinogens are usually taken orally, although some forms are smoked (e.g., DMT, salvia) or (rarely) taken intranasally or by injection (e.g., ecstasy). Duration of effects varies across types of hallucinogens. Some of these substances (i.e., LSD, MDMA) have a long half-life and extended duration such that users may spend hours to days using and/or recovering from the effects of these drugs. However, other hallucinogenic drugs (e.g., DMT, salvia) are short acting. Tolerance to hallucinogens develops with repeated use and has been reported to have both autonomic and psychological effects.

MDMA/ecstasy as a hallucinogen may have distinctive effects attributable to both its hallucinogenic and its stimulant properties. Ecstasy users have a higher risk of developing a hallucinogen use disorder than those using other hallucinogens. Among both adolescent and adult ecstasy users and users of other hallucinogens, the most frequently reported hallucinogen use disorder criteria are tolerance, hazardous use, use despite emotional or health problems, giving up activities in favor of use, and spending a lot of time obtaining, using, or recovering from the effects of use. As found for other substances, diagnostic criteria for other hallucinogen use disorder are arrayed along a single continuum of severity.

Given that a clinically significant withdrawal syndrome has not been consistently documented in humans, the diagnosis of hallucinogen withdrawal syndrome is not included in this manual and therefore is not part of the hallucinogen use disorder diagnostic criteria. However, there may be evidence of withdrawal from MDMA, with endorsement of any two or more withdrawal symptoms (e.g., malaise, appetite disturbance, mood changes [anxious, depressed, irritable], poor concentration, sleep disruption) or withdrawal avoidance observed in more than half of individuals in diverse samples of ecstasy users in the United States and internationally.

Associated Features

The characteristic symptom features of use of some hallucinogens can aid in diagnosis if urine or blood toxicology results are not available. For example, individuals who use LSD tend to experience visual hallucinations that can be frightening.

Prevalence

Other hallucinogen use disorder is rare. In the U.S. general population, about 0.1% of individuals age 12 or older endorsed the symptoms of past 12-month hallucinogen use disorder in 2018. The rate was 0.2% among those ages 12–17, 0.4% among those ages 18–25,

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and < 0.1% among those age 26 and older. Prevalence is higher in U.S. clinical samples (e.g., 19% in adolescents in treatment), and among select groups of individuals who use hallucinogens frequently (e.g., recent heavy ecstasy use) in the United States and Australia, 73.5% of adults and 77% of adolescents have a problematic pattern of use that may meet other hallucinogen use disorder criteria.

Development and Course

Prevalence of other hallucinogen use disorder by age among adolescents is unknown. Among U.S. adults age 18 years and older, most (90%) of those with other hallucinogen use disorder are ages 18–29, suggesting that the disorder is not often persistent and is concentrated in young adults.

Risk and Prognostic Factors

Temperamental. The use of specific hallucinogens (i.e., ecstasy, salvia) has been linked with high

sensation-seeking.

Environmental. On the basis of research in the United States, environmental risk factors of other hallucinogen use disorder include higher income, lower education, being never married, and residing in urban areas. Early onset of hallucinogen use has also been linked to transition to hallucinogen use disorder. Peer use of other drugs is also highly associated with ecstasy and salvia use.

Genetic and physiological. Among male twins, total variance due to additive genetics has been estimated to range from 26% to 79%, with inconsistent evidence for shared environmental influences.

Culture-Related Diagnostic Issues

Historically, hallucinogens have been used as part of established religious or spiritual practices, such as the use of peyote in the Native American Church and in Mexico. Ritual use by Indigenous populations of psilocybin obtained from certain types of mushrooms has occurred in South America, Mexico, and some areas in the United States, or of ayahuasca in the Santo Daime and União de Vegetal religious groups.

Sex- and Gender-Related Diagnostic Issues

Among U.S. adolescents, boys have greater 12-month prevalence rates of other hallucinogen use than girls, and these gender differences extend to specific hallucinogens, including LSD, MDMA, psilocybin, and salvia divinorum. Among U.S. adults, 60% of individuals with other hallucinogen use disorder are men. International research suggests that women administered MDMA may have greater subjective effects, such as altered state of consciousness, anxiety, and depression. No information from international studies is available regarding gender differences for other hallucinogen use disorder.

Diagnostic Markers

Laboratory testing can be useful in distinguishing among the different hallucinogens. However, because some agents (e.g., LSD) are so potent that as little as 75 micrograms can produce severe reactions, typical toxicological examination will not always reveal which substance has been used.

Functional Consequences of Other Hallucinogen Use Disorder

Although insufficient information exists to clearly note the functional consequences of other hallucinogen use disorder, complications of use of these substances have been

identified. Adverse effects of other hallucinogen use include those related to intoxication, such as hyperthermia, cardiac tachyarrhythmias, pneumothorax hypernatremia, motor incoordination, nystagmus, restlessness, hallucinations/delusions, mydriasis, increased alertness, and high blood pressure. Other more serious reactions related to consequences of repeated use of other hallucinogens include renal failure, hepatic failure, seizures, cerebral infarction, rhabdomyolysis,

cardiac complications, and hepatotoxicity.

There is evidence for persisting neurotoxic effects of MDMA/ecstasy use, including impairments in memory, psychological function, and neuroendocrine function; serotonin system dysfunction; and sleep disturbance; as well as adverse effects on brain microvasculature, white matter maturation, and damage to axons.

Differential Diagnosis

Other substance disorders. The effects of hallucinogen use must be distinguished from those of other substances (e.g., amphetamine use disorder, alcohol or sedative withdrawal), especially because contamination of the hallucinogens with other drugs is relatively common.

Hallucinogen intoxication and hallucinogen-induced mental disorders. Hallucinogen use disorder is differentiated from hallucinogen intoxication and hallucinogen-induced mental disorders (e.g., hallucinogen-induced psychotic disorder) in that hallucinogen use disorder describes a problematic pattern of hallucinogen use that involves impaired control over hallucinogen use, social impairment attributable to hallucinogen use, risky hallucinogen use (e.g., driving while intoxicated), and pharmacological symptoms (the development of tolerance), whereas hallucinogen intoxication and hallucinogen-induced mental disorders describe psychiatric syndromes that occur in the context of heavy use. Hallucinogen intoxication and hallucinogen-induced mental disorders occur frequently in individuals with hallucinogen use disorder. In such cases, a diagnosis of hallucinogen intoxication or a hallucinogen-induced mental disorder should be given in addition to a diagnosis of hallucinogen use disorder, the presence of which is indicated in the diagnostic code.

Independent mental disorders. Some of the effects of hallucinogen use may resemble symptoms of independent psychiatric disorders, such as schizophrenia and depressive and bipolar disorders. Discerning whether symptoms occurred before the intake of the drug is important in the differentiation of acute drug effects from a preexisting mental disorder. In particular, schizophrenia should be ruled out, as some affected individuals (e.g., individuals with schizophrenia who exhibit paranoia) may falsely attribute their symptoms to use of hallucinogens.

Comorbidity

Other hallucinogen use disorder is highly associated with cocaine use disorder, stimulant use disorder, other substance use disorder, tobacco (nicotine) use disorder, any personality disorder, posttraumatic stress disorder, and panic attacks.

Phencyclidine Intoxication

Diagnostic Criteria

- A. Recent use of phencyclidine (or a pharmacologically similar substance).
- B. Clinically significant problematic behavioral changes (e.g., belligerence,

assaultiveness, impulsiveness, unpredictability, psychomotor agitation, impaired judgment) that developed during, or shortly after, phencyclidine use.

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C. Within 1 hour, two (or more) of the following signs or symptoms:

Note: When the drug is smoked, “snorted,” or used intravenously, the onset may be particularly rapid.

1. Vertical or horizontal nystagmus.
2. Hypertension or tachycardia.
3. Numbness or diminished responsiveness to pain.
4. Ataxia.
5. Dysarthria.
6. Muscle rigidity.
7. Seizures or coma.
8. Hyperacusis.

D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication with another substance.

Coding note: The ICD-10-CM code depends on whether there is a comorbid phencyclidine use disorder. If a mild phencyclidine use disorder is comorbid, the ICD-10-CM code is **F16.120**, and if a moderate or severe phencyclidine use disorder is comorbid, the ICD-10-CM code is **F16.220**. If there is no comorbid phencyclidine use disorder, then the ICD-10-CM code is **F16.920**.

Note: In addition to the section “Functional Consequences of Phencyclidine Intoxication,” see the corresponding section in Phencyclidine Use Disorder.

Diagnostic Features

Phencyclidine intoxication reflects the clinically significant behavioral changes that occur shortly after ingestion of this substance (or a pharmacologically similar substance). The most common clinical presentations of phencyclidine intoxication include disorientation; confusion without hallucinations; nystagmus; numbness or diminished responsiveness to pain; ataxia; dysarthria; muscle rigidity; hyperacusis; and coma of varying severity. Other clinically significant behavioral changes associated with phencyclidine intoxication include violent behavior, extreme agitation, persecutory delusions, euphoria, retrograde amnesia, and hypertension.

Prevalence

Use of phencyclidine or related substances (e.g., ketamine) may be taken as an estimate of the prevalence of intoxication. Phencyclidine use is rare, with <0.1% of the U.S. population age 12 and older reporting past 12-month use in 2018. In surveys of U.S. students and young adults followed up from high school, past 12-month prevalence of ketamine use, which is assessed

separately from other substances, was estimated at about 1.2% among 12th graders and 0.5% among young adults, ages 19–28 years.

Diagnostic Markers

Laboratory testing may be useful, as phencyclidine is detectable in urine for up to 8 days following use, although the levels are only weakly associated with an individual's clinical presentation and may therefore not be useful for case management. Creatine phosphokinase and aspartate aminotransferase levels may be elevated.

Functional Consequences of Phencyclidine Intoxication

Phencyclidine intoxication produces extensive cardiovascular and neurological (e.g., seizures, dystonias, dyskinesias, catalepsy, hypothermia or hyperthermia) toxicity.

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Differential Diagnosis

In particular, in the absence of intact reality testing (i.e., without insight that the perceptual abnormalities are drug induced), an additional diagnosis of phencyclidine-induced psychotic disorder should be considered.

Other substance intoxication. Phencyclidine intoxication should be differentiated from intoxication due to other substances, including other hallucinogens; amphetamine, cocaine, or other stimulants; and anticholinergics, as well as withdrawal from benzodiazepines. Nystagmus and bizarre and violent behavior may distinguish intoxication due to phencyclidine from that due to other substances. Toxicological tests may be useful in making this distinction. However, the weak correlation between quantitative toxicology levels of phencyclidine and clinical presentation may diminish the utility of the laboratory findings for patient management.

Phencyclidine-induced mental disorders. Phencyclidine intoxication is distinguished from phencyclidine-induced mental disorders (e.g., phencyclidine-induced depressive disorder, with onset during intoxication) because the symptoms (e.g., depressed mood) in the latter disorders are in excess of those usually associated with phencyclidine intoxication, predominate in the clinical presentation, and are severe enough to warrant clinical attention.

Other medical conditions. Medical conditions to be considered include certain metabolic disorders like hypoglycemia and hyponatremia, central nervous system tumors, seizure disorders, sepsis, neuroleptic malignant syndrome, and vascular insults.

Comorbidity

Given the typical overlap of phencyclidine intoxication with phencyclidine use disorder, see “Comorbidity” under Phencyclidine Use Disorder for more details about co-occurring conditions that are likely to be encountered.

Other Hallucinogen Intoxication

Diagnostic Criteria

- A. Recent use of a hallucinogen (other than phencyclidine).
- B. Clinically significant problematic behavioral or psychological changes (e.g., marked anxiety or depression, ideas of reference, fear of “losing one’s mind,” paranoid ideation, impaired judgment) that developed during, or shortly after, hallucinogen use.
- C. Perceptual changes occurring in a state of full wakefulness and alertness (e.g., subjective intensification of perceptions, depersonalization, derealization, illusions, hallucinations, synesthesias) that developed during, or shortly after, hallucinogen use.
- D. Two (or more) of the following signs developing during, or shortly after, hallucinogen use:
 - 1. Pupillary dilation.
 - 2. Tachycardia.
 - 3. Sweating.
 - 4. Palpitations.
 - 5. Blurring of vision.
 - 6. Tremors.
 - 7. Incoordination.
- E. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication with another substance.

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Coding note: The ICD-10-CM code depends on whether there is a comorbid hallucinogen use disorder. If a mild hallucinogen use disorder is comorbid, the ICD-10-CM code is **F16.120**, and if a moderate or severe hallucinogen use disorder is comorbid, the ICD-10-CM code is **F16.220**. If there is no comorbid hallucinogen use disorder, then the ICD-10-CM code is **F16.920**.

Note: For information on Associated Features and Culture-Related Diagnostic Issues, see the corresponding sections in Other Hallucinogen Use Disorder.

Diagnostic Features

Other hallucinogen intoxication reflects the clinically significant behavioral or psychological changes that occur shortly after ingestion of a hallucinogen. Depending on the specific hallucinogen, the intoxication may last only minutes (e.g., for salvia) or several hours or longer (e.g., for LSD [lysergic acid diethylamide] or MDMA [3,4-methylenedioxymethamphetamine]).

Prevalence

The prevalence of other hallucinogen intoxication is not fully known but may be approximated based on the prevalence of use of the substances. In 2018, 1.5% of individuals ages 12–17 years in the United States reported use of hallucinogens in the past year; among individuals ages 18–25, the rate was 6.9%, and among those age 26 or older, the rate was 1.3%. Rates were consistently higher for boys and men than for girls and women in every age group.

Association With Suicidal Thoughts or Behavior

Other hallucinogen intoxication may lead to increased suicidal thoughts or behavior, although suicide is rare among individuals who use hallucinogens. Of note, a study of more than 135,000 randomly selected U.S. adults, including more than 19,000 individuals who use psychedelics, did not find evidence, after adjustment for sociodemographics, other drug use, and childhood depression, that lifetime psychedelic use is an independent risk factor for mental health problems, suicidal thoughts, or suicide attempts. In addition, one large U.S. population survey found that a lifetime history of hallucinogen use was associated with lower odds of mental distress and suicidal thoughts or behavior, although a causal relationship between hallucinogenic drugs and lower distress cannot be inferred from this study. On the basis of these findings, the relationship of other hallucinogen use to suicidal thoughts and behaviors is uncertain.

Functional Consequences of Other Hallucinogen Intoxication

Other hallucinogen intoxication can have serious consequences. The perceptual disturbances and impaired judgment associated with other hallucinogen intoxication can result in injuries or fatalities from automobile crashes, physical fights, or unintentional self-injury (e.g., cuts or falls from impaired depth perception). When other hallucinogens are consumed in combination with other drugs (including alcohol), coma can occur, with the duration and profundity of coma greater than when other hallucinogens are taken alone. Continued use of hallucinogens, particularly MDMA, has also been linked with neurotoxic effects. Adverse effects of other hallucinogen use include hyperthermia, cardiac tachyarrhythmias, pneumothorax hypernatremia, motor incoordination, nystagmus, restlessness, hallucinations/delusions, mydriasis, increased alertness, and high blood pressure. More serious reactions include renal failure, hepatic failure, seizures, cerebral infarction, rhabdomyolysis, cardiac complications, and hepatotoxicity.

Differential Diagnosis

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Other substance intoxication. Other hallucinogen intoxication should be differentiated from intoxication with amphetamine-type substances, cocaine, or other stimulants; anticholinergics, inhalants, and phencyclidine. Toxicological tests are useful in making this distinction, and determining the route of administration may also be useful.

Other conditions. Other disorders and conditions to be considered include schizophrenia, depression, withdrawal from other drugs (e.g., sedatives, alcohol), certain metabolic disorders (e.g., hypoglycemia), seizure disorders, tumors of the central nervous system, and vascular insults.

Hallucinogen persisting perception disorder. Other hallucinogen intoxication is distinguished from

hallucinogen persisting perception disorder because the symptoms in the latter continue episodically or continuously for weeks (or longer) after the most recent intoxication.

Hallucinogen-induced mental disorders. Other hallucinogen intoxication is distinguished from hallucinogen-induced mental disorders (e.g., hallucinogen-induced anxiety disorder, with onset during intoxication) because the symptoms (e.g., anxiety) in these latter disorders are in excess of those usually associated with other hallucinogen intoxication, predominate in the clinical presentation, and are severe enough to warrant independent clinical attention.

Comorbidity

Given the typical overlap of other hallucinogen intoxication with other hallucinogen use disorder, see “Comorbidity” under Other Hallucinogen Use Disorder for more details about co-occurring conditions that are likely to be encountered.

Hallucinogen Persisting Perception Disorder

Diagnostic Criteria	F16.983
A. Following cessation of use of a hallucinogen, the reexperiencing of one or more of the perceptual symptoms that were experienced while intoxicated with the hallucinogen (e.g., geometric hallucinations, false perceptions of movement in the peripheral visual fields, flashes of color, intensified colors, trails of images of moving objects, positive afterimages, halos around objects, macropsia and micropsia). B. The symptoms in Criterion A cause clinically significant distress or impairment in social, occupational, or other important areas of functioning. C. The symptoms are not attributable to another medical condition (e.g., anatomical lesions and infections of the brain, visual epilepsies) and are not better explained by another mental disorder (e.g., delirium, major neurocognitive disorder, schizophrenia) or hypnopompic hallucinations.	

Diagnostic Features

The hallmark of hallucinogen persisting perception disorder is the reexperiencing, when the individual is sober, of the perceptual disturbances that were experienced while the individual was intoxicated with the hallucinogen (Criterion A). The symptoms may include any perceptual perturbations, but visual disturbances tend to be predominant. Typical of the abnormal visual perceptions are geometric hallucinations, false perceptions of

movement in the peripheral visual fields, flashes of color, intensified colors, trails of images of moving objects (i.e., images left suspended in the path of a moving object as seen in stroboscopic

photography), perceptions of entire objects, visual snow, positive afterimages (i.e., a same-colored or complementary-colored “shadow” of an object remaining after removal of the object), halos around objects, or misperception of images as too large (macropsia) or too small (micropsia). Duration of the visual disturbances may be episodic or nearly continuous and must cause clinically significant distress or impairment in social, occupational, or other important areas of functioning (Criterion B). The disturbances may last for weeks, months, or years. Other explanations for the disturbances (e.g., brain lesions, preexisting psychosis, seizure disorders, migraine aura without headaches) must be ruled out (Criterion C).

Hallucinogen persisting perception disorder occurs primarily after LSD (lysergic acid diethylamide) use, but not exclusively. There does not appear to be a strong correlation between hallucinogen persisting perception disorder and number of occasions of hallucinogen use, with some instances of hallucinogen persisting perception disorder occurring in individuals with minimal exposure to hallucinogens. Some instances of hallucinogen persisting perception disorder may be triggered by use of other substances (e.g., cannabis or alcohol), adaptation to dark environments, exercise, and exposure to noise and photophobia.

Associated Features

Reality testing remains intact in individuals with hallucinogen persisting perception disorder (i.e., the individual is aware that the disturbance is linked to the effect of the drug). If this is not the case, another disorder might better explain the abnormal perceptions.

Prevalence

Prevalence estimates of hallucinogen persisting perception disorder are unknown. Initial prevalence estimates of the disorder among individuals who use hallucinogens is approximately 4.2%.

Development and Course

Little is known about the development of hallucinogen persisting perception disorder. Its course, as suggested by its name, is persistent, lasting for weeks, months, or even years in certain individuals.

Risk and Prognostic Factors

There is little evidence regarding risk factors for hallucinogen persisting perception disorder, although genetic factors have been suggested as a possible explanation underlying the susceptibility to LSD effects in this condition.

Functional Consequences of Hallucinogen Persisting Perception Disorder

Although hallucinogen persisting perception disorder remains a chronic condition in some cases, many individuals with the disorder are able to suppress the disturbances and continue to function normally.

Differential Diagnosis

Conditions to be ruled out include schizophrenia, other drug effects, neurodegenerative disorders, stroke, brain tumors, infections, and head trauma. Neuroimaging results in hallucinogen persisting perception disorder cases are typically negative. As noted earlier, reality testing remains intact (i.e., the individual is aware that the disturbance is linked to the

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effect of the drug); if this is not the case, another disorder (e.g., psychotic disorder, another medical condition) might better explain the abnormal perceptions.

Comorbidity

Common comorbid mental disorders accompanying hallucinogen persisting perception disorder are panic disorder, alcohol use disorder, major depressive disorder, bipolar I disorder, and schizophrenia spectrum disorders.

Phencyclidine-Induced Mental Disorders

Other phencyclidine-induced mental disorders are described in other chapters of the manual with disorders with which they share phenomenology (see the substance/medication-induced mental disorders in these chapters): phencyclidine-induced psychotic disorder (“Schizophrenia Spectrum and Other Psychotic Disorders”); phencyclidine-induced bipolar and related disorder (“Bipolar and Related Disorders”); phencyclidine-induced depressive disorder (“Depressive Disorders”); and phencyclidine-induced anxiety disorder (“Anxiety Disorders”). For phencyclidine-induced intoxication delirium and delirium induced by ketamine taken as prescribed, see the criteria and discussion of delirium in the chapter “Neurocognitive Disorders.” These phencyclidine-induced mental disorders are diagnosed instead of phencyclidine intoxication only when the symptoms are sufficiently severe to warrant independent clinical attention.

Hallucinogen-Induced Mental Disorders

The following other hallucinogen-induced mental disorders are described in other chapters of the manual with disorders with which they share phenomenology (see the substance/medication-induced mental disorders in these chapters): other hallucinogen–induced psychotic disorder (“Schizophrenia Spectrum and Other Psychotic Disorders”); other hallucinogen–induced bipolar and related disorder (“Bipolar and Related Disorders”); other hallucinogen–induced depressive disorder (“Depressive Disorders”); and other hallucinogen–induced anxiety disorder (“Anxiety Disorders”). For other hallucinogen intoxication delirium and delirium induced by other hallucinogens taken as prescribed, see the criteria and discussion of delirium in the chapter “Neurocognitive Disorders.” These hallucinogen-induced mental disorders are diagnosed instead of other hallucinogen intoxication only when the symptoms are sufficiently severe to warrant independent clinical attention.

Unspecified Phencyclidine-Related Disorder

F16.99

This category applies to presentations in which symptoms characteristic of a phencyclidine-related disorder that cause clinically significant distress or impairment in social, occupational, or other important areas of functioning predominate but do not meet the full criteria for any specific phencyclidine-related disorder or any of the disorders in the substance-related and addictive disorders diagnostic class.

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Unspecified Hallucinogen-Related Disorder

F16.99

This category applies to presentations in which symptoms characteristic of a hallucinogen-related disorder that cause clinically significant distress or impairment in social, occupational, or other important areas of functioning predominate but do not meet the full criteria for any specific hallucinogen-related disorder or any of the disorders in the substance-related and addictive disorders diagnostic class.

Inhalant-Related Disorders

Inhalant Use Disorder

Inhalant Intoxication

Inhalant-Induced Mental Disorders

Unspecified Inhalant-Related Disorder

Inhalant Use Disorder

Diagnostic Criteria

- A. A problematic pattern of use of a hydrocarbon-based inhalant substance leading to clinically significant impairment or distress, as manifested by at least two of the following, occurring within a 12-month period:
1. The inhalant substance is often taken in larger amounts or over a longer period than was intended.
 2. There is a persistent desire or unsuccessful efforts to cut down or control use

of the inhalant substance.

3. A great deal of time is spent in activities necessary to obtain the inhalant substance, use it, or recover from its effects.
4. Craving, or a strong desire or urge to use the inhalant substance.
5. Recurrent use of the inhalant substance resulting in a failure to fulfill major role obligations at work, school, or home.
6. Continued use of the inhalant substance despite having persistent or recurrent social or interpersonal problems caused or exacerbated by the effects of its use.
7. Important social, occupational, or recreational activities are given up or reduced because of use of the inhalant substance.
8. Recurrent use of the inhalant substance in situations in which it is physically hazardous.
9. Use of the inhalant substance is continued despite knowledge of having a persistent or recurrent physical or psychological problem that is likely to have been caused or exacerbated by the substance.
10. Tolerance, as defined by either of the following:
 - a. A need for markedly increased amounts of the inhalant substance to achieve intoxication or desired effect.
 - b. A markedly diminished effect with continued use of the same amount of the inhalant substance.

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Specify the particular inhalant: When possible, the particular substance involved should be named (e.g., “solvent use disorder”).

Specify if:

In early remission: After full criteria for inhalant use disorder were previously met, none of the criteria for inhalant use disorder have been met for at least 3 months but for less than 12 months (with the exception that Criterion A4, “Craving, or a strong desire or urge to use the inhalant substance,” may be met).

In sustained remission: After full criteria for inhalant use disorder were previously met, none of the criteria for inhalant use disorder have been met at any time during a period of 12 months or longer (with the exception that Criterion A4, “Craving, or a strong desire or urge to use the inhalant substance,” may be met).

Specify if:

In a controlled environment: This additional specifier is used if the individual is in an environment where access to inhalant substances is restricted.

Code based on current severity/remission: If an inhalant intoxication or another inhalant-induced mental disorder is also present, do not use the codes below for inhalant use disorder. Instead, the comorbid inhalant use disorder is indicated in the

4th character of the inhalant-induced disorder code (see the coding note for inhalant intoxication or a specific inhalant-induced mental disorder). For example, if there is comorbid inhalant-induced depressive disorder and inhalant use disorder, only the inhalant-induced depressive disorder code is given, with the 4th character indicating whether the comorbid inhalant use disorder is mild, moderate, or severe: F18.14 for mild inhalant use disorder with inhalant-induced depressive disorder or F18.24 for a moderate or severe inhalant use disorder with inhalant-induced depressive disorder.

Specify current severity/remission:

F18.10 Mild: Presence of 2–3 symptoms.

F18.11 Mild, In early remission

F18.11 Mild, In sustained remission

F18.20 Moderate: Presence of 4–5 symptoms.

F18.21 Moderate, In early remission

F18.21 Moderate, In sustained remission

F18.20 Severe: Presence of 6 or more symptoms.

F18.21 Severe, In early remission

F18.21 Severe, In sustained remission

Specifiers

“In a controlled environment” applies as a further specifier of remission if the individual is both in remission and in a controlled environment (i.e., in early remission in a controlled environment or in sustained remission in a controlled environment). Examples of these environments are closely supervised and substance-free jails, therapeutic communities, and locked hospital units.

The severity of individuals’ inhalant use disorder is assessed by the number of diagnostic criteria endorsed. Changing severity of individuals’ inhalant use disorder across time is reflected by reductions in the frequency (e.g., days used per month) and/or dose (e.g., tubes of glue per day) used, as assessed by the individual’s self-report, report of others, clinician’s observations, and biological testing (when practical).

Diagnostic Features

Examples of inhalant substances include volatile hydrocarbons, which comprise toxic gases from glues, fuels, paints, and other volatile compounds. When possible, the

particular substance involved should be named (e.g., “toluene use disorder”). However, most compounds that are inhaled are a mixture of several substances that can produce psychoactive effects, and it is often difficult to ascertain the exact substance responsible for the disorder. Unless there is clear evidence that a single, unmixed substance has been used, the general term *inhalant* should be used in recording the diagnosis. Disorders arising from inhalation of nitrous oxide or of amyl-, butyl-, or isobutyl nitrite are considered as other (or unknown) substance use

disorder.

Features of inhalant use disorder include repeated use of an inhalant substance despite the individual's knowing that the substance is causing serious problems for the individual (Criterion A9). Those problems are reflected in the diagnostic criteria.

Missing work or school or inability to perform typical responsibilities at work or school (Criterion A5), and continued use of the inhalant substance even though it causes arguments with family or friends, fights, and other social or interpersonal problems (Criterion A6), may be seen in inhalant use disorder. Limiting family contact, work or school obligations, or recreational activities (e.g., sports, games, hobbies) may also occur (Criterion A7). Use of inhalants when driving or operating dangerous equipment (Criterion A8) is also seen.

Tolerance (Criterion A10) is reported by about 10% of individuals who use inhalants. Because a clinically significant withdrawal syndrome has not been established with inhalant use, neither a diagnosis of inhalant withdrawal nor a corresponding diagnostic criterion for withdrawal complaints for inhalant use disorder is included. However, withdrawal symptoms may occur among inhalant users and individuals with moderate to severe inhalant use disorder, and these symptoms appear to be similar in frequency to withdrawal symptoms among those with moderate to severe cocaine use disorder.

Associated Features

A diagnosis of inhalant use disorder is supported by recurring episodes of intoxication with negative results in standard drug screens (which do not detect inhalants); possession, or lingering odors, of inhalant substances; peri-oral or peri-nasal "glue-sniffer's rash"; association with other individuals known to use inhalants; membership in groups with prevalent inhalant use (e.g., some native or aboriginal communities, homeless children in street gangs); easy access to certain inhalant substances; paraphernalia possession; presence of the disorder's characteristic medical complications (e.g., brain white matter pathology, rhabdomyolysis); and the presence of multiple other substance use disorders. Individuals with inhalant use disorder may present with symptoms of pernicious anemia, subacute combined degeneration of the spinal cord, major or mild neurocognitive disorder, brain atrophy, leukoencephalopathy, and many other nervous system disorders.

Prevalence

About 2.3% of American youth ages 12–17 years have used inhalants in the past 12 months, with 0.1% having a pattern of use that meets criteria for inhalant use disorder. Among U.S. adults, age 18 years and older, past 12-month prevalence of inhalant use is about 0.21%, with 0.04% having a pattern of use that meets criteria for an inhalant use disorder. Among youth, the prevalence of past 12-month inhalant use is highest among non-Hispanic Whites and individuals reporting more than one racialized identity and lowest among American Indians/Alaska Natives. Twelve-month prevalence rates of inhalant use and inhalant use disorder among adults are highest among non-Hispanic Whites and lowest among non-Hispanic Blacks and American Indians/Alaska Natives.

Development and Course

The declining prevalence in the United States of inhalant use and inhalant use disorder after adolescence (from 2.3% during adolescence to 0.1% in early adulthood for inhalant use and from 0.1% to 0.04% for inhalant use disorder) indicates that the disorder usually

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remits in early adulthood. Inhalant use disorder is rare in prepubertal children, most common in adolescents and young adults, and uncommon in older persons. Calls to poison-control centers for “intentional abuse” of inhalants peak with calls involving individuals at age 14 years. Those with inhalant use disorder extending into adulthood demonstrate earlier onset of inhalant use, use of multiple inhalants, and more frequent inhalant use.

Risk and Prognostic Factors

Temperamental. Predictors of inhalant use disorder include sensation seeking and impulsivity.

Environmental. Inhalant gases are widely and legally available, increasing the risk of misuse. Childhood maltreatment or trauma also is associated with youthful progression from inhalant non-use to inhalant use disorder.

Genetic and physiological. *Behavioral disinhibition* is a highly heritable general propensity to not constrain behavior in socially acceptable ways, to break social norms and rules, and to take dangerous risks, pursuing rewards excessively despite dangers of adverse consequences. Youths with strong behavioral disinhibition show risk factors for inhalant use disorder: early-onset substance use disorder, multiple substance involvement, and early conduct problems. Because behavioral disinhibition is under strong genetic influence, youths in families with substance use and antisocial behaviors are at elevated risk for inhalant use disorder.

Culture-Related Diagnostic Issues

Internationally, certain isolated Indigenous communities have experienced a high prevalence of inhalant problems. Also, in some low- and middle-income countries, groups of homeless children living on the streets have extensive inhalant use problems because of the effects of poverty and the availability and affordability of the substances, and as a way to cope with homelessness.

Sex- and Gender-Related Diagnostic Issues

Although the past 12-month prevalence of inhalant use disorder in the United States is almost identical among adolescent boys and girls, the disorder is very rare among adult women.

Diagnostic Markers

Urine, breath, or saliva tests may be valuable for assessing concurrent use of non-inhalant substances by individuals with inhalant use disorder. However, technical problems and the considerable expense of analyses make frequent biological testing for inhalants themselves impractical.

Association With Suicidal Thoughts or Behavior

In the United States, adolescent and adult inhalant use and inhalant use disorder are associated with suicidal thoughts and behavior, especially among individuals reporting symptoms of anxiety and depression and histories of trauma.

Functional Consequences of Inhalant Use Disorder

Because of inherent toxicity, use of inhalants can be fatal. Death can occur from anoxia, cardiac dysfunction, extreme allergic reaction, severe injury to the lungs, vomiting, accidents or injury, or central nervous system depression. Moreover, any inhaled volatile hydrocarbons may produce “sudden sniffing death” from cardiac arrhythmia. Inhalant use impairs neurobehavioral function and causes various neurological, gastrointestinal, cardiovascular, and pulmonary problems.

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Long-term inhalant users are at increased risk for tuberculosis, HIV/AIDS, sexually transmitted diseases, depression, anxiety, bronchitis, asthma, and sinusitis.

Differential Diagnosis

Inhalant exposure (unintentional) from industrial or other accidents. A diagnosis of inhalant use disorder only applies if the inhalant exposure is intentional.

Inhalant intoxication, without meeting criteria for inhalant use disorder. Inhalant intoxication occurs frequently during inhalant use disorder but also may occur among individuals whose use does not meet criteria for inhalant use disorder.

Inhalant intoxication meeting criteria for inhalant use disorder, and inhalant-induced mental disorders.

Inhalant use disorder is differentiated from inhalant intoxication and inhalant-induced mental disorders (e.g., inhalant-induced depressive disorder) in that inhalant use disorder describes a problematic pattern of inhalant use that involves impaired control over inhalant use, social impairment attributable to inhalant use, risky inhalant use (e.g., inhalant use despite medical complications), and pharmacological symptoms (the development of tolerance), whereas inhalant intoxication and inhalant-induced mental disorders describe psychiatric syndromes that develop in the context of heavy use. Inhalant intoxication and inhalant-induced mental disorders occur frequently in individuals with inhalant use disorder. In such cases, a diagnosis of inhalant intoxication or an inhalant-induced mental disorder should be given in addition to a diagnosis of inhalant use disorder, the presence of which is indicated in the diagnostic code.

Other substance use disorders, especially those involving sedating substances (e.g., alcohol, benzodiazepines, barbiturates).

Inhalant use disorder commonly co-occurs with other substance use disorders, and the symptoms of the disorders may be similar and overlapping. To disentangle symptom patterns, it is helpful to inquire about which symptoms persisted during periods when some of the substances were not being used.

Comorbidity

Individuals with inhalant use disorder receiving clinical care often have numerous other substance use, mood, anxiety, and personality disorders. Inhalant use disorder commonly co-

occurs with conduct disorder in adolescents and with antisocial personality disorder. Individuals with inhalant use disorder may have comorbid symptoms of hepatic or renal damage, rhabdomyolysis, methemoglobinemia, or symptoms of other gastrointestinal, cardiovascular, or pulmonary diseases.

Inhalant Intoxication

Diagnostic Criteria

- A. Recent intended or unintended short-term, high-dose exposure to inhalant substances, including volatile hydrocarbons such as toluene or gasoline.
- B. Clinically significant problematic behavioral or psychological changes (e.g., belligerence, assaultiveness, apathy, impaired judgment) that developed during, or shortly after, exposure to inhalants.
- C. Two (or more) of the following signs or symptoms developing during, or shortly after, inhalant use or exposure:
 - 1. Dizziness.
 - 2. Nystagmus.
 - 3. Incoordination.

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- 4. Slurred speech.
 - 5. Unsteady gait.
 - 6. Lethargy.
 - 7. Depressed reflexes.
 - 8. Psychomotor retardation.
 - 9. Tremor.
 - 10. Generalized muscle weakness.
 - 11. Blurred vision or diplopia.
 - 12. Stupor or coma.
 - 13. Euphoria.
- D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication with another substance.

Coding note: The ICD-10-CM code depends on whether there is a comorbid inhalant use disorder. If a mild inhalant use disorder is comorbid, the ICD-10-CM code is **F18.120**, and if a moderate or severe inhalant use disorder is comorbid, the ICD-10-CM code is **F18.220**. If there is no comorbid inhalant use disorder, then the ICD-10-CM code is **F18.920**.

Note: For information on Development and Course, Risk and Prognostic Factors, Culture-Related Diagnostic Issues, and Diagnostic Markers, see the corresponding sections in Inhalant Use Disorder.

Diagnostic Features

The essential feature of inhalant intoxication is the presence of clinically significant problematic behavioral or psychological changes that develop during, or immediately after, intended or unintended inhalation of a volatile hydrocarbon substance. When possible, the particular substance involved should be named (e.g., toluene intoxication). Intoxication clears within a few minutes to a few hours after the exposure ends. Thus, inhalant intoxication usually occurs in brief episodes that may recur with further inhalant use.

Associated Features

Inhalant intoxication may be indicated by evidence of possession, or lingering odors, of inhalant substances (e.g., glue, paint thinner, gasoline, butane lighters); other features may include euphoria, relaxation, headache, rapid heartbeat, confusion, talkativeness, blurred vision, amnesia, slurred speech, irritability, nausea, fatigue, burning in eyes or throat, grandiosity, chest pain, auditory or visual hallucinations, and dissociation.

Prevalence

The prevalence of actual episodes of inhalant intoxication in the general population is unknown, but it is probable that a majority of inhalant users would at some time exhibit behavioral or psychological changes and symptoms that would meet criteria for inhalant intoxication. Therefore, the prevalence of inhalant use and the prevalence of inhalant intoxication are likely similar. In 2017, inhalant use in the past year was reported by 0.6% of all Americans older than 12 years; the prevalence was highest in younger age groups (2.3% for individuals ages 12–17 years, 1.6% for individuals ages 18–25 years, and 0.3% for individuals age 26 and older).

Sex- and Gender-Related Diagnostic Issues

Gender differences in the prevalence of inhalant intoxication in the general population are unknown. Regarding gender differences in the prevalence of inhalant use in the United

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States, 0.8% of boys/men older than 12 years and 0.5% of girls/women older than 12 years have used inhalants in the previous year, but in the younger age groups differences are minimal or girls may have slightly higher prevalence (e.g., among adolescents ages 12–17 years, 2.4% of girls and 2.2% of boys have used inhalants in the past year).

Functional Consequences of Inhalant Intoxication

Use of inhaled substances in a closed container, such as a plastic bag over the head, may lead to unconsciousness, anoxia, and death. Separately, “sudden sniffing death,” likely from cardiac arrhythmia or arrest, may occur with various volatile inhalants. The enhanced toxicity of certain

volatile inhalants, such as butane or propane, also causes fatalities. Although inhalant intoxication itself is of short duration, it may produce persisting medical and neurological problems, especially if the intoxications are frequent. Clinically significant correlates of inhalant intoxication include reckless behaviors (e.g., taking foolish risks, getting into fights, having unprotected sex), antisocial behaviors (cruelty, damaging property, arrests), and having serious accidents.

Differential Diagnosis

Intoxication from other substances, especially from sedating substances (e.g., alcohol, benzodiazepines, barbiturates).

These disorders may have similar signs and symptoms, but intoxication attributable to other intoxicants may be identified via a toxicology screen. Differentiating the source of the intoxication may involve discerning evidence of inhalant exposure as described for inhalant use disorder. A diagnosis of inhalant intoxication may be suggested by possession or lingering odors of inhalant substances (e.g., glue, paint thinner, gasoline, butane lighters); paraphernalia possession (e.g., rags or bags for concentrating glue fumes); perioral or perinasal “glue-sniffer’s rash”; reports from family or friends that the intoxicated individual possesses or uses inhalants; or apparent intoxication despite negative results on standard drug screens (which usually fail to identify inhalants).

Inhalant-induced mental disorders. Inhalant intoxication is distinguished from inhalant-induced mental disorders (e.g., inhalant-induced anxiety disorder, with onset during intoxication) because the symptoms (e.g., anxiety) in these latter disorders are in excess of those usually associated with inhalant intoxication, predominate in the clinical presentation, and are severe enough to warrant independent clinical attention.

Other toxic, metabolic, traumatic, neoplastic, or infectious disorders that impair brain function and cognition.

Numerous neurological and other medical conditions may produce the clinically significant behavioral or psychological changes (e.g., belligerence, assaultiveness, apathy, impaired judgment) that also characterize inhalant intoxication.

Comorbidity

Given the typical overlap of inhalant intoxication with inhalant use disorder, see “Comorbidity” under Inhalant Use Disorder for more details about co-occurring conditions that are likely to be encountered.

Inhalant-Induced Mental Disorders

The following inhalant-induced mental disorders are described in other chapters of the manual with disorders with which they share phenomenology (see the substance/medication-induced mental disorders in these chapters): inhalant-induced psychotic disorder (“Schizophrenia Spectrum and Other Psychotic Disorders”); inhalant-induced depressive

disorder (“Depressive Disorders”); inhalant-induced anxiety disorder (“Anxiety Disorders”); and

inhalant-induced major or mild neurocognitive disorder (“Neurocognitive Disorders”). For inhalant intoxication delirium, see the criteria and discussion of delirium in the chapter “Neurocognitive Disorders.” These inhalant-induced mental disorders are diagnosed instead of inhalant intoxication only when symptoms are sufficiently severe to warrant independent clinical attention.

Unspecified Inhalant-Related Disorder

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This category applies to presentations in which symptoms characteristic of an inhalant-related disorder that cause clinically significant distress or impairment in social, occupational, or other important areas of functioning predominate but do not meet the full criteria for any specific inhalant-related disorder or any of the disorders in the substance-related and addictive disorders diagnostic class.

Opioid-Related Disorders

Opioid Use Disorder

Opioid Intoxication

Opioid Withdrawal

Opioid-Induced Mental Disorders

Unspecified Opioid-Related Disorder

Opioid Use Disorder

Diagnostic Criteria

- A. A problematic pattern of opioid use leading to clinically significant impairment or distress, as manifested by at least two of the following, occurring within a 12-month period:
1. Opioids are often taken in larger amounts or over a longer period than was intended.
 2. There is a persistent desire or unsuccessful efforts to cut down or control opioid use.
 3. A great deal of time is spent in activities necessary to obtain the opioid, use the opioid, or recover from its effects.

4. Craving, or a strong desire or urge to use opioids.
5. Recurrent opioid use resulting in a failure to fulfill major role obligations at work, school, or home.
6. Continued opioid use despite having persistent or recurrent social or interpersonal problems caused or exacerbated by the effects of opioids.
7. Important social, occupational, or recreational activities are given up or reduced because of opioid use.

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8. Recurrent opioid use in situations in which it is physically hazardous.
9. Continued opioid use despite knowledge of having a persistent or recurrent physical or psychological problem that is likely to have been caused or exacerbated by the substance.
10. Tolerance, as defined by either of the following:
 - a. A need for markedly increased amounts of opioids to achieve intoxication or desired effect.
 - b. A markedly diminished effect with continued use of the same amount of an opioid.

Note: This criterion is not considered to be met for those taking opioids solely under appropriate medical supervision.

11. Withdrawal, as manifested by either of the following:
 - a. The characteristic opioid withdrawal syndrome (refer to Criteria A and B of the criteria set for opioid withdrawal).
 - b. Opioids (or a closely related substance) are taken to relieve or avoid withdrawal symptoms.

Note: This criterion is not considered to be met for those individuals taking opioids solely under appropriate medical supervision.

Specify if:

In early remission: After full criteria for opioid use disorder were previously met, none of the criteria for opioid use disorder have been met for at least 3 months but for less than 12 months (with the exception that Criterion A4, "Craving, or a strong desire or urge to use opioids," may be met).

In sustained remission: After full criteria for opioid use disorder were previously met, none of the criteria for opioid use disorder have been met at any time during a period of 12 months or longer (with the exception that Criterion A4, "Craving, or a strong desire or urge to use opioids," may be met).

Specify if:

On maintenance therapy: This additional specifier is used if the individual is taking a prescribed agonist medication such as methadone or buprenorphine and none of the criteria for opioid use disorder have been met for that class of

medication (except tolerance to, or withdrawal from, the agonist). This category also applies to those individuals being maintained on a partial agonist, an agonist/antagonist, or a full antagonist such as oral naltrexone or depot naltrexone.

In a controlled environment: This additional specifier is used if the individual is in an environment where access to opioids is restricted.

Code based on current severity/remission: If an opioid intoxication, opioid withdrawal, or another opioid-induced mental disorder is also present, do not use the codes below for opioid use disorder. Instead, the comorbid opioid use disorder is indicated in the 4th character of the opioid-induced disorder code (see the coding note for opioid intoxication, opioid withdrawal, or a specific opioid-induced mental disorder). For example, if there is comorbid opioid-induced depressive disorder and opioid use disorder, only the opioid-induced depressive disorder code is given, with the 4th character indicating whether the comorbid opioid use disorder is mild, moderate, or severe: F11.14 for mild opioid use disorder with opioid-induced depressive disorder or F11.24 for a moderate or severe opioid use disorder with opioid-induced depressive disorder.

Specify current severity/remission:

F11.10 Mild: Presence of 2–3 symptoms.

F11.11 Mild, In early remission

F11.11 Mild, In sustained remission

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F11.20 Moderate: Presence of 4–5 symptoms.

F11.21 Moderate, In early remission

F11.21 Moderate, In sustained remission

F11.20 Severe: Presence of 6 or more symptoms.

F11.21 Severe, In early remission

F11.21 Severe, In sustained remission

Specifiers

The “on maintenance therapy” specifier applies as a further specifier of remission if the individual is both in remission and receiving maintenance therapy. “In a controlled environment” applies as a further specifier of remission if the individual is both in remission and in a controlled environment (i.e., in early remission in a controlled environment or in sustained remission in a controlled environment). Examples of these environments are closely supervised and substance-free jails, therapeutic communities, and locked hospital units.

Changing severity across time in an individual is also reflected by reductions in the frequency (e.g., days of use per month) and/or dose (e.g., injections or number of pills) of an opioid, as assessed by the individual’s self-report, report of knowledgeable others, clinician’s

observations, and biological testing.

Diagnostic Features

The opioids include natural opioids (e.g., morphine, codeine), semisynthetics (e.g., heroin, oxycodone, hydrocodone, hydromorphone, oxymorphone), and synthetics with morphine-like action (e.g., methadone, meperidine, tramadol, fentanyl, carfentanil). Medications such as pentazocine and buprenorphine that have both opiate agonist and antagonist effects are also included in this class because, especially at lower doses, their agonist properties produce similar physiological and behavioral effects as classic opioid agonists. Opioids are prescribed as analgesics, anesthetics, antidiarrheal agents, or cough suppressants. Heroin is one of the most commonly misused drugs of this class and is usually taken by injection, although it can be smoked or “snorted,” especially when very pure heroin is available. Fentanyl is typically injected, both medically and nonmedically, and is used medically in transdermal and transmucosal formulations, whereas cough suppressants and antidiarrheal agents are taken orally. The other opioids are generally taken both by injection and orally.

Opioid use disorder can arise from prescription opioids or illicit opioids (e.g., heroin and, especially in recent years, fentanyl-related synthetic opioids). Opioid use disorder consists of signs and symptoms reflecting compulsive, prolonged self-administration of opioid substances either for a purpose other than a legitimate medical one or for use in a “non-medical” manner (i.e., greatly exceeding the amount prescribed for a medical condition). For example, an individual with adequate doses of prescribed analgesic opioid medication for pain relief who uses significantly more of the medication than prescribed, and not only because of persistent pain, is engaging in nonmedical opioid use and may have an opioid use disorder. Most individuals with opioid use disorder have tolerance and experience withdrawal on abrupt cessation or reduction in opioid use. Similar to processes that occur with other psychoactive substances, individuals with opioid use disorder often develop conditioned responses to drug-related stimuli (e.g., cue-reactive craving on seeing drug images or paraphernalia). These responses probably contribute to relapse, are difficult to extinguish, and typically persist long after withdrawal is completed.

Individuals with opioid use disorder tend to develop such regular patterns of compulsive drug use that daily activities are planned around obtaining and administering

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opioids. Prescription opioids used nonmedically can be obtained from family or friends, from physicians by falsifying or exaggerating medical problems, by receiving simultaneous prescriptions from several physicians, or via purchase on the illegal market. Health care professionals with opioid use disorder can obtain opioids by writing prescriptions for themselves or by diverting opioids that have been prescribed for individuals or from pharmacy supplies.

Associated Features

An attempt to achieve opioid intoxication may result in fatal or nonfatal opioid overdose. Opioid overdose is characterized by unconsciousness, respiratory depression, and pinpoint pupils. However, opioid overdoses can also occur in the absence of intoxication-seeking drug use.

Opioid overdoses have increased exponentially in the United States since 1999. Up to 2009, opioid overdoses were mainly due to prescribed opioids, but since 2010, overdoses due to heroin began a sharp rise, and additionally, since 2015, fatal overdoses due to synthetic opioids other than methadone (generally fentanyl) have outnumbered overdoses due to prescribed opioids.

Opioid use disorder can be associated with a history of drug-related crimes (e.g., possession or distribution of drugs, forgery, burglary, robbery, larceny, receiving stolen goods). Among health care professionals and individuals who have ready access to controlled substances, a different pattern of illegal activities may involve problems with state licensing boards, professional staffs of hospitals, or other administrative agencies. Marital difficulties (including divorce), unemployment, and irregular employment can be associated with opioid use disorder at all socioeconomic levels.

Prevalence

The prevalence of nonmedical prescription opioid use among U.S. adults age 18 and older is 4.1%–4.7%, with rates of use higher in adults ages 18–25 than in those age 26 and older (5.5% vs. 3.4%, respectively). The prevalence of heroin use in the United States is 0.3%–0.4% and is higher among adults ages 18–25 (0.5%–0.7%) than in other age groups. In U.S. adolescents ages 12–17, 2.8%–3.9% use prescription opioids nonmedically, with higher rates in older adolescents than in younger adolescents. Heroin use in adolescents is quite low (< 0.05%–0.1%).

The prevalence of prescription opioid use disorder among U.S. adults age 18 and older (DSM-IV or DSM-5 criteria) is 0.6%–0.9%, and the prevalence of heroin use disorder (DSM-IV or DSM-5 criteria) is 0.1%–0.3%. Among those ages 12–17, prevalence of prescription opioid use disorder is 0.4%, and heroin use disorder is rare (essentially 0%). In the United States, rates of opioid use disorder (prescription opioids and heroin) are higher among men than women, among young adults than older adults, and among those with lower income or education. Among U.S. adults in 2012–2013, the prevalence of nonmedical prescription opioid use disorder varied by ethnoracial group: 1.42% in Native Americans, 1.04% in African Americans, 0.96% in non-Latinx Whites, 0.70% in Latinx, and 0.16% in Asian Americans or Pacific Islanders. Rates based on household surveys may underestimate national prevalence by omitting individuals in institutions and jail or prison, whose rates are likely to be much higher.

Globally in 2016, there were 26.8 million cases of DSM-IV opioid dependence, with an age-standardized prevalence of 353.0 cases per 100,000 people; prevalence of opioid dependence across geographic regions ranged from 0.14% to 0.46%.

Development and Course

Opioid use disorder can begin at any age. In the United States, problems associated with opioid use are most commonly first observed in the late teens or early 20s, with a longer interval between first opioid use and onset of disorder for prescription opioids than for

heroin. Early use can reflect a desire for relief from life stressors or psychological pain. Long-term studies show that once an opioid use disorder that requires treatment develops, it can continue over many years, with brief periods of abstinence in some individuals but long-term

abstinence only in a minority. An exception occurred among U.S. soldiers who became dependent on opioids while serving in the Vietnam War; over 90% had long-term abstinence from opioids after returning to the United States, although many subsequently experienced problems with alcohol, amphetamines, or suicidal thoughts or behavior.

Risk and Prognostic Factors

In addition to an association with more frequent nonmedical prescription opioid use, adult prescription opioid use disorder is associated with most other substance use disorders. Opioid use disorder is highly associated with externalizing traits such as novelty-seeking, impulsivity, and disinhibition. Family, peer, and social environmental factors all increase the risk for opioid use disorder. Family and twin studies also indicate a strong genetic contribution to the risk for opioid use disorders, although identifying the specific genetic variants contributing to genetic risk has been slow. Peer factors may relate to genetic predisposition in terms of how individuals select their environments, including their peers.

Culture-Related Diagnostic Issues

Individuals from socially oppressed ethnoracial groups were historically overrepresented among individuals with opioid use disorder. However, over time, opioid use disorder has become more common among White individuals, suggesting that the widespread availability of opioids and other social factors (e.g., changes in rates of poverty and unemployment) have an impact on prevalence. Consistent with these factors, despite small variations between ethnoracial groups in the psychometric performance of opioid use disorder criterion items, the criteria for opioid use disorder perform equally well across ethnoracial groups.

Sex- and Gender-Related Diagnostic Issues

Women with opioid use disorder appear more likely than men to have initiated opioid use in response to sexual abuse and violence, and they are more likely than men to be introduced to the drug by a partner. There is substantial evidence of telescoping among women in that they progress to a use disorder more quickly than men after first use; women also appear to be more ill when entering treatment facilities than are men, as noted in a large sample of heroin users in Italy.

Diagnostic Markers

Routine urine toxicology test results are often positive for opioid drugs in individuals with opioid use disorder. Urine test results remain positive for most opioids (e.g., heroin, morphine, codeine, oxycodone, propoxyphene) for 12–36 hours after administration. Some opioids, such as fentanyl and oxycodone, are not detected by standard urine tests (which test for morphine), but can be identified by more specialized procedures for several days after use. Similarly, methadone and buprenorphine (or buprenorphine/naloxone combinations) will not cause a positive result on routine tests for opiates; they require specific tests that can detect these substances for several days up to more than 1 week.

Although not specific markers of opioid use disorder, laboratory evidence of the presence of other substances (e.g., cocaine, marijuana, alcohol, amphetamines, benzodiazepines) is common

in heroin users. In addition, screening test results for hepatitis A, B, and C virus are often positive in injection opioid users, either for hepatitis antigen (signifying active infection) or for hepatitis antibody (signifying past infection). Mildly elevated liver

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function test results are common, either as a result of resolving hepatitis or from toxic injury to the liver due to contaminants that have been mixed with the injected opioid. HIV is also prevalent in injection opioid users. Subtle changes in cortisol secretion patterns and body temperature regulation have been observed for up to 6 months following opioid withdrawal.

Association With Suicidal Thoughts or Behavior

Opioid use disorder is associated with a heightened risk for suicide attempts and suicide. Some suicide risk factors overlap with risk factors for an opioid use disorder. In addition, repeated opioid intoxication or withdrawal may be associated with severe depressions that although temporary can be intense enough to lead to suicide attempts and suicide. Nonfatal accidental opioid overdose and attempted suicide are distinct phenomena that can be difficult to differentiate but should not be mistaken for each other, if possible.

Findings from the Global Burden of Disease Study 2010 showed that among drugs of abuse, suicide is a common cause of death among regular users of opioids. Evidence suggests that suicides are undercounted or often misclassified in opioid-poisoning data. In a study of the Veterans Health Administration (VHA) national medical records, after adjustment for psychiatric comorbidity, opioid use disorder elevated the risk for suicide mortality, with greater increase in risk among women than among men. In another study also using VHA national medical records, among veterans prescribed opioids for chronic pain, suicide mortality increased with higher opioid doses, even after demographic and clinical factors were taken into account. A follow-up of a U.S. national cohort of adults with a history of an opioid overdose found that the standardized mortality ratio (SMR; the ratio between the observed number of deaths in a study population and the number of deaths that would be expected) was 25.9 for suicide, with a higher SMR for women than for men. A review posited that the reasons for the increased risk for suicide among opioid users were related to shared risk factors, namely, comorbid mental disorders and pain.

Functional Consequences of Opioid Use Disorder

Physiologically, opioid use is associated with a lack of mucous membrane secretions, causing dry mouth and nose. Slowing of gastrointestinal activity and a decrease in gut motility can produce severe constipation. Visual acuity may be impaired as a result of pupillary constriction with acute administration. In individuals who inject opioids, sclerosed veins (“tracks”) and puncture marks on the lower portions of the upper extremities are common. Veins sometimes become so severely sclerosed that peripheral edema develops, and individuals switch to injecting in veins in the legs, neck, or groin. When these veins become unusable, individuals often inject directly into their subcutaneous tissue (“skin-popping”), resulting in cellulitis, abscesses, and circular-appearing scars from healed skin lesions. Tetanus and *Clostridium botulinum* infections are rare but serious consequences of injecting opioids, especially with contaminated needles.

Infections may also occur in other organs and include bacterial endocarditis, hepatitis, and HIV infection. Hepatitis C infections, for example, may occur in up to 90% of individuals who inject opioids. In addition, the prevalence of HIV infection is high among individuals who inject drugs, a large proportion of whom are individuals with opioid use disorder. For example, HIV infection rates are as high as 60% among heroin users in some areas of the United States and the Russian Federation. However, the incidence may be much lower in areas where access to clean injection material and paraphernalia is facilitated.

Tuberculosis is a particularly serious problem among individuals who use drugs intravenously, especially those who are dependent on heroin; infection is usually asymptomatic and evident only by the presence of a positive tuberculin skin test or tuberculosis blood test (interferon gamma release assay). However, many cases of active tuberculosis have been found, especially among those who are infected with HIV. These individuals

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often have a newly acquired infection but also are likely to experience reactivation of a prior infection because of impaired immune function.

Individuals who sniff heroin or other opioids into the nose (insufflation, or “snorting”) often develop irritation of the nasal mucosa, sometimes accompanied by perforation of the nasal septum. Difficulties in sexual functioning are common. Males often experience erectile dysfunction during intoxication or chronic use. Females commonly have disturbances of reproductive function and irregular menses.

Although acute opioid use produces analgesia, chronic use can produce hyperalgesia (opioid-induced hyperalgesia), a condition characterized by increased sensitivity to pain. Physiological dependence on opioids may occur in about half of the infants born to females with opioid use disorder. This can produce a severe withdrawal syndrome in the neonate requiring medical treatment and has increased markedly in prevalence.

The mortality rate in individuals with opioid use disorder is 6–20 times greater than in the general population. Fatal overdoses due to prescription opioids increased dramatically in the United States since 1999, with almost 400,000 such deaths occurring since then, and the rate of such overdoses is now five times higher than in 1999. Fatal overdoses due to heroin began a sharp increase in 2010, and since 2013, fatal overdoses due to synthetic opioids (e.g., fentanyl) increased so sharply that these rates were almost double the rates for prescription opioid or heroin overdoses by 2017. Nonfatal opioid overdoses resulting in hospitalization and emergency department visits have increased as well. Although not all risk factors for opioid use disorder and opioid overdose are the same, substantial overlap exists, making the risk for overdose one of the most serious potential consequences of opioid use disorder. Individuals with opioid use disorder are also at increased risk for mortality from many medical conditions (e.g., hepatitis, HIV infection, tuberculosis, cardiovascular disease). Death can also result from accidents, injuries, or other general medical complications.

Differential Diagnosis

Opioid intoxication, opioid withdrawal, and opioid-induced mental disorders. Opioid use disorder is differentiated from opioid intoxication, opioid withdrawal, and opioid-induced mental disorders

(e.g., opioid-induced depressive disorder) in that opioid use disorder describes a problematic pattern of opioid use that involves impaired control over opioid use, social impairment attributable to opioid use, risky opioid use (e.g., continued opioid use despite medical complications), and pharmacological symptoms (the development of tolerance or withdrawal), whereas opioid intoxication, opioid withdrawal, and opioid-induced mental disorders describe psychiatric syndromes that occur in the context of heavy use. Opioid intoxication, opioid withdrawal, and opioid-induced mental disorders occur frequently in individuals with opioid use disorder. In such cases, a diagnosis of opioid intoxication, opioid withdrawal, or opioid-induced mental disorder should be given in addition to a diagnosis of opioid use disorder, the presence of which is indicated in the diagnostic code.

Other substance intoxication. Alcohol intoxication and sedative, hypnotic, or anxiolytic intoxication can cause a clinical picture that resembles that of opioid intoxication. A diagnosis of alcohol or sedative, hypnotic, or anxiolytic intoxication can usually be made based on the absence of pupillary constriction or the lack of a response to naloxone challenge. In some cases, intoxication may be due both to opioids and to alcohol or other sedatives. In these cases, the naloxone challenge will not reverse all of the sedative effects.

Other withdrawal disorders. The anxiety and restlessness associated with opioid withdrawal resemble symptoms seen in sedative-hypnotic withdrawal. However, opioid withdrawal is also accompanied by rhinorrhea, lacrimation, and pupillary dilation, which are not seen in sedative-type withdrawal. Dilated pupils are also seen in hallucinogen intoxication and stimulant intoxication. However, other signs or symptoms of opioid

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withdrawal, such as nausea, vomiting, diarrhea, abdominal cramps, rhinorrhea, or lacrimation, are not present.

Independent mental disorders. Some of the effects of opioid use may resemble symptoms (e.g., depressed mood) of an independent mental disorder (e.g., persistent depressive disorder). Opioids are less likely to produce symptoms of mental disturbance than are most other drugs of abuse.

Comorbidity

Other than overdose, the most common medical comorbidities associated with opioid use disorder are viral (e.g., HIV, hepatitis C virus) and bacterial infections, particularly among injection heroin users. These infections are less common in prescription opioid use disorder.

Research with nationally representative samples of the U.S. population has found that opioid use disorder is often associated with other substance use disorders, especially those involving tobacco, alcohol, cannabis, stimulants, and benzodiazepines. Individuals with opioid use disorder are at risk for the development of persistent depressive disorder or major depressive disorder. These symptoms may represent an opioid-induced depressive disorder or an exacerbation of a preexisting independent depressive disorder. Periods of depression are especially common during chronic intoxication or in association with physical or psychosocial stressors related to the opioid use disorder. Insomnia is also common, especially during withdrawal. Opioid use disorder is also associated with bipolar I disorder, posttraumatic stress disorder, and antisocial, borderline,

and schizotypal personality disorders. A history of conduct disorder in childhood or adolescence has also been identified as a significant risk factor for substance-related disorders, especially opioid use disorder. Further, prescription opioid use disorder and heroin use disorder are generally associated with *serious mental illness*, defined as a mental disorder other than a substance use disorder that results in serious functional impairment substantially limiting or interfering with major life activities.

Opioid Intoxication

Diagnostic Criteria

- A. Recent use of an opioid.
- B. Clinically significant problematic behavioral or psychological changes (e.g., initial euphoria followed by apathy, dysphoria, psychomotor agitation or retardation, impaired judgment) that developed during, or shortly after, opioid use.
- C. Pupillary constriction (or pupillary dilation due to anoxia from severe overdose) and one (or more) of the following signs or symptoms developing during, or shortly after, opioid use:
 - 1. Drowsiness or coma.
 - 2. Slurred speech.
 - 3. Impairment in attention or memory.
- D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication with another substance.

Specify if:

With perceptual disturbances: This specifier may be noted in the rare instance in which hallucinations with intact reality testing or auditory, visual, or tactile illusions occur in the absence of a delirium.

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Coding note: The ICD-10-CM code depends on whether or not there is a comorbid opioid use disorder and whether or not there are perceptual disturbances.

For opioid intoxication, without perceptual disturbances: If a mild opioid use disorder is comorbid, the ICD-10-CM code is **F11.120**, and if a moderate or severe opioid use disorder is comorbid, the ICD-10-CM code is **F11.220**. If there is no comorbid opioid use disorder, then the ICD-10-CM code is **F11.920**.

For opioid intoxication, with perceptual disturbances: If a mild opioid use disorder is comorbid, the ICD-10-CM code is **F11.122**, and if a moderate or severe opioid use disorder is comorbid, the ICD-10-CM code is **F11.222**. If there is no comorbid opioid use disorder, then the ICD-10-CM code is **F11.922**.

Diagnostic Features

The essential feature of opioid intoxication is the presence of clinically significant problematic behavioral or psychological changes (e.g., initial euphoria followed by apathy, dysphoria, psychomotor agitation or retardation, impaired judgment) that develop during, or shortly after, opioid use (Criteria A and B). Intoxication is accompanied by pupillary constriction (unless there has been a severe overdose with consequent anoxia and pupillary dilation) and one or more of the following signs: drowsiness (described as being “on the nod”), slurred speech, and impairment in attention or memory (Criterion C); drowsiness may progress to coma. Individuals with opioid intoxication may demonstrate inattention to the environment, even to the point of ignoring potentially harmful events. The signs or symptoms of opioid intoxication must not be attributable to another medical condition and are not better explained by another mental disorder (Criterion D).

Up to 2009, opioid overdoses were mainly due to prescribed opioids, but starting in 2010, overdoses due to heroin began a sharp rise, and additionally, since 2015, fatal overdoses due to synthetic opioids other than methadone (generally fentanyl) have outnumbered overdoses due to prescribed opioids.

Associated Features

Opioid intoxication can include decreases in respiratory rate and blood pressure, and mild hypothermia. The duration of opioid intoxication can vary as a function of the pharmacokinetics of the opioid ingested. Opioid intoxication may result in fatal or nonfatal opioid overdose. Opioid overdose is characterized by unconsciousness, respiratory depression, and pinpoint pupils. Fatal opioid overdoses have increased exponentially in the United States since 1999.

Development and Course

Opioid intoxication can occur in an individual who is opioid naïve, an individual who uses opioids sporadically, and an individual who is physically dependent on opioids. The dose of opioid consumed relative to the likelihood of experiencing opioid intoxication will vary as a function of the status and history of the individual’s opioid exposure (i.e., tolerance). Individuals often report that the qualitative pleasurable experience of opioid intoxication diminishes after repeated use of an opioid.

Differential Diagnosis

Other substance intoxication. Alcohol intoxication and sedative-hypnotic intoxication can cause a clinical picture that resembles opioid intoxication. A diagnosis of alcohol or sedative-hypnotic intoxication can usually be made based on the absence of pupillary constriction or the lack of a response to a naloxone challenge. In some cases, intoxication may be due both to opioids and to alcohol or other sedatives. In these cases, naloxone administration will not reverse all of the sedative effects. While response to administration of

naloxone can support the diagnosis of opioid intoxication, nonresponse may be due to the co-ingestion of an opioid with another drug (e.g., a benzodiazepine, alcohol) or to ingestion of a higher dose of and/or higher-potency opioid (e.g., fentanyl).

Opioid-induced mental disorders. Opioid intoxication is distinguished from opioid-induced mental disorders (e.g., opioid-induced depressive disorder, with onset during intoxication) because the symptoms (e.g., depressed mood) in the latter disorders are in excess of those usually associated with opioid intoxication, predominate in the clinical presentation, and are severe enough to warrant clinical attention.

Comorbidity

Given the typical overlap of opioid intoxication with opioid use disorder, see “Comorbidity” under Opioid Use Disorder for more details about co-occurring conditions that are likely to be encountered.

Opioid Withdrawal

Diagnostic Criteria

- A. Presence of either of the following:
 - 1. Cessation of (or reduction in) opioid use that has been heavy and prolonged (i.e., several weeks or longer).
 - 2. Administration of an opioid antagonist after a period of opioid use.
- B. Three (or more) of the following developing within minutes to several days after Criterion A:
 - 1. Dysphoric mood.
 - 2. Nausea or vomiting.
 - 3. Muscle aches.
 - 4. Lacrimation or rhinorrhea.
 - 5. Pupillary dilation, piloerection, or sweating.
 - 6. Diarrhea.
 - 7. Yawning.
 - 8. Fever.
 - 9. Insomnia.
- C. The signs or symptoms in Criterion B cause clinically significant distress or impairment in social, occupational, or other important areas of functioning.
- D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication or withdrawal from another substance.

Coding note: The ICD-10-CM code depends on whether or not there is a comorbid

opioid use disorder. If a mild opioid use disorder is comorbid, the ICD-10-CM code is **F11.13**, and if a moderate or severe opioid use disorder is comorbid, the ICD-10-CM code is **F11.23**. For opioid withdrawal occurring in the absence of an opioid use disorder (e.g., in a patient taking opioids solely under appropriate medical supervision), the ICD-10-CM code is **F11.93**.

Diagnostic Features

The essential feature of opioid withdrawal is the presence of a characteristic withdrawal syndrome that develops after the cessation of (or reduction in) prolonged opioid use

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(Criterion A1). The opioids used may be illicit or licitly obtained drugs prescribed to treat pain. A withdrawal syndrome can also be precipitated by administration of an opioid antagonist (e.g., naloxone, naltrexone, nalmefene) after a period of opioid use (Criterion A2); it can also occur after administration of an opioid partial agonist (e.g., buprenorphine) to an individual currently using a full opioid agonist.

Opioid withdrawal has a characteristic pattern of signs and symptoms. The first of these are subjective and consist of complaints of anxiety, restlessness, and an “achy feeling” that is often located in the back and legs, along with irritability and increased sensitivity to pain. Three or more of the following must be present to make a diagnosis of opioid withdrawal: dysphoric mood; nausea or vomiting; muscle aches; lacrimation or rhinorrhea; pupillary dilation, piloerection, or increased sweating; diarrhea; yawning; fever; and insomnia (Criterion B). Piloerection and fever are associated with more severe withdrawal and are not often seen in routine clinical practice because individuals with opioid use disorder usually obtain substances before withdrawal becomes that far advanced. These symptoms of opioid withdrawal must cause clinically significant distress or impairment in social, occupational, or other important areas of functioning (Criterion C). The symptoms must not be attributable to another medical condition and are not better explained by another mental disorder (Criterion D). Having symptoms that meet diagnostic criteria for opioid withdrawal alone is not sufficient for a diagnosis of opioid use disorder, but concurrent symptoms of craving and drug-seeking behavior are suggestive of comorbid opioid use disorder.

Associated Features

Opioid withdrawal may occur in any individual after cessation of repeated use of an opioid, whether in the setting of medical management of pain, during opioid agonist therapy for opioid use disorder, in the context of illicit use, or following attempts to self-treat symptoms of mental disorders with opioids. Opioid withdrawal is a distinct condition from opioid addiction or opioid use disorder and does not necessarily require the drug-seeking behaviors associated with opioid use disorder to be diagnosed. Thus, opioid withdrawal may occur in individuals without opioid use disorder and should not be confused with it. Males with opioid withdrawal may experience piloerection, sweating, and spontaneous ejaculations while awake.

Prevalence

Among individuals from various U.S. clinical settings, opioid withdrawal occurred in 60% of individuals who had used heroin at least once in the prior 12 months. Individuals regularly using opioids (e.g., prescription opioids for pain, illicit opioids) for a period of time are at risk for developing physical dependence, including withdrawal, on cessation or marked reduction in use.

Development and Course

The speed and severity of withdrawal associated with opioids depend on the half-life of the opioid used. Most individuals who are physiologically dependent on short-acting drugs such as heroin begin to have withdrawal symptoms within 6–12 hours after the last dose. Symptoms may take 2–4 days to emerge in the case of longer-acting drugs such as methadone or buprenorphine. Acute withdrawal symptoms for a short-acting opioid such as heroin usually peak within 1–3 days and gradually subside over a period of 5–7 days. More chronic symptoms (e.g., anxiety, dysphoria, anhedonia, craving, insomnia) can last for weeks to months. The severity of opioid withdrawal also varies depending on the duration of opioid use. Opioid withdrawal symptoms among individuals receiving long-term prescription opioid treatment for pain can be minimized by tapering the drug slowly.

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Among those with an opioid use disorder, opioid withdrawal and attempts to relieve withdrawal are typical. The course of withdrawal can be part of an escalating pattern in which an opioid is used to reduce withdrawal symptoms, in turn leading to recurrent episodes of withdrawal at a later time.

Differential Diagnosis

Other withdrawal disorders. The anxiety and restlessness associated with opioid withdrawal resemble symptoms seen in sedative-hypnotic withdrawal. However, opioid withdrawal is also accompanied by rhinorrhea, lacrimation, and pupillary dilation, which are not seen in sedative-type withdrawal.

Other substance intoxication. Dilated pupils are also seen in hallucinogen intoxication and stimulant intoxication. However, other signs or symptoms of opioid withdrawal, such as nausea, vomiting, diarrhea, abdominal cramps, rhinorrhea, and lacrimation, are not present.

Opioid-induced mental disorders. Opioid withdrawal is distinguished from opioid-induced mental disorders (e.g., opioid-induced depressive disorder, with onset during withdrawal) because the symptoms (e.g., depressed mood) in these latter disorders are in excess of those usually associated with opioid withdrawal, predominate in the clinical presentation, and are severe enough to warrant clinical attention.

Comorbidity

Given the typical overlap of opioid withdrawal with opioid use disorder, see “Comorbidity” under Opioid Use Disorder for more details about co-occurring conditions that are likely to be encountered.

Opiod-Induced Mental Disorders

The following opiod-induced mental disorders are described in other chapters of the manual with disorders with which they share phenomenology (see the substance/medication-induced mental disorders in these chapters): opiod-induced depressive disorder (“Depressive Disorders”); opiod-induced anxiety disorder (“Anxiety Disorders”); opiod-induced sleep disorder (“Sleep-Wake Disorders”); and opiod-induced sexual dysfunction (“Sexual Dysfunctions”). For opiod intoxication delirium, opiod withdrawal delirium, and delirium induced by opiods taken as prescribed, see the criteria and discussion of delirium in the chapter “Neurocognitive Disorders.” These opiod-induced mental disorders are diagnosed instead of opiod intoxication or opiod withdrawal only when the symptoms are sufficiently severe to warrant independent clinical attention.

Unspecified Opioid-Related Disorder

F11.99

This category applies to presentations in which symptoms characteristic of an opioid-related disorder that cause clinically significant distress or impairment in social, occupational, or other important areas of functioning predominate but do not meet the full criteria for any specific opioid-related disorder or any of the disorders in the substance-related and addictive disorders diagnostic class.

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Sedative-, Hypnotic-, or Anxiolytic-Related Disorders

Sedative, Hypnotic, or Anxiolytic Use Disorder

Sedative, Hypnotic, or Anxiolytic Intoxication

Sedative, Hypnotic, or Anxiolytic Withdrawal

Sedative-, Hypnotic-, or Anxiolytic-Induced Mental Disorders

Unspecified Sedative-, Hypnotic-, or Anxiolytic-Related Disorder

Sedative, Hypnotic, or Anxiolytic Use Disorder

Diagnostic Criteria

- A. A problematic pattern of sedative, hypnotic, or anxiolytic use leading to clinically significant impairment or distress, as manifested by at least two of the following, occurring within a 12-month period:
1. Sedatives, hypnotics, or anxiolytics are often taken in larger amounts or over a longer period than was intended.
 2. There is a persistent desire or unsuccessful efforts to cut down or control sedative, hypnotic, or anxiolytic use.
 3. A great deal of time is spent in activities necessary to obtain the sedative, hypnotic, or anxiolytic; use the sedative, hypnotic, or anxiolytic; or recover from its effects.
 4. Craving, or a strong desire or urge to use the sedative, hypnotic, or anxiolytic.
 5. Recurrent sedative, hypnotic, or anxiolytic use resulting in a failure to fulfill major role obligations at work, school, or home (e.g., repeated absences from work or poor work performance related to sedative, hypnotic, or anxiolytic use; sedative-, hypnotic-, or anxiolytic-related absences, suspensions, or expulsions from school; neglect of children or household).
 6. Continued sedative, hypnotic, or anxiolytic use despite having persistent or recurrent social or interpersonal problems caused or exacerbated by the effects of sedatives, hypnotics, or anxiolytics (e.g., arguments with a spouse about consequences of intoxication; physical fights).
 7. Important social, occupational, or recreational activities are given up or reduced because of sedative, hypnotic, or anxiolytic use.
 8. Recurrent sedative, hypnotic, or anxiolytic use in situations in which it is physically hazardous (e.g., driving an automobile or operating a machine when impaired by sedative, hypnotic, or anxiolytic use).
 9. Sedative, hypnotic, or anxiolytic use is continued despite knowledge of having a persistent or recurrent physical or psychological problem that is likely to have been caused or exacerbated by the sedative, hypnotic, or anxiolytic.
 10. Tolerance, as defined by either of the following:
 - a. A need for markedly increased amounts of the sedative, hypnotic, or anxiolytic to achieve intoxication or desired effect.
 - b. A markedly diminished effect with continued use of the same amount of the sedative, hypnotic, or anxiolytic.

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Note: This criterion is not considered to be met for individuals taking sedatives, hypnotics, or anxiolytics under medical supervision.

11. Withdrawal, as manifested by either of the following:
 - a. The characteristic withdrawal syndrome for sedatives, hypnotics, or anxiolytics (refer to Criteria A and B of the criteria set for sedative, hypnotic, or anxiolytic withdrawal).

- b. Sedatives, hypnotics, or anxiolytics (or a closely related substance, such as alcohol) are taken to relieve or avoid withdrawal symptoms.

Note: This criterion is not considered to be met for individuals taking sedatives, hypnotics, or anxiolytics under medical supervision.

Specify if:

In early remission: After full criteria for sedative, hypnotic, or anxiolytic use disorder were previously met, none of the criteria for sedative, hypnotic, or anxiolytic use disorder have been met for at least 3 months but for less than 12 months (with the exception that Criterion A4, "Craving, or a strong desire or urge to use the sedative, hypnotic, or anxiolytic," may be met).

In sustained remission: After full criteria for sedative, hypnotic, or anxiolytic use disorder were previously met, none of the criteria for sedative, hypnotic, or anxiolytic use disorder have been met at any time during a period of 12 months or longer (with the exception that Criterion A4, "Craving, or a strong desire or urge to use the sedative, hypnotic, or anxiolytic," may be met).

Specify if:

In a controlled environment: This additional specifier is used if the individual is in an environment where access to sedatives, hypnotics, or anxiolytics is restricted.

Code based on current severity/remission: If a sedative, hypnotic, or anxiolytic intoxication; sedative, hypnotic, or anxiolytic withdrawal; or another sedative-, hypnotic-, or anxiolytic-induced mental disorder is also present, do not use the codes below for sedative, hypnotic, or anxiolytic use disorder. Instead, the comorbid sedative, hypnotic, or anxiolytic use disorder is indicated in the 4th character of the sedative-, hypnotic-, or anxiolytic-induced disorder (see the coding note for sedative, hypnotic, or anxiolytic intoxication; sedative, hypnotic, or anxiolytic withdrawal; or specific sedative-, hypnotic-, or anxiolytic-induced mental disorder). For example, if there is comorbid sedative-, hypnotic-, or anxiolytic-induced depressive disorder and sedative, hypnotic, or anxiolytic use disorder, only the sedative-, hypnotic-, or anxiolytic-induced depressive disorder code is given, with the 4th character indicating whether the comorbid sedative, hypnotic, or anxiolytic use disorder is mild, moderate, or severe: F13.14 for mild sedative, hypnotic, or anxiolytic use disorder with sedative-, hypnotic-, or anxiolytic-induced depressive disorder or F13.24 for a moderate or severe sedative, hypnotic, or anxiolytic use disorder with sedative-, hypnotic-, or anxiolytic-induced depressive disorder.

Specify current severity/remission:

F13.10 Mild: Presence of 2–3 symptoms.

F13.11 Mild, In early remission

F13.11 Mild, In sustained remission

F13.20 Moderate: Presence of 4–5 symptoms.

F13.21 Moderate, In early remission

F13.21 Moderate, In sustained remission

F13.20 Severe: Presence of 6 or more symptoms.

F13.21 Severe, In early remission

F13.21 Severe, In sustained remission

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Specifiers

“In a controlled environment” applies as a further specifier of remission if the individual is both in remission and in a controlled environment (i.e., in early remission in a controlled environment or in sustained remission in a controlled environment). Examples of these environments are closely supervised and substance-free jails, therapeutic communities, and locked hospital units.

Diagnostic Features

Sedative, hypnotic, or anxiolytic substances include benzodiazepines, benzodiazepine-like drugs (e.g., zolpidem, zaleplon), carbamates (e.g., glutethimide, meprobamate), barbiturates (e.g., secobarbital), and barbiturate-like hypnotics (e.g., glutethimide, methaqualone, propofol). This class of substances includes most prescription sleeping medications and most prescription antianxiety medications. Nonbenzodiazepine antianxiety agents (e.g., buspirone, gepirone) are not included in this class because they do not appear to be associated with significant misuse.

Like alcohol, these agents are brain depressants and can produce similar substance/medication-induced and substance use disorders. Sedative, hypnotic, or anxiolytic substances are available both by prescription and illegally. Some individuals who obtain these substances by prescription will develop a sedative, hypnotic, or anxiolytic use disorder, while others who misuse these substances or use them for intoxication will not develop a use disorder. In particular, sedatives, hypnotics, or anxiolytics with rapid onset or short to intermediate lengths of action may be taken for intoxication purposes, although longer-acting substances in this class may be taken for intoxication as well.

Craving (Criterion A4), either during periods of active use or during periods of abstinence, is a typical feature of sedative, hypnotic, or anxiolytic use disorder. Misuse of substances from this class may occur in conjunction with use of other substances. For example, individuals may use intoxicating doses of sedatives or benzodiazepines to “come down” from cocaine or amphetamines or use high doses of benzodiazepines in combination with methadone to “boost” its effects.

Repeated absences or poor work performance, school absences, suspensions or expulsions, and neglect of children or household (Criterion A5) may be related to sedative, hypnotic, or anxiolytic use disorder; the continued use of the substances despite arguments with a spouse about consequences of intoxication or despite physical fights (Criterion A6) may also occur. Limiting contact with family or friends, avoiding work or school, or stopping participation in hobbies, sports, or games (Criterion A7) and recurrent sedative, hypnotic, or anxiolytic use when driving an automobile or operating machinery when impaired by such use (Criterion A8) are also seen in sedative, hypnotic, or anxiolytic use disorder.

Very significant levels of tolerance and withdrawal can develop to sedative, hypnotic, or

anxiolytic substances. There may be evidence of tolerance and withdrawal in the absence of a diagnosis of a sedative, hypnotic, or anxiolytic use disorder in an individual who has abruptly discontinued use of benzodiazepines that were taken for long periods of time at prescribed and therapeutic doses. In these cases, an additional diagnosis of sedative, hypnotic, or anxiolytic use disorder is made only if other criteria are met. That is, sedative, hypnotic, or anxiolytic medications may be prescribed for appropriate medical purposes, and depending on the dose regimen, these drugs may then produce tolerance and withdrawal. If these drugs are prescribed or recommended for appropriate medical purposes, and if they are used as prescribed, the resulting tolerance or withdrawal does not count toward the diagnosis of a substance use disorder. However, it is necessary to determine whether the drugs were inappropriately prescribed and used (e.g., falsifying medical symptoms to obtain the medication; using more medication than prescribed; obtaining the medication from several doctors without informing them).

Given the unidimensional nature of the symptoms of sedative, hypnotic, or anxiolytic use disorder, severity is based on the number of criteria endorsed.

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Associated Features

Research with nationally representative samples of the U.S. population has found that sedative, hypnotic, or anxiolytic use disorder is often associated with other substance use disorders (e.g., alcohol, cannabis, opioid, stimulant use disorders). Sedatives are often used to alleviate the unwanted effects of these other substances. With repeated use of the sedative, hypnotic, or anxiolytic, tolerance develops to the sedative effects, and a progressively higher dose is used. However, tolerance to brain stem depressant effects develops much more slowly, and as the individual takes more substance to achieve euphoria or other desired effects, there may be a sudden onset of respiratory depression and hypotension, which may result in death. Intense or repeated sedative, hypnotic, or anxiolytic intoxication may be associated with severe depression that although temporary can lead to suicide attempt and suicide.

Prevalence

The 12-month prevalence of DSM-IV sedative, hypnotic, or anxiolytic use disorder in the United States is estimated to be 0.3% among adolescents ages 12–17 years and adults age 18 years and older, and this prevalence has remained stable nationally despite increases in rates of prescription of these medications. Rates of DSM-IV sedative, hypnotic, or anxiolytic use disorder in the United States have not been shown to vary consistently by gender, but data from other countries have generally found higher rates among girls and women than boys and men. The 12-month prevalence of DSM-IV sedative, hypnotic, or anxiolytic use disorder in the United States decreases as a function of age and is greatest among individuals ages 18–29 years (0.5%) and lowest among individuals 65 years and older (0.04%).

Twelve-month prevalence of sedative, hypnotic, or anxiolytic use, misuse (e.g., use without a prescription), or disorder varies across U.S. ethnoracial groups. For instance, 12-month prevalence estimates for sedative, hypnotic, or anxiolytic misuse across ethnoracial groups range from 0.6% to 2.5% for adolescents ages 12–17 years and 0.7% to 10.1% for adults.

Development and Course

The usual course of sedative, hypnotic, or anxiolytic use disorder involves individuals in their teens or 20s who escalate their occasional use of sedative, hypnotic, or anxiolytic agents to the point at which they develop problems that meet criteria for a diagnosis. This pattern may be especially likely among individuals who have other substance use disorders (e.g., alcohol, opioids, stimulants). An initial pattern of intermittent use socially (e.g., at parties) can lead to daily use and high levels of tolerance. Once this occurs, an increasing level of interpersonal difficulties can be expected, as well as increasingly severe episodes of cognitive dysfunction and physiological withdrawal.

The second and less frequently observed clinical course begins with an individual who originally obtained the medication by prescription from a physician, usually for the treatment of anxiety, insomnia, or somatic complaints. As either tolerance or a need for higher doses of the medication develops, there is a gradual increase in the dose and frequency of self-administration. The individual is likely to continue to justify use on the basis of original anxiety or insomnia symptoms, but substance-seeking behavior becomes more prominent, and the individual may seek out multiple physicians to obtain sufficient supplies of the medication. Tolerance can reach high levels, and withdrawal (including seizures and withdrawal delirium) may occur.

As with many substance use disorders, sedative, hypnotic, or anxiolytic use disorder generally has an onset during adolescence or early adult life. Although the risk for misuse and use disorder decreases with age after about age 30, side effects associated with psychoactive substances may increase as individuals age. In particular, cognitive impairment increases as a side effect with age, and the metabolism of sedatives, hypnotics, or

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anxiolytics decreases with age among older individuals. Both acute and chronic toxic effects of these substances, especially effects on cognition, memory, and motor coordination, are likely to increase with age as a consequence of pharmacodynamic and pharmacokinetic age-related changes. Individuals with major neurocognitive disorder are more likely to develop intoxication and impaired physiological functioning at lower doses. Because sedatives, hypnotics, and anxiolytics are often used in combination with other psychoactive substances, it can be difficult to ascertain whether the functional consequences are attributable to a single substance (e.g., sedative) or to the use of multiple substances.

Deliberate intoxication to achieve a “high” is most likely to be observed in teenagers and individuals in their 20s. Problems associated with sedatives, hypnotics, or anxiolytics are also seen in individuals in their 40s and older who escalate the dose of prescribed medications. In older individuals, intoxication can resemble a progressive major neurocognitive disorder.

Risk and Prognostic Factors

Temperamental. Impulsivity and novelty seeking are individual temperaments that relate to the propensity to develop a substance use disorder but may themselves be genetically determined. Personality disorders can also increase the risk of sedative, hypnotic, or anxiolytic misuse or use disorder.

Environmental. Because sedatives, hypnotics, or anxiolytics are all medications, a key risk factor

relates to availability of the substances, both through an individual's own prescriptions and from prescriptions dispensed to family and friends. In the United States, the historical patterns of sedative, hypnotic, or anxiolytic misuse relate to broad prescribing patterns. For instance, a marked decrease in prescription of barbiturates was associated with an increase in benzodiazepine prescriptions. Peer factors may relate to genetic predisposition in terms of how individuals select their environment. Other individuals at heightened risk might include those with alcohol use disorder who may receive repeated prescriptions in response to their complaints of alcohol-related anxiety or insomnia.

Genetic and physiological. As with other substance use disorders, the risk for sedative, hypnotic, or anxiolytic use disorder has been found in U.S.-based twin registry studies to be related to individual, family, peer, social, and environmental factors. Within these domains, genetic factors play a particularly important role both directly and indirectly. Overall, across development, genetic factors seem to play a larger role in the onset of sedative, hypnotic, or anxiolytic use disorder as individuals age through puberty into adult life.

Course modifiers. In nationally representative U.S. studies, early onset of use is associated with greater likelihood for developing a sedative, hypnotic, or anxiolytic use disorder.

Culture-Related Diagnostic Issues

Prescription patterns (and availability) of this class of substances vary across countries and populations, which may lead to variations in prevalence of sedative, hypnotic, or anxiolytic use disorder. In the United States, use of benzodiazepines has been more frequently reported by non-Latinx Whites than Latinx or African Americans. However, risk of the disorder may vary within populations exposed to these substances. For example, the 12-month prevalence of DSM-IV benzodiazepine use disorder among U.S. individuals who used benzodiazepines was higher among African Americans (3.0%) and non-Latinx "others" (2.6%) than among non-Latinx Whites (1.3%).

Sex- and Gender-Related Diagnostic Issues

Although estimates from individual studies vary, there appear to be no gender differences in the prevalence of sedative, hypnotic, or anxiolytic use disorder.

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Diagnostic Markers

Almost all sedative, hypnotic, or anxiolytic substances can be identified through laboratory evaluations of urine or blood (the latter of which can quantify the amounts of these agents in the body). Urine test results are likely to remain positive for up to approximately 1 week after the use of long-acting substances, such as diazepam or flurazepam.

Association With Suicidal Thoughts or Behavior

U.S. epidemiological studies show that hypnotics are associated with suicide, but it is unclear if this association is attributable to underlying psychiatric conditions such as depression and insomnia, which are themselves risk factors for suicide.

Functional Consequences of Sedative, Hypnotic, or Anxiolytic Use Disorder

The social and interpersonal consequences of sedative, hypnotic, or anxiolytic use disorder mimic those of alcohol in terms of the potential for disinhibited behavior. Accidents, interpersonal difficulties, and interference with work or school performance are common outcomes. The disinhibiting effects of these agents, like alcohol, may potentially contribute to overly aggressive behavior and arguments or fights, with subsequent interpersonal and legal problems. Physical examination is likely to reveal evidence of a mild decrease in most aspects of autonomic nervous system functioning, including a slower pulse, a slightly decreased respiratory rate, and a slight drop in blood pressure (most likely to occur with postural changes).

Acute intoxication can result in accidental injuries and automobile accidents. There may be consequences of trauma (e.g., internal bleeding, a subdural hematoma) from accidents that occur while intoxicated. For elderly individuals, even short-term use of these sedating medications at prescribed doses may be associated with an increased risk for cognitive problems and falls. The association of sedative, hypnotic, or anxiolytic medications with increased risk of major neurocognitive disorder remains unclear.

At high doses, sedative, hypnotic, or anxiolytic substances can be lethal, particularly when mixed with other central nervous system depressants, such as opioids or alcohol, although the lethal dosage varies considerably among the specific substances. Intravenous use of these substances can result in medical complications related to the use of contaminated needles (e.g., hepatitis, HIV).

Accidental or deliberate overdoses, similar to those observed for alcohol use disorder or repeated alcohol intoxication, can occur. Overdoses may be associated with a deterioration in vital signs that signals an impending medical emergency (e.g., respiratory arrest from barbiturates). In contrast to their wide margin of safety when used alone, benzodiazepines taken in combination with opioids and alcohol can be particularly dangerous, and accidental overdoses are reported commonly in U.S. data. Accidental overdoses have also been reported in individuals who deliberately misuse barbiturates and other nonbenzodiazepine sedatives (e.g., methaqualone), but because these agents are much less available than the benzodiazepines, the frequency of overdosing is low in most settings.

Differential Diagnosis

Sedative, hypnotic, or anxiolytic intoxication; sedative, hypnotic, or anxiolytic withdrawal; and sedative-, hypnotic-, or anxiolytic-induced mental disorders.

Sedative, hypnotic, or anxiolytic use disorder is differentiated from sedative, hypnotic, or anxiolytic intoxication; sedative, hypnotic, or anxiolytic withdrawal; and sedative-, hypnotic-, or anxiolytic-induced mental disorders (e.g., sedative-, hypnotic-, or anxiolytic-induced depressive disorder) in that sedative, hypnotic, or anxiolytic use disorder describes a problematic pattern

of sedative, hypnotic, or anxiolytic use that involves impaired control over such use; social impairment attributable to this use; risky sedative, hypnotic, or anxiolytic use (e.g., driving while intoxicated); and pharmacological symptoms (the development of tolerance or withdrawal); whereas sedative, hypnotic, or anxiolytic intoxication; sedative, hypnotic, or anxiolytic

withdrawal; and sedative-, hypnotic-, or anxiolytic-induced mental disorders describe psychiatric syndromes that occur in the context of heavy use. Sedative, hypnotic, or anxiolytic intoxication; sedative, hypnotic, or anxiolytic withdrawal; and sedative-, hypnotic-, or anxiolytic-induced mental disorders occur frequently in individuals with sedative, hypnotic, or anxiolytic use disorder. In such cases, a diagnosis of sedative, hypnotic, or anxiolytic intoxication; sedative, hypnotic, or anxiolytic withdrawal; or a sedative-, hypnotic-, or anxiolytic-induced mental disorder should be given in addition to a diagnosis of sedative, hypnotic, and anxiolytic use disorder, the presence of which is indicated in the diagnostic code.

Other medical conditions. The slurred speech, incoordination, and other associated features characteristic of sedative, hypnotic, or anxiolytic intoxication could be the result of another medical condition (e.g., multiple sclerosis) or of a prior head trauma (e.g., a subdural hematoma).

Alcohol use disorder. Sedative, hypnotic, or anxiolytic use disorder must be differentiated from alcohol use disorder. The differential diagnosis is determined mostly through clinical history, although liver damage and other potential signs of chronic alcohol toxicity (e.g., cardiomyopathy) can also be more suggestive of alcohol use disorder than of sedative, hypnotic, or anxiolytic use disorder.

Clinically appropriate use of sedative, hypnotic, or anxiolytic medications. Individuals may continue to take benzodiazepine medication according to a physician's direction for a legitimate medical indication over extended periods of time. Even if physiological signs of tolerance or withdrawal are manifested, many of these individuals do not develop symptoms that meet the criteria for sedative, hypnotic, or anxiolytic use disorder because they are not preoccupied with obtaining the substance and its use does not interfere with their performance of usual social or occupational roles.

Comorbidity

Nonmedical use of sedative, hypnotic, or anxiolytic agents is associated with alcohol use disorder, tobacco use disorder, and, generally, illicit drug use. There may also be an overlap between sedative, hypnotic, or anxiolytic use disorder and antisocial personality disorder; depressive, bipolar, and anxiety disorders; and other substance use disorders, such as alcohol use disorder and illicit drug use disorders. Antisocial behavior and antisocial personality disorder are especially associated with sedative, hypnotic, or anxiolytic use disorder when the substances are obtained illegally. Comorbidity with other substance use disorders and other psychiatric disorders increases the risk of transition from sedative, hypnotic, or anxiolytic use to use disorder and decreases the probability of remission.

Sedative, Hypnotic, or Anxiolytic Intoxication

Diagnostic Criteria

- A. Recent use of a sedative, hypnotic, or anxiolytic.
- B. Clinically significant maladaptive behavioral or psychological changes (e.g.,

inappropriate sexual or aggressive behavior, mood lability, impaired judgment) that developed during, or shortly after, sedative, hypnotic, or anxiolytic use.

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- C. One (or more) of the following signs or symptoms developing during, or shortly after, sedative, hypnotic, or anxiolytic use:
1. Slurred speech.
 2. Incoordination.
 3. Unsteady gait.
 4. Nystagmus.
 5. Impairment in cognition (e.g., attention, memory).
 6. Stupor or coma.
- D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication with another substance.

Coding note: The ICD-10-CM code depends on whether there is a comorbid sedative, hypnotic, or anxiolytic use disorder. If a mild sedative, hypnotic, or anxiolytic use disorder is comorbid, the ICD-10-CM code is **F13.120**, and if a moderate or severe sedative, hypnotic, or anxiolytic use disorder is comorbid, the ICD-10-CM code is **F13.220**. If there is no comorbid sedative, hypnotic, or anxiolytic use disorder, then the ICD-10-CM code is **F13.920**.

Note: For information on Development and Course; Risk and Prognostic Factors; Culture-Related Diagnostic Issues; Diagnostic Markers; Functional Consequences of Sedative, Hypnotic, or Anxiolytic Intoxication; and Comorbidity, see the corresponding sections in Sedative, Hypnotic, or Anxiolytic Use Disorder.

Diagnostic Features

The essential feature of sedative, hypnotic, or anxiolytic intoxication is the presence of clinically significant maladaptive behavioral or psychological changes (e.g., inappropriate sexual or aggressive behavior, mood lability, impaired judgment, impaired social or occupational functioning) that develop during, or shortly after, use of a sedative, hypnotic, or anxiolytic (Criteria A and B). As with other brain depressants, such as alcohol, these behaviors may be accompanied by slurred speech, incoordination (at levels that can interfere with driving abilities and with performing usual activities to the point of causing falls or automobile accidents), an unsteady gait, nystagmus, impairment in cognition (e.g., attentional or memory problems), and stupor or coma (Criterion C). Memory impairment is a prominent feature of sedative, hypnotic, or anxiolytic intoxication and is most often characterized by an anterograde amnesia that resembles “alcoholic blackouts,” which can be disturbing to the individual. The symptoms must not be attributable to another medical condition and are not better explained by another mental disorder (Criterion D). Intoxication may occur in individuals who are receiving these substances by prescription, are borrowing the medication from friends or relatives, or are deliberately taking

the substance to achieve intoxication. Because sedatives, hypnotics, and anxiolytics are often used in combination with other psychoactive substances, it can be difficult to ascertain whether the functional consequences are attributable to a sedative, hypnotic, or anxiolytic or to the use of multiple substances.

Associated Features

Associated features include taking more medication than prescribed, taking multiple different medications, or mixing sedative, hypnotic, or anxiolytic agents with alcohol, which can markedly increase the effects of these agents.

Prevalence

The prevalence of sedative, hypnotic, or anxiolytic intoxication in the general population is unknown. However, it is probable that most nonmedical users of sedatives, hypnotics,

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or anxiolytics would at some time have signs or symptoms that meet criteria for sedative, hypnotic, or anxiolytic intoxication; if so, then the prevalence of nonmedical sedative, hypnotic, or anxiolytic use in the general population may be similar to the prevalence of sedative, hypnotic, or anxiolytic intoxication. For example, in 2018, tranquilizers or sedative were used nonmedically in the United States by 2.4% of individuals age 12 or older and 4.9% of those ages 18–25.

Differential Diagnosis

Alcohol use disorder. Because the clinical presentations may be identical, distinguishing sedative, hypnotic, or anxiolytic intoxication from alcohol use disorder requires evidence for recent ingestion of sedative, hypnotic, or anxiolytic medications by self-report, informant report, or toxicological testing. Many individuals who misuse sedatives, hypnotics, or anxiolytics may also misuse alcohol and other substances, and so multiple intoxication diagnoses are possible.

Alcohol intoxication. Alcohol intoxication may be distinguished from sedative, hypnotic, or anxiolytic intoxication by the smell of alcohol on the breath. Otherwise, the features of the two disorders may be similar.

Sedative-, hypnotic-, or anxiolytic-induced mental disorders. Sedative, hypnotic, or anxiolytic intoxication is distinguished from sedative-, hypnotic-, or anxiolytic-induced mental disorders (e.g., sedative-, hypnotic-, or anxiolytic-induced anxiety disorder, with onset during withdrawal) because the symptoms (e.g., anxiety) in the latter disorders are in excess of those usually associated with sedative, hypnotic, or anxiolytic intoxication; predominate in the clinical presentation; and are severe enough to warrant clinical attention.

Neurocognitive disorders. In situations of cognitive impairment, traumatic brain injury, and delirium from other causes, sedatives, hypnotics, or anxiolytics may be intoxicating at quite low dosages. The differential diagnosis in these complex settings is based on the predominant syndrome. An additional diagnosis of sedative, hypnotic, or anxiolytic intoxication may be appropriate even if the substance has been ingested at a low dosage in the setting of these other

(or similar) co-occurring conditions.

Comorbidity

Given the typical overlap of sedative, hypnotic, or anxiolytic intoxication with sedative, hypnotic, or anxiolytic use disorder, see “Comorbidity” under Sedative, Hypnotic, or Anxiolytic Use Disorder for more details about co-occurring conditions that are likely to be encountered.

Sedative, Hypnotic, or Anxiolytic Withdrawal

Diagnostic Criteria

- A. Cessation of (or reduction in) sedative, hypnotic, or anxiolytic use that has been prolonged.
- B. Two (or more) of the following, developing within several hours to a few days after the cessation of (or reduction in) sedative, hypnotic, or anxiolytic use described in Criterion A:
 - 1. Autonomic hyperactivity (e.g., sweating or pulse rate greater than 100 bpm).
 - 2. Hand tremor.
 - 3. Insomnia.
 - 4. Nausea or vomiting.
 - 5. Transient visual, tactile, or auditory hallucinations or illusions.
 - 6. Psychomotor agitation.
 - 7. Anxiety.
 - 8. Grand mal seizures.
- C. The signs or symptoms in Criterion B cause clinically significant distress or impairment in social, occupational, or other important areas of functioning.
- D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication or withdrawal from another substance.

Specify if:

With perceptual disturbances: This specifier may be noted when hallucinations with intact reality testing or auditory, visual, or tactile illusions occur in the absence of a delirium.

Coding note: The ICD-10-CM code depends on whether or not there is a comorbid sedative, hypnotic, or anxiolytic use disorder and whether or not there are perceptual disturbances.

For sedative, hypnotic, or anxiolytic withdrawal, without perceptual

disturbances: If a mild sedative, hypnotic, or anxiolytic use disorder is comorbid, the ICD-10-CM code is **F13.130**, and if a moderate or severe sedative, hypnotic, or anxiolytic use disorder is comorbid, the ICD-10-CM code is **F13.230**. If there is no comorbid sedative, hypnotic, or anxiolytic use disorder (e.g., in a patient taking sedatives, hypnotics, or anxiolytics solely under appropriate medical supervision), then the ICD-10-CM code is **F13.930**.

For sedative, hypnotic, or anxiolytic withdrawal, with perceptual disturbances: If a mild sedative, hypnotic, or anxiolytic use disorder is comorbid, the ICD-10-CM code is **F13.132**, and if a moderate or severe sedative, hypnotic, or anxiolytic use disorder is comorbid, the ICD-10-CM code is **F13.232**. If there is no comorbid sedative, hypnotic, or anxiolytic use disorder (e.g., in a patient taking sedatives, hypnotics, or anxiolytics solely under appropriate medical supervision), then the ICD-10-CM code is **F13.932**.

Note: For information on Development and Course; Risk and Prognostic Factors; Culture-Related Diagnostic Issues; Functional Consequences of Sedative, Hypnotic, or Anxiolytic Withdrawal; and Comorbidity, see the corresponding sections in Sedative, Hypnotic, or Anxiolytic Use Disorder.

Diagnostic Features

The essential feature of sedative, hypnotic, or anxiolytic withdrawal is the presence of a characteristic syndrome that develops after a marked decrease in or cessation of intake after several weeks or more of regular use (Criteria A and B). This withdrawal syndrome is characterized by two or more symptoms (similar to alcohol withdrawal) that include autonomic hyperactivity (e.g., increases in heart rate, respiratory rate, blood pressure, or body temperature, along with sweating); a tremor of the hands; insomnia; nausea, sometimes accompanied by vomiting; anxiety; and psychomotor agitation. A grand mal seizure may occur in perhaps as many as 20%–30% of individuals undergoing untreated withdrawal from these substances. In severe withdrawal, visual, tactile, or auditory hallucinations or illusions can occur but are usually in the context of a withdrawal delirium. If the individual’s reality testing is intact (i.e., knows the substance is causing the hallucinations) and the illusions occur in a clear sensorium, the specifier “with perceptual disturbances”

can be noted. When hallucinations occur in the absence of intact reality testing, a diagnosis of substance/medication-induced psychotic disorder should be considered. The symptoms cause clinically significant distress or impairment in social, occupational, or other important areas of functioning (Criterion C). The symptoms must not be attributable to another medical condition and are not better explained by another mental disorder (e.g., alcohol withdrawal, generalized anxiety disorder) (Criterion D). Relief of withdrawal symptoms with administration of any sedative-hypnotic agent would support a diagnosis of sedative, hypnotic, or anxiolytic withdrawal.

Associated Features

The timing and severity of the withdrawal syndrome will differ depending on the specific substance and its pharmacokinetics and pharmacodynamics. For example, withdrawal from shorter-acting substances that are rapidly absorbed and that have no active metabolites (e.g., triazolam) can begin within hours after the substance is stopped; withdrawal from substances with long-acting metabolites (e.g., diazepam) may not begin for 1–2 days or longer. The withdrawal syndrome produced by substances in this class may be characterized by the development of a delirium that can be life-threatening. There may be evidence of tolerance and withdrawal in the absence of a diagnosis of a benzodiazepine use disorder in an individual who has abruptly discontinued benzodiazepines that were taken for long periods of time at prescribed and therapeutic doses.

The time course of the withdrawal syndrome is generally predicted by the half-life of the substance. Medications whose actions typically last about 10 hours or less (e.g., lorazepam, oxazepam, temazepam) produce withdrawal symptoms within 6–8 hours of decreasing blood levels that peak in intensity on the second day and improve markedly by the fourth or fifth day. For substances with longer half-lives (e.g., diazepam), symptoms may not develop for more than 1 week, peak in intensity during the second week, and decrease markedly during the third or fourth week. There may be additional longer-term symptoms at a much lower level of intensity that persist for several months.

The longer the substance has been taken and the higher the dosages used, the more likely there will be severe withdrawal. However, withdrawal has been reported with as little as 15 mg of diazepam (or its equivalent in other benzodiazepines) when taken daily for several months. Doses of approximately 40 mg of diazepam (or its equivalent) daily are more likely to produce clinically relevant withdrawal symptoms, and even higher doses (e.g., 100 mg of diazepam) are more likely to be followed by withdrawal seizures or delirium. Sedative, hypnotic, or anxiolytic withdrawal delirium is characterized by disturbances in consciousness and cognition, with visual, tactile, or auditory hallucinations. When present, sedative, hypnotic, or anxiolytic withdrawal delirium should be diagnosed instead of withdrawal.

Prevalence

The prevalence of sedative, hypnotic, or anxiolytic withdrawal is unknown.

Diagnostic Markers

Seizures and autonomic instability in the setting of a history of prolonged exposure to sedative, hypnotic, or anxiolytic medications suggest a high likelihood of sedative, hypnotic, or anxiolytic withdrawal.

Differential Diagnosis

Other medical conditions. The symptoms of sedative, hypnotic, or anxiolytic withdrawal may be mimicked by other medical conditions (e.g., hypoglycemia, diabetic ketoacidosis). If seizures are a feature of the sedative, hypnotic, or anxiolytic withdrawal, the

differential diagnosis includes the various causes of seizures (e.g., infections, head injury, poisonings).

Essential tremor. Essential tremor, a neurological condition that frequently runs in families, may erroneously suggest the tremulousness associated with sedative, hypnotic, or anxiolytic withdrawal.

Alcohol withdrawal. Alcohol withdrawal produces a syndrome very similar to that of sedative, hypnotic, or anxiolytic withdrawal. The differential diagnosis is determined mostly through clinical history, although liver damage and other potential signs of chronic alcohol toxicity (e.g., cardiomyopathy) can also be more suggestive of alcohol withdrawal than of sedative, hypnotic, or anxiolytic withdrawal.

Sedative-, hypnotic-, or anxiolytic-induced mental disorders. Sedative, hypnotic, or anxiolytic withdrawal is distinguished from sedative-, hypnotic-, or anxiolytic-induced mental disorders (e.g., sedative-, hypnotic-, or anxiolytic-induced anxiety disorder, with onset during withdrawal) because the symptoms (e.g., anxiety) in the latter disorders are in excess of those usually associated with sedative, hypnotic, or anxiolytic withdrawal; predominate in the clinical presentation; and are severe enough to warrant clinical attention.

Anxiety disorders. Recurrence or worsening of an underlying anxiety disorder produces a syndrome similar to sedative, hypnotic, or anxiolytic withdrawal, although the most extreme manifestations of withdrawal, such as delirium tremens or true seizures, are not symptoms of any anxiety disorder. Withdrawal would be suspected with an abrupt reduction in the dosage of a sedative, hypnotic, or anxiolytic medication. When a taper is under way, distinguishing the withdrawal syndrome from the underlying anxiety disorder can be difficult. As with alcohol, lingering withdrawal symptoms (e.g., anxiety, moodiness, trouble sleeping) can be mistaken for independent anxiety or depressive disorders (e.g., generalized anxiety disorder).

Comorbidity

Given the typical overlap of sedative, hypnotic, or anxiolytic withdrawal with sedative, hypnotic, or anxiolytic use disorder, see “Comorbidity” under Sedative, Hypnotic, or Anxiolytic Use Disorder for more details about co-occurring conditions that are likely to be encountered.

Sedative-, Hypnotic-, or Anxiolytic-Induced Mental Disorders

The following sedative-, hypnotic-, or anxiolytic-induced mental disorders are described in other chapters of the manual with disorders with which they share phenomenology (see the substance/medication-induced mental disorders in these chapters): sedative-, hypnotic-, or anxiolytic-induced psychotic disorder (“Schizophrenia Spectrum and Other Psychotic Disorders”); sedative-, hypnotic-, or anxiolytic-induced bipolar and related disorder (“Bipolar and Related Disorders”); sedative-, hypnotic-, or anxiolytic-induced depressive disorder (“Depressive Disorders”); sedative-, hypnotic-, or anxiolytic-induced anxiety disorder (“Anxiety Disorders”); sedative-, hypnotic-, or anxiolytic-induced sleep disorder (“Sleep-Wake Disorders”); sedative-, hypnotic-, or anxiolytic-induced sexual dysfunction (“Sexual

Dysfunctions”); and sedative-, hypnotic-, or anxiolytic-induced major or mild neurocognitive disorder (“Neurocognitive Disorders”). For sedative, hypnotic, or anxiolytic intoxication delirium; sedative, hypnotic, or anxiolytic withdrawal delirium; and delirium induced by sedatives, hypnotics, or anxiolytics taken as prescribed, see the

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criteria and discussion of delirium in the chapter “Neurocognitive Disorders.” These sedative-, hypnotic-, or anxiolytic-induced mental disorders are diagnosed instead of sedative, hypnotic, or anxiolytic intoxication or sedative, hypnotic, or anxiolytic withdrawal only when the symptoms are sufficiently severe to warrant independent clinical attention.

Unspecified Sedative-, Hypnotic-, or Anxiolytic-Related Disorder

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This category applies to presentations in which symptoms characteristic of a sedative-, hypnotic-, or anxiolytic-related disorder that cause clinically significant distress or impairment in social, occupational, or other important areas of functioning predominate but do not meet the full criteria for any specific sedative-, hypnotic-, or anxiolytic-related disorder or any of the disorders in the substance-related and addictive disorders diagnostic class.

Stimulant-Related Disorders

Stimulant Use Disorder

Stimulant Intoxication

Stimulant Withdrawal

Stimulant-Induced Mental Disorders

Unspecified Stimulant-Related Disorder

Stimulant Use Disorder

Diagnostic Criteria

- A. A pattern of amphetamine-type substance, cocaine, or other stimulant use leading to clinically significant impairment or distress, as manifested by at least

two of the following, occurring within a 12-month period:

1. The stimulant is often taken in larger amounts or over a longer period than was intended.
2. There is a persistent desire or unsuccessful efforts to cut down or control stimulant use.
3. A great deal of time is spent in activities necessary to obtain the stimulant, use the stimulant, or recover from its effects.
4. Craving, or a strong desire or urge to use the stimulant.
5. Recurrent stimulant use resulting in a failure to fulfill major role obligations at work, school, or home.
6. Continued stimulant use despite having persistent or recurrent social or interpersonal problems caused or exacerbated by the effects of the stimulant.
7. Important social, occupational, or recreational activities are given up or reduced because of stimulant use.

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8. Recurrent stimulant use in situations in which it is physically hazardous.
9. Stimulant use is continued despite knowledge of having a persistent or recurrent physical or psychological problem that is likely to have been caused or exacerbated by the stimulant.
10. Tolerance, as defined by either of the following:
 - a. A need for markedly increased amounts of the stimulant to achieve intoxication or desired effect.
 - b. A markedly diminished effect with continued use of the same amount of the stimulant.

Note: This criterion is not considered to be met for those taking stimulant medications solely under appropriate medical supervision, such as medications for attention-deficit/hyperactivity disorder or narcolepsy.

11. Withdrawal, as manifested by either of the following:
 - a. The characteristic withdrawal syndrome for the stimulant (refer to Criteria A and B of the criteria set for stimulant withdrawal).
 - b. The stimulant (or a closely related substance) is taken to relieve or avoid withdrawal symptoms.

Note: This criterion is not considered to be met for those taking stimulant medications solely under appropriate medical supervision, such as medications for attention-deficit/hyperactivity disorder or narcolepsy.

Specify if:

In early remission: After full criteria for stimulant use disorder were previously met, none of the criteria for stimulant use disorder have been met for at least 3 months but for less than 12 months (with the exception that Criterion A4,

“Craving, or a strong desire or urge to use the stimulant,” may be met).

In sustained remission: After full criteria for stimulant use disorder were previously met, none of the criteria for stimulant use disorder have been met at any time during a period of 12 months or longer (with the exception that Criterion A4, “Craving, or a strong desire or urge to use the stimulant,” may be met).

Specify if:

In a controlled environment: This additional specifier is used if the individual is in an environment where access to stimulants is restricted.

Code based on current severity/remission: If an amphetamine-type substance intoxication, amphetamine-type substance withdrawal, or amphetamine-type substance-induced mental disorder is also present, do not use the codes below for amphetamine-type substance use disorder. Instead, the comorbid amphetamine-type substance use disorder is indicated in the 4th character of the amphetamine-type substance-induced disorder code (see the coding note for amphetamine-type substance intoxication, amphetamine-type substance withdrawal, or a specific amphetamine-type substance-induced mental disorder). For example, if there is comorbid amphetamine-induced depressive disorder and amphetamine use disorder, only the amphetamine-induced depressive disorder code is given, with the 4th character indicating whether the comorbid amphetamine use disorder is mild, moderate, or severe: F15.14 for mild amphetamine use disorder with amphetamine-induced depressive disorder or F15.24 for a moderate or severe amphetamine use disorder with amphetamine-induced depressive disorder. (The instructions for amphetamine-type substance also apply to other or unspecified stimulant intoxication, other or unspecified stimulant withdrawal, and other or unspecified stimulant-induced mental disorder.) Similarly, if there is comorbid cocaine-induced depressive disorder and cocaine use disorder, only the cocaine-induced depressive disorder code is given, with the 4th character indicating whether the comorbid cocaine use disorder is mild, moderate,

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or severe: F14.14 for a mild cocaine use disorder with cocaine-induced depressive disorder or F14.24 for a moderate or severe cocaine use disorder with cocaine-induced depressive disorder.

Specify current severity/remission:

Mild: Presence of 2–3 symptoms.

F15.10 Amphetamine-type substance

F14.10 Cocaine

F15.10 Other or unspecified stimulant

Mild, In early remission

F15.11 Amphetamine-type substance

F14.11 Cocaine

F15.11 Other or unspecified stimulant

Mild, In sustained remission

F15.11 Amphetamine-type substance

F14.11 Cocaine

F15.11 Other or unspecified stimulant

Moderate: Presence of 4–5 symptoms.

F15.20 Amphetamine-type substance

F14.20 Cocaine

F15.20 Other or unspecified stimulant

Moderate, In early remission

F15.21 Amphetamine-type substance

F14.21 Cocaine

F15.21 Other or unspecified stimulant

Moderate, In sustained remission

F15.21 Amphetamine-type substance

F14.21 Cocaine

F15.21 Other or unspecified stimulant

Severe: Presence of 6 or more symptoms.

F15.20 Amphetamine-type substance

F14.20 Cocaine

F15.20 Other or unspecified stimulant

Severe, In early remission

F15.21 Amphetamine-type substance

F14.21 Cocaine

F15.21 Other or unspecified stimulant

Severe, In sustained remission

F15.21 Amphetamine-type substance

F14.21 Cocaine

F15.21 Other or unspecified stimulant

Specifiers

“In a controlled environment” applies as a further specifier of remission if the individual is both in remission and in a controlled environment (i.e., in early remission in a controlled environment or in sustained remission in a controlled environment). Examples of these environments are closely supervised and substance-free jails, therapeutic communities, and locked hospital units.

Diagnostic Features

Stimulants are a type of psychoactive substance that increases activity in the brain and can temporarily elevate alertness, mood, and awareness. Stimulants covered in this chapter include amphetamine and prescription stimulants with similar effects (e.g., methylphenidate) and cocaine. Substance-related disorders involving certain other substances with stimulant properties are classified in other sections of this chapter. These include caffeine (in caffeine-related disorders), nicotine (in tobacco-related disorders), and MDMA (3,4-methylenedioxymethamphetamine; in other hallucinogen-related disorders), which has both stimulant and hallucinogenic effects.

Given that the effects of amphetamine-type substances are similar to those of cocaine, amphetamine-related disorders and cocaine-related disorders are grouped under the single rubric “stimulant-related disorders.” Amphetamine-type substances (and other or unspecified stimulants) and cocaine have different ICD-10-CM codes (e.g., F15.10 mild amphetamine-type substance use disorder, F14.10 mild cocaine use disorder). The particular stimulant used by the individual is recorded in the diagnosis (e.g., “methamphetamine withdrawal,” “methylphenidate use disorder,” “cocaine intoxication”).

The amphetamine-type substances include stimulants with a substituted phenylethylamine structure, such as amphetamine, dextroamphetamine, and methamphetamine. Also included are substances that are structurally different but have similar effects, such as methylphenidate, modafinil, and armodafinil. These amphetamine-type substances are usually taken orally or intravenously, although methamphetamine is also taken by the nasal route. In addition to the synthetic amphetamine-type compounds, there are naturally occurring, plant-derived stimulants such as *khât*, as well as synthetic chemical *khât* analogs, called *cathinones*.

Amphetamines and other stimulants may be obtained by prescription for the treatment of obesity, attention-deficit/hyperactivity disorder, and narcolepsy. Consequently, prescribed stimulants may be diverted into the illegal market.

Cocaine, a naturally occurring substance produced by the coca plant, is consumed in several preparations (e.g., coca leaves, coca paste, cocaine hydrochloride, and cocaine alkaloids such as freebase and crack) that differ in potency because of varying levels of purity and speed of onset. However, in all of the forms, cocaine is the active ingredient. Cocaine hydrochloride powder is usually “snorted” through the nostrils or dissolved in water and injected intravenously. Crack and other cocaine alkaloids are easily vaporized and inhaled, and thus their effects have an extremely rapid onset.

Individuals exposed to amphetamine-type substances or cocaine can develop stimulant use disorder as rapidly as 1 week, although the onset is not always this rapid. Regardless of the route of administration, tolerance occurs with repeated use. Withdrawal symptoms, particularly hypersomnia, increased appetite, and dysphoria, can occur and can enhance craving. Most individuals with stimulant use disorder have experienced tolerance or withdrawal.

Use patterns and course are similar for disorders involving amphetamine-type substances and cocaine, as both are potent central nervous system stimulants with similar psychoactive and sympathomimetic effects. Amphetamine-type substances are longer acting than cocaine and thus are used fewer times per day. Usage may be chronic or episodic, with binges punctuated by brief non-use periods. Aggressive or violent behavior is common when high doses are smoked, ingested, or administered intravenously. Intense temporary anxiety resembling panic disorder or

generalized anxiety disorder, as well as paranoid ideation and psychotic episodes that resemble schizophrenia, is seen with high-dose use.

Withdrawal states are associated with temporary but intense depressive symptoms that can resemble a major depressive episode; the depressive symptoms usually resolve

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within 1 week. Tolerance to amphetamine-type substances develops and leads to escalation of the dose. Conversely, some users of amphetamine-type substances develop sensitization, characterized by enhanced effects.

Associated Features

When injected or smoked, stimulants typically produce an instant feeling of well-being, confidence, and euphoria. Dramatic behavioral changes can rapidly develop with stimulant use disorder. Chaotic behavior, social isolation, aggressive behavior, and sexual dysfunction can result from long-term stimulant use disorder.

Individuals with acute intoxication may present with rambling speech, headache, transient ideas of reference, and tinnitus. There may be paranoid ideation, auditory hallucinations in a clear sensorium, and tactile hallucinations, which the individual usually recognizes as drug effects. Threats or acting out of aggressive behavior may occur. Depression, suicidal thoughts, irritability, anhedonia, emotional lability, or disturbances in attention and concentration commonly occur during withdrawal. Mental disturbances associated with cocaine use usually resolve hours to days after cessation of use but can persist for 1 month. Physiological changes during stimulant withdrawal are opposite to those of the intoxication phase, sometimes including bradycardia. Temporary depressive symptoms may meet symptomatic and duration criteria for major depressive episode. Histories consistent with repeated panic attacks, social anxiety disorder-like behavior, and generalized anxiety-like syndromes are common, as are eating disorders. One extreme instance of stimulant toxicity is stimulant-induced psychotic disorder, a disorder that resembles schizophrenia, with delusions and hallucinations.

Individuals with stimulant use disorder often develop conditioned responses to drug-related stimuli (e.g., craving on seeing any white powderlike substance). These responses contribute to relapse, are difficult to extinguish, and persist after detoxification.

Depressive symptoms with suicidal thoughts or behavior can occur and are generally the most serious problems seen during stimulant withdrawal.

Prevalence

Stimulant use disorder: amphetamine-type substances. Estimated 12-month prevalence of amphetamine-type substance use disorder in the United States is 0.4% among individuals 12 years and older. Twelve-month prevalence is 0.1% among individuals ages 12–17 years, 0.5% among those ages 18–25, and 0.4% among those age 26 and older. Rates are 0.5% for men and 0.2% for women, overall. Rates are approximately 0.4% among Hispanics and non-Hispanic Whites and 0.1% among African Americans and Asian Americans. Prevalence estimates for American Indian/Alaskan Natives and Native Hawaiian/Pacific Islander populations are difficult to determine, given small sample sizes, but there is some evidence for higher rates in American

Indians/Alaskan Natives.

Among U.S. adults, 6.6% (annual average) used prescription stimulants overall; 4.5% used without misuse, 1.9% misused without use disorders, and 0.2% had use disorders. While non-Hispanic Whites are more likely to use prescription stimulants nonmedically, Hispanics tend to use them more frequently and have higher rates of prescription stimulant use disorder.

Stimulant use disorder: cocaine. Estimated 12-month prevalence of cocaine use disorder in the United States is 0.4% among individuals 12 years and older. Rates are 0.1% among individuals ages 12–17 years, 0.7% among those ages 18–25 years, and 0.3% among those age 26 and older. Rates are 0.5% for men and 0.2% for women, overall. Rates are 0.4% among African Americans and non-Hispanic Whites, 0.3% in Hispanics, and < 0.1% among Asian Americans.

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Development and Course

In the United States, stimulant use disorder occurs throughout all levels of society and is more common among individuals ages 18–25 years compared with individuals ages 12–17 or 26 years and older. On average, first regular use among individuals in treatment occurs at approximately age 23 years. For primary methamphetamine treatment admissions, the average age is 34 years, and for primary cocaine treatment admissions, the average age is 44 years for smoked cocaine and 37 years for other routes.

Some persons begin stimulant use to control weight or to improve performance in school, work, or athletics. Initial use may include obtaining medications such as methylphenidate or amphetamine salts prescribed to others for the treatment of attention-deficit/hyperactivity disorder. Among primary treatment admissions for amphetamine-type substance use in the United States, 61% reported smoking, 26% reported injecting, and 9% reported snorting, suggesting that stimulant use disorder can develop from multiple modes of administration.

Patterns of stimulant administration include episodic or daily (or almost daily) use. Episodic use (e.g., intense use over a weekend or on one or more weekdays) tends to be separated by 2 or more days of nonuse. “Binges” involve continuous high-dose use over hours or days and are often associated with physical dependence. Binges usually terminate only when stimulant supplies are depleted or exhaustion ensues. Chronic daily use may involve high or low doses, often with an increase in dose over time.

Stimulant smoking and intravenous use are associated with rapid progression to severe-level stimulant use disorder, often occurring over weeks to months. Intranasal use of cocaine and oral use of amphetamine-type substances result in more gradual progression occurring over months to years. With continued use, there is a diminution of pleasurable effects because of tolerance and an increase in dysphoric effects.

Risk and Prognostic Factors

Temperamental. Comorbid bipolar disorder, schizophrenia, antisocial personality disorder, and other substance use disorders are risk factors for developing stimulant use disorder and for relapse to cocaine use in treatment samples. Higher stress reactivity has been correlated with frequency of cocaine use in some U.S. treatment samples. Conduct disorder in childhood and antisocial personality disorder are associated with the development of stimulant-related

disorders. In the United States, previous use of another substance, being male, having a Cluster B personality disorder, family history of substance use disorder, and being separated, divorced, or widowed all result in increased risk of using cocaine. Men who have sex with men are also at higher risk for methamphetamine use.

Environmental. Predictors of cocaine use among a cohort of U.S. teenagers include prenatal cocaine exposure, postnatal cocaine use by parents, and exposure to community violence during childhood. Research in industrialized countries suggests that exposure to intimate partner violence or childhood mistreatment often co-occurs with stimulant use, especially in women. In a cohort of U.S. women followed up longitudinally, socioeconomic status, including food insecurity, had a dose-dependent effect on risk of stimulant use. For youth, especially girls, risk factors include living in an unstable home environment, having a psychiatric condition, criminal behavior, and associating with dealers and users.

Culture-Related Diagnostic Issues

The prevalence of cocaine use in the United States increased between 2001–2002 and 2012–2013 among non-Latinx Whites, African Americans, and Latinx, but the prevalence of cocaine use disorder increased only among Whites. Despite small variations, cocaine and other stimulant use disorder diagnostic criteria perform equally across gender and

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ethnoracial groups. In limited data on prevalence estimates, it appears that American Indian/Alaskan Native populations are at higher risk for methamphetamine use disorder, and, to a lesser degree, cocaine use disorder, than are non-Hispanic Whites, while native Hawaiian/Pacific Islanders appear to have similar risks to non-Hispanic Whites.

Approximately 64% of individuals admitted to publicly funded substance abuse treatment programs for primary methamphetamine/amphetamine-related disorders are non-Hispanic White, followed by 20% of Hispanic origin, 3% Asian and Pacific Islander, and 6% non-Hispanic Black. Among individuals admitted for primary treatment related to smoked cocaine, 51% were non-Hispanic Black, 35% non-Hispanic White, 8% Hispanic, and 1% Asian/Pacific Islander. For admissions related to other routes of cocaine administration, 47% were non-Hispanic White, 31% were non-Hispanic Black, 17% were of Hispanic origin, and 1% were Asian/Pacific Islander. Rates of disorders in clinical samples should be interpreted with caution because they may be affected by differential access to and utilization of services, pathways to care, criminalization, stigma, and racial bias in diagnosis and referral for treatment.

Sex- and Gender-Related Diagnostic Issues

In the United States, women with cocaine use disorder more frequently have comorbid psychiatric disorders, such as depression and posttraumatic stress disorder (PTSD), compared with men. Gonadal hormones affect a male's responses to cocaine. Females with cocaine use disorder and higher levels of progesterone have lower stress-induced and cue-induced cocaine craving and lower cue-induced changes in blood pressure than females with cocaine use disorder and lower levels of progesterone. This may explain why use of cocaine in pregnant females is lower than in nonpregnant females.

Diagnostic Markers

Benzoyllecgonine, a metabolite of cocaine, typically remains in the urine for 1–3 days after a single dose and may be present for 7–12 days in individuals using repeated high doses. Mildly elevated liver function tests can be present in cocaine injectors or users with concomitant alcohol use. There are no neurobiological markers of diagnostic utility. Discontinuation of chronic cocaine use may be associated with electroencephalographic changes, suggesting persistent abnormalities; alterations in secretion patterns of prolactin; and downregulation of dopamine receptors.

Short-half-life amphetamine-type substances (e.g., methamphetamine) can be detected for 1–3 days, and possibly up to 4 days depending on dosage and metabolism. Hair samples can be used to detect presence of amphetamine-type substances for up to 90 days. Other laboratory findings, as well as physical findings and other medical conditions (e.g., weight loss, malnutrition; poor hygiene), are similar for both cocaine and amphetamine-type substance use disorder.

Association With Suicidal Thoughts or Behavior

Few data on the association of stimulant use disorders and suicide are available because most studies examining suicidal thoughts and behavior examine use of stimulants rather than stimulant use disorders. One systematic review found that regular or problem amphetamine use (examining primarily individuals who inject amphetamines and/or individuals admitted to treatment for use of amphetamines) is associated with increased suicide mortality. A general population study of adults in the United States found an association of prescription stimulant use disorder with suicidal thoughts. In a study of individuals admitted to substance use treatment, those with cocaine use disorder were much more likely to report suicidal thoughts than those with other substance use disorders. In a study of both men and women in the U.S. Veterans Administration health care system,

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cocaine and amphetamine use disorders were each associated with increased rates of suicide deaths.

Functional Consequences of Stimulant Use Disorder

Various medical conditions may occur depending on the route of administration. Intranasal users often develop sinusitis, irritation, bleeding of the nasal mucosa, and a perforated nasal septum. Individuals who smoke stimulants are at increased risk for respiratory problems (e.g., coughing, bronchitis, and pneumonitis). Injectors have puncture marks and “tracks,” most commonly on their forearms. Risk of HIV and hepatitis C infection increases with frequent intravenous injections and unsafe sexual activity. Other sexually transmitted diseases, hepatitis B, and tuberculosis and other lung infections are also seen. Weight loss and malnutrition are common.

Chest pain may be a common symptom during stimulant intoxication. Myocardial infarction, palpitations and arrhythmias, sudden death from respiratory or cardiac arrest, and stroke have been associated with stimulant use among young and otherwise healthy individuals. Pneumothorax can result from performing Valsalva-like maneuvers done to better absorb inhaled

smoke. Cocaine use is associated with irregularities in placental blood flow, abruptio placentae, premature labor and delivery, and an increased prevalence of infants with very low birth weights.

Individuals with stimulant use disorder may become involved in theft, prostitution, or drug dealing in order to acquire drugs or money for drugs. Traumatic injuries due to violent behavior are common among individuals trafficking drugs.

Neurocognitive impairment is common among both methamphetamine and cocaine users, including deficits related to attention, impulsivity, verbal learning/memory, working memory, and executive functioning. Transient psychosis and seizure have also been reported with chronic use of either cocaine or methamphetamine, possibly related to patterns of use or the exacerbation of preexisting vulnerabilities. Amphetamine use can cause toxic effects related to elevated body temperature, and there is some evidence that chronic use causes neuroinflammation and neurotoxicity in dopaminergic neurons. Oral health problems include “meth mouth” with gum disease, tooth decay, and mouth sores related to the toxic effects of smoking the drug and to bruxism while intoxicated. Adverse pulmonary effects appear to be less common for amphetamine-type substances because they are smoked fewer times per day, although methamphetamine use is still associated with a risk of pulmonary arterial hypertension. Emergency department visits are common for stimulant-related mental disorder symptoms, injury, skin infections, and dental pathology. In the United States, diagnosis of a stimulant use disorder is associated with a 20% increase in 30-day readmission rates in assessment of follow-up after hospitalization for “any cause” (a standard measure of overall hospital quality of care).

Differential Diagnosis

Phencyclidine intoxication. Intoxication with phencyclidine (PCP or “angel dust”) or synthetic “designer drugs” such as mephedrone (known by different names, including “bath salts”) may cause a similar clinical picture and can only be distinguished from stimulant intoxication by the presence of cocaine or amphetamine-type substance metabolites in a urine or plasma sample.

Stimulant intoxication, stimulant withdrawal, and stimulant-induced mental disorders. Stimulant use disorder is differentiated from stimulant intoxication, stimulant withdrawal, and stimulant-induced mental disorders (e.g., stimulant-induced depressive disorder) in that stimulant use disorder describes a problematic pattern of stimulant use that involves impaired control over stimulant use, social impairment attributable to stimulant use, risky stimulant use (e.g., continued stimulant use despite medical complications), and pharmacological symptoms (the development of tolerance or withdrawal), whereas stimulant

intoxication, stimulant withdrawal, and stimulant-induced mental disorders describe psychiatric syndromes that occur in the context of heavy use. Stimulant intoxication, stimulant withdrawal, and stimulant-induced mental disorders occur frequently in individuals with stimulant use disorder. In such cases, a diagnosis of stimulant intoxication, stimulant withdrawal, or a stimulant-induced mental disorder should be given in addition to a diagnosis of stimulant use disorder, the presence of which is indicated in the diagnostic code.

Independent mental disorders. Some of the effects of stimulant use may resemble symptoms of independent mental disorders, such as psychosis (schizophrenia) and low mood (major

depressive disorder). Discerning whether these behaviors occurred before the intake of the drug is important in the differentiation of acute drug effects from a preexisting mental disorder.

Comorbidity

Stimulant-related disorders often co-occur with other substance use disorders, especially those involving substances with sedative properties, which are often taken to reduce insomnia, nervousness, and other unpleasant side effects. Individuals admitted to treatment for cocaine use are likely to also use heroin, PCP, or alcohol, and individuals admitted for amphetamine-type substance use disorder are likely to use marijuana, heroin, or alcohol. Stimulant use disorder may be associated with posttraumatic stress disorder, antisocial personality disorder, attention-deficit/hyperactivity disorder, and gambling disorder. Cardiopulmonary problems are often present in individuals seeking treatment for cocaine-related problems, with chest pain being the most common. Medical problems occur in response to adulterants used as “cutting” agents. Cocaine users who ingest cocaine cut with levamisole, an antimicrobial and veterinary medication, may experience agranulocytosis and febrile neutropenia.

Stimulant Intoxication

Diagnostic Criteria

- A. Recent use of an amphetamine-type substance, cocaine, or other stimulant.
- B. Clinically significant problematic behavioral or psychological changes (e.g., euphoria or affective blunting; changes in sociability; hypervigilance; interpersonal sensitivity; anxiety, tension, or anger; stereotyped behaviors; impaired judgment) that developed during, or shortly after, use of a stimulant.
- C. Two (or more) of the following signs or symptoms, developing during, or shortly after, stimulant use:
 - 1. Tachycardia or bradycardia.
 - 2. Pupillary dilation.
 - 3. Elevated or lowered blood pressure.
 - 4. Perspiration or chills.
 - 5. Nausea or vomiting.
 - 6. Evidence of weight loss.
 - 7. Psychomotor agitation or retardation.
 - 8. Muscular weakness, respiratory depression, chest pain, or cardiac arrhythmias.
 - 9. Confusion, seizures, dyskinesias, dystonias, or coma.
- D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication with another substance.

Specify the particular intoxicant (i.e., amphetamine-type substance, cocaine, or other stimulant).

Specify if:

With perceptual disturbances: This specifier may be noted when hallucinations with intact reality testing or auditory, visual, or tactile illusions occur in the absence of a delirium.

Coding note: The ICD-10-CM code depends on whether the stimulant is an amphetamine-type substance, cocaine, or other stimulant; whether there is a comorbid amphetamine-type substance, cocaine, or other stimulant use disorder; and whether or not there are perceptual disturbances.

For amphetamine-type substance, cocaine, or other stimulant intoxication, without perceptual disturbances: If a mild amphetamine-type substance or other stimulant use disorder is comorbid, the ICD-10-CM code is **F15.120**, and if a moderate or severe amphetamine-type substance or other stimulant use disorder is comorbid, the ICD-10-CM code is **F15.220**. If there is no comorbid amphetamine-type substance or other stimulant use disorder, then the ICD-10-CM code is **F15.920**. Similarly, if a mild cocaine use disorder is comorbid, the ICD-10-CM code is **F14.120**, and if a moderate or severe cocaine use disorder is comorbid, the ICD-10-CM code is **F14.220**. If there is no comorbid cocaine use disorder, then the ICD-10-CM code is **F14.920**.

For amphetamine-type substance, cocaine, or other stimulant intoxication, with perceptual disturbances: If a mild amphetamine-type substance or other stimulant use disorder is comorbid, the ICD-10-CM code is **F15.122**, and if a moderate or severe amphetamine-type substance or other stimulant use disorder is comorbid, the ICD-10-CM code is **F15.222**. If there is no comorbid amphetamine-type substance or other stimulant use disorder, then the ICD-10-CM code is **F15.922**. Similarly, if a mild cocaine use disorder is comorbid, the ICD-10-CM code is **F14.122**, and if a moderate or severe cocaine use disorder is comorbid, the ICD-10-CM code is **F14.222**. If there is no comorbid cocaine use disorder, then the ICD-10-CM code is **F14.922**.

Diagnostic Features

The essential feature of stimulant intoxication, related to amphetamine-type substances and cocaine, is the presence of clinically significant behavioral or psychological changes that develop during, or shortly after, use of stimulants (Criteria A and B). Auditory hallucinations may be prominent, as may paranoid ideation, and these symptoms must be distinguished from an independent psychotic disorder such as schizophrenia. Stimulant intoxication usually begins with a “high” feeling and includes one or more of the following: euphoria with enhanced vigor, gregariousness, hyperactivity, restlessness, hypervigilance, interpersonal sensitivity, talkativeness, anxiety, tension, alertness, grandiosity, stereotyped and repetitive behavior, anger, impaired judgment, and, in the case of chronic intoxication, affective blunting with fatigue or

sadness and social withdrawal. These behavioral and psychological changes are accompanied by two or more of the following signs and symptoms that develop during or shortly after stimulant use: tachycardia or bradycardia; pupillary dilation; elevated or lowered blood pressure; perspiration or chills; nausea or vomiting; evidence of weight loss; psychomotor agitation or retardation; muscular weakness, respiratory depression, chest pain, or cardiac arrhythmias; and confusion, seizures, dyskinesias, dystonias, or coma (Criterion C). Intoxication, either acute or chronic, is often associated with impaired social or occupational functioning. Severe intoxication can lead to convulsions, cardiac arrhythmias, hyperpyrexia, and death. For the diagnosis of stimulant intoxication to be made, the symptoms must not be attributable to another medical condition and are not better explained by another mental disorder

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(Criterion D). While stimulant intoxication occurs in individuals with stimulant use disorders, intoxication is not a criterion for stimulant use disorder, which is confirmed by the presence of 2 of the 11 diagnostic criteria for use disorder.

Associated Features

The magnitude and direction of the behavioral and physiological changes depend on many variables, including the dose used and the characteristics of the individual using the substance or the context (e.g., tolerance, rate of absorption, chronicity of use, context in which taken). Stimulant effects such as euphoria, increased pulse and blood pressure, and psychomotor activity are most commonly seen. Depressant effects such as sadness, bradycardia, decreased blood pressure, and decreased psychomotor activity are less common and generally emerge only with chronic high-dose use.

Prevalence

Although prevalence of stimulant intoxication is not known, prevalence of stimulant use can be used as a proxy. Many individuals who use stimulants may not have symptoms that fully meet the criteria for stimulant intoxication, which requires “clinically significant problematic behavioral or psychological changes.” Thus, rates of stimulant use can be considered the upper bounds of the likely prevalence of stimulant intoxication.

Estimated 12-month prevalence of cocaine use in the United States is 2.2% for individuals age 12 and older (0.5% among individuals ages 12–17-years, 6.2% among individuals ages 18–25 years, and 1.7% among individuals age 26 and older); 3% of men/boys and 1.4% of women/girls used cocaine in the last 12 months. Twelve-month prevalence of cocaine use is 2.3% among Whites, 2.2% among Hispanics, 1.7% among African Americans, and 1% among Asian Americans.

Estimated 12-month prevalence of methamphetamine use in the United States is 0.6% for individuals age 12 and older (0.2% among individuals ages 12–17 years, 1.1% among individuals ages 18–25 years, and 0.6% among individuals age 26 and older). Twelve-month prevalence of methamphetamine use is 0.8% among men/boys and 0.4% among women/girls. Twelve-month prevalence of methamphetamine use is 0.7% among Whites, 0.6% among Hispanics, 0.2% among African Americans, and 0.1% among Asian Americans. Small sample

sizes make estimating rates among American Indians/Alaskan Natives difficult.

Differential Diagnosis

Stimulant-induced mental disorders. Stimulant intoxication is distinguished from stimulant-induced mental disorders (e.g., stimulant-induced anxiety disorder, with onset during intoxication) because the symptoms (e.g., anxiety) in the latter disorders are in excess of those usually seen in stimulant intoxication, predominate in the clinical presentation, and meet full criteria for the relevant disorder.

Independent mental disorders. Salient mental disturbances associated with stimulant intoxication should be distinguished from the symptoms of schizophrenia, bipolar and depressive disorders, generalized anxiety disorder, and panic disorder as described in this manual.

Comorbidity

Given the typical overlap of stimulant intoxication with stimulant use disorder, see “Comorbidity” under Stimulant Use Disorder for more details about co-occurring conditions that are likely to be encountered.

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Stimulant Withdrawal

Diagnostic Criteria

- A. Cessation of (or reduction in) prolonged amphetamine-type substance, cocaine, or other stimulant use.
- B. Dysphoric mood and two (or more) of the following physiological changes, developing within a few hours to several days after Criterion A:
 - 1. Fatigue.
 - 2. Vivid, unpleasant dreams.
 - 3. Insomnia or hypersomnia.
 - 4. Increased appetite.
 - 5. Psychomotor retardation or agitation.
- C. The signs or symptoms in Criterion B cause clinically significant distress or impairment in social, occupational, or other important areas of functioning.
- D. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication or withdrawal from another substance.

Specify the particular substance that causes the withdrawal syndrome (i.e., amphetamine-type substance, cocaine, or other stimulant).

Coding note: The ICD-10-CM code depends on whether the stimulant is an

amphetamine-type substance, cocaine, or other stimulant and on whether or not there is a comorbid amphetamine-type substance, cocaine, or other stimulant use disorder. If mild amphetamine-type substance or other stimulant use disorder is comorbid, the ICD-10-CM code is **F15.13**. If moderate or severe amphetamine-type substance or other stimulant use disorder is comorbid, the ICD-10-CM code is **F15.23**. For amphetamine-type substance or other stimulant withdrawal occurring in the absence of amphetamine-type substance or other stimulant use disorder (e.g., in a patient taking amphetamine solely under appropriate medical supervision), the ICD-10-CM code is **F15.93**. If mild cocaine use disorder is comorbid, the ICD-10-CM code is **F14.13**. If moderate or severe cocaine use disorder is comorbid, the ICD-10-CM code is **F14.23**. For cocaine withdrawal occurring in the absence of a cocaine use disorder, the ICD-10-CM code is **F14.93**.

Diagnostic Features

The essential feature of stimulant withdrawal is the presence of a characteristic withdrawal syndrome that develops within a few hours to several days after the cessation of (or marked reduction in) stimulant use (generally high dose) that has been prolonged (Criterion A). The withdrawal syndrome is characterized by the development of dysphoric mood accompanied by two or more of the following physiological changes: fatigue, vivid and unpleasant dreams, insomnia or hypersomnia, increased appetite, and psychomotor retardation or agitation (Criterion B). Bradycardia is often present and is a reliable measure of stimulant withdrawal.

Anhedonia and drug craving can often be present but are not part of the diagnostic criteria. These symptoms cause clinically significant distress or impairment in social, occupational, or other important areas of functioning (Criterion C). The symptoms must not be attributable to another medical condition and are not better explained by another mental disorder (Criterion D).

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Associated Features

Acute withdrawal symptoms (“a crash”) are often seen after periods of repetitive high-dose use (“runs” or “binges”). These symptoms are characterized by intense and unpleasant feelings of lassitude and depression and increased appetite, generally requiring several days of rest and recuperation. Depressive symptoms with suicidal thoughts or behavior can occur and are generally the most serious problems seen during “crashing” or other forms of stimulant withdrawal. Many individuals with stimulant use disorder may experience a withdrawal syndrome at some point.

Differential Diagnosis

Stimulant-induced mental disorders. Stimulant withdrawal is distinguished from stimulant-induced mental disorders (e.g., stimulant-induced depressive disorder, with onset during withdrawal) because the symptoms (e.g., depressed mood) in these latter disorders are in excess of those usually associated with stimulant withdrawal, predominate in the clinical presentation, and are severe enough to warrant clinical attention.

Comorbidity

Given the typical overlap of stimulant withdrawal with stimulant use disorder, see “Comorbidity” under Stimulant Use Disorder for more details about co-occurring conditions that are likely to be encountered.

Stimulant-Induced Mental Disorders

The following stimulant-induced mental disorders (which include amphetamine-type substance-, cocaine-, and other stimulant-induced mental disorders) are described in other chapters of the manual with disorders with which they share phenomenology (see the substance/medication-induced mental disorders in these chapters): stimulant-induced psychotic disorder (“Schizophrenia Spectrum and Other Psychotic Disorders”); stimulant-induced bipolar and related disorder (“Bipolar and Related Disorders”); stimulant-induced depressive disorder (“Depressive Disorders”); stimulant-induced anxiety disorder (“Anxiety Disorders”); stimulant-induced obsessive-compulsive disorder (“Obsessive-Compulsive and Related Disorders”); stimulant-induced sleep disorder (“Sleep-Wake Disorders”); stimulant-induced sexual dysfunction (“Sexual Dysfunctions”); and stimulant-induced mild neurocognitive disorder (“Neurocognitive Disorders”). For stimulant intoxication delirium and delirium induced by stimulants taken as prescribed, see the criteria and discussion of delirium in the chapter “Neurocognitive Disorders.” These stimulant-induced mental disorders are diagnosed instead of stimulant intoxication or stimulant withdrawal only when the symptoms are sufficiently severe to warrant independent clinical attention.

Unspecified Stimulant-Related Disorder

This category applies to presentations in which symptoms characteristic of a stimulant-related disorder that cause clinically significant distress or impairment in social, occupational, or other important areas of functioning predominate but do not meet the full criteria for any specific stimulant-related disorder or any of the disorders in the substance-related and addictive disorders diagnostic class.

Coding note: The ICD-10-CM code depends on whether the stimulant is an amphetamine-type substance, cocaine, or other stimulant. The ICD-10-CM code for an unspecified amphetamine-type substance or other stimulant-related disorder is **F15.99**. The ICD-10-CM code for an unspecified cocaine-related disorder is **F14.99**.

Tobacco-Related Disorders

Tobacco Use Disorder

Tobacco Withdrawal

Tobacco Use Disorder

Diagnostic Criteria

- A. A problematic pattern of tobacco use leading to clinically significant impairment or distress, as manifested by at least two of the following, occurring within a 12-month period:
1. Tobacco is often taken in larger amounts or over a longer period than was intended.
 2. There is a persistent desire or unsuccessful efforts to cut down or control tobacco use.
 3. A great deal of time is spent in activities necessary to obtain or use tobacco.
 4. Craving, or a strong desire or urge to use tobacco.
 5. Recurrent tobacco use resulting in a failure to fulfill major role obligations at work, school, or home (e.g., interference with work).
 6. Continued tobacco use despite having persistent or recurrent social or interpersonal problems caused or exacerbated by the effects of tobacco (e.g., arguments with others about tobacco use).
 7. Important social, occupational, or recreational activities are given up or reduced because of tobacco use.
 8. Recurrent tobacco use in situations in which it is physically hazardous (e.g., smoking in bed).
 9. Tobacco use is continued despite knowledge of having a persistent or recurrent physical or psychological problem that is likely to have been caused or exacerbated by tobacco.
 10. Tolerance, as defined by either of the following:
 - a. A need for markedly increased amounts of tobacco to achieve the desired effect.
 - b. A markedly diminished effect with continued use of the same amount of tobacco.
 11. Withdrawal, as manifested by either of the following:
 - a. The characteristic withdrawal syndrome for tobacco (refer to Criteria A and B of the criteria set for tobacco withdrawal).
 - b. Tobacco (or a closely related substance, such as nicotine) is taken to relieve or avoid withdrawal symptoms.

Specify if:

In early remission: After full criteria for tobacco use disorder were previously met, none of the criteria for tobacco use disorder have been met for at least 3 months but for less than 12 months (with the exception that Criterion A4, “Craving, or a strong desire or urge to use tobacco,” may be met).

In sustained remission: After full criteria for tobacco use disorder were previously met, none of the criteria for tobacco use disorder have been met at any time during a period of 12 months or longer (with the exception that Criterion A4, “Craving, or a strong desire or urge to use tobacco,” may be met).

Specify if:

On maintenance therapy: The individual is taking a long-term maintenance medication, such as nicotine replacement medication, and no criteria for tobacco use disorder have been met for that class of medication (except tolerance to, or withdrawal from, the nicotine replacement medication).

In a controlled environment: This additional specifier is used if the individual is in an environment where access to tobacco is restricted.

Code based on current severity/remission: If a tobacco withdrawal or tobacco-induced sleep disorder is also present, do not use the codes below for tobacco use disorder. Instead, the comorbid tobacco use disorder is indicated in the 4th character of the tobacco-induced disorder code (see the coding note for tobacco withdrawal or tobacco-induced sleep disorder). For example, if there is comorbid tobacco-induced sleep disorder and tobacco use disorder, only the tobacco-induced sleep disorder code is given, with the 4th character indicating whether the comorbid tobacco use disorder is moderate or severe: F17.208 for moderate or severe tobacco use disorder with tobacco-induced sleep disorder. It is not permissible to code a comorbid mild tobacco use disorder with a tobacco-induced sleep disorder.

Specify current severity/remission:

Z72.0 Mild: Presence of 2–3 symptoms.

F17.200 Moderate: Presence of 4–5 symptoms.

F17.201 Moderate, In early remission

F17.201 Moderate, In sustained remission

F17.200 Severe: Presence of 6 or more symptoms.

F17.201 Severe, In early remission

F17.201 Severe, In sustained remission

Specifiers

“On maintenance therapy” applies as a specifier to be added to “in remission” if the individual is both in remission and on maintenance therapy. “In a controlled environment” applies as a further specifier of remission if the individual is both in remission and in a controlled environment (i.e., in early remission in a controlled environment or in sustained remission in a controlled

environment). Examples of these environments are closely supervised and substance-free jails, therapeutic communities, and locked hospital units.

Diagnostic Features

Tobacco use disorder can develop with use of all forms of tobacco (e.g., cigarettes, chewing tobacco, snuff, pipes, cigars, electronic nicotine delivery devices such as electronic cigarettes [e-cigarettes]) and with prescription nicotine-containing medications (nicotine gum and patch). The relative ability of these products to produce tobacco use disorder or to induce withdrawal is associated with the rapidity of the route of administration (smoked over oral over transdermal) and the nicotine content of the product. The name of this substance category was changed from “nicotine” in prior editions of DSM to “tobacco” in DSM-5 on the basis of harms from addiction being associated mostly with tobacco and much less with nicotine.

Tobacco use disorder is common among individuals who use cigarettes and smokeless tobacco daily, is less common among individuals who use e-cigarettes, and is uncommon

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among those who do not use tobacco daily or use nicotine medications. Tolerance to tobacco is exemplified by the disappearance of nausea and dizziness after intake and by a more intense effect of tobacco the first time it is used during the day. Cessation of tobacco use can produce a well-defined withdrawal syndrome. Many individuals with tobacco use disorder use tobacco to relieve or to avoid withdrawal symptoms (e.g., after being in a situation where use is restricted). Many individuals with tobacco use disorder have tobacco-related physical symptoms or diseases and continue to smoke. The large majority report craving when they do not smoke for several hours. Spending excessive time using tobacco can be exemplified by chain-smoking (i.e., smoking one cigarette after another with no time between cigarettes). Because tobacco sources are readily and legally available, and because tobacco intoxication is very rare, spending a great deal of time attempting to procure tobacco or recovering from its effects is uncommon. Giving up important social, occupational, or recreational activities can occur when an individual forgoes an activity because it occurs in tobacco use-restricted areas. Use of tobacco rarely results in failure to fulfill major role obligations (e.g., interference with work or home responsibilities), but persistent social or interpersonal problems (e.g., having arguments with others about tobacco use, avoiding social situations because of others’ disapproval of tobacco use) or use that is physically hazardous (e.g., smoking in bed, smoking around flammable chemicals) occur at an intermediate prevalence. Although these criteria are less often endorsed by tobacco users, if endorsed, they can indicate a more severe disorder.

Associated Features

Smoking within 30 minutes of waking, smoking daily, smoking more cigarettes per day, and waking at night to smoke are associated with tobacco use disorder. Environmental cues can evoke craving and withdrawal. Serious medical conditions often occur, including lung and other cancers, cardiac and pulmonary disease, perinatal problems, cough, shortness of breath, and accelerated skin aging.

Prevalence

Although cigarettes are the most commonly used tobacco product, use of other tobacco products (especially e-cigarettes) has become more common. In the United States, 19% of adults used a tobacco product in the last year, 19% used more than one product, 14% used cigarettes, 4% used cigars, 3% used e-cigarettes, and 2% used smokeless tobacco. One fourth (24%) of current U.S. smokers are nondaily smokers.

The 12-month prevalence of DSM-5 tobacco use disorder in the United States in 2012–2013 was 20% among adults age 18 years and older and 29.6% among Native Americans, 22.3% among non-Latinx Whites, 20.1% among African Americans, 12.2% among Latinx, and 11.2% among Asian Americans and Pacific Islanders. Prevalence was higher among men; those who were young, unmarried, less educated, poor, or residing in the southern United States; and those with almost any psychiatric disorder. The prevalence among current daily smokers is approximately 50%.

Global comparisons show that in all geographic regions of the world, the age-standardized prevalence of daily tobacco smoking is higher in men than in women, but the gender ratio varies greatly, from 16.9:1 in East Asia to 1.2:1 in Australasia.

Development and Course

About 20% of U.S. high school seniors report having ever smoked cigarettes, and about 5% have used in the past 30 days. Among adolescents who smoke cigarettes at least monthly, most of these individuals will become daily tobacco users in the future. Initiation of smoking after age 21 years is rare. Some of the tobacco use disorder criteria symptoms (e.g., craving) occur soon after beginning tobacco use, suggesting the addiction process begins with initial use; however, fulfilling DSM criteria usually occurs over several years.

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Nondaily smoking has become more prevalent since the late 1990s in the United States, especially among individuals ages 18–34 years, Blacks, Hispanics, and individuals with at least a college education.

Risk and Prognostic Factors

Temperamental. Individuals with externalizing personality traits are more likely to initiate tobacco use. Children with attention-deficit/hyperactivity disorder or conduct disorder, and adults with depressive, bipolar, anxiety, personality, psychotic, or other substance use disorders, are at higher risk for starting and continuing tobacco use and of tobacco use disorder.

Environmental. Persons with low incomes and low educational levels are more likely to initiate tobacco use and are less likely to stop.

Genetic and physiological. Genetic factors contribute to the onset of tobacco use, the continuation of tobacco use, and the development of tobacco use disorder, with a degree of heritability equivalent to that observed with other substance use disorders (i.e., about 50%). Some of this risk is specific to tobacco, and some is common with the vulnerability to developing any substance use disorder.

Culture-Related Diagnostic Issues

Acceptance of tobacco use varies across cultural contexts. Age-standardized prevalence of daily tobacco smoking varies greatly by geographic region, ranging from 4.7% in Western Sub-Saharan Africa to 24.2% in Eastern Europe. The degree to which these geographic differences are the result of income, education, and tobacco control activities in a country is unclear. Prevalence of tobacco use in the United States varies by age, gender, and ethnoracial background, with lower rates of smoking onset and progression to daily smoking among Black youth, especially young women. Liver enzyme polymorphisms that vary across ethnoracial groups can affect nicotine metabolism, contributing to variation in smoking behavior. Higher tobacco use disorder prevalence is also associated with exposure to racism and ethnic discrimination. Prevalence of DSM-IV nicotine dependence is higher among adult lesbian, gay, and bisexual individuals than among heterosexuals, possibly also due to an association with exposure to sexual orientation–related discrimination. Among individuals with DSM-IV nicotine dependence, lower income and education are associated with disorder persistence.

Sex- and Gender-Related Diagnostic Issues

The ratio of men to women among U.S. smokers is approximately 1.4:1 and has been stable between 2004 and 2014. This ratio is generally consistent across various income and educational levels. The ratio diminishes in older age groups as fewer men are smoking as age increases. The literature from several U.S. settings suggests that negative reinforcement (i.e., that smoking relieves negative affect) is a greater motivator in women than in men. Menstrual cycle effects on smoking are found inconsistently, but tobacco withdrawal appears worse in the luteal than the follicular phase of the cycle. Pregnant females smoke at a lower rate than nonpregnant females but relapse back to smoking rapidly after delivery.

Diagnostic Markers

The following biomarkers can be used to measure the extent of tobacco or nicotine use: carbon monoxide in the breath and nicotine and its metabolite cotinine in blood, saliva, or urine; however, these are only weakly associated with tobacco use disorder.

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Association With Suicidal Thoughts or Behavior

National U.S. survey data show that past-year cigarette use is associated with a two- to threefold increased risk of suicidal thoughts and behavior, with earlier age at first tobacco use increasing risk. Evidence from the U.S. Veterans Health Administration shows that even after adjustment for covariates, tobacco use disorder is associated with an increased risk of suicide. A large study of twins in Finland found that the relationship between tobacco use and suicide increased in a dose-response manner, and that for identical twins discordant for tobacco use, tobacco use was associated with a sixfold increased risk for suicide.

Functional Consequences of Tobacco Use Disorder

Medical consequences of tobacco use often begin when tobacco users are in their 40s and usually

become progressively more debilitating over time. One-half of smokers who do not stop using tobacco will die early from a tobacco-related illness, and smoking-related morbidity occurs in more than one-half of tobacco users. Most medical conditions result from exposure to carbon monoxide, tars, and other non-nicotine components of tobacco. The major predictor of reversibility is duration of smoking. Secondhand smoke increases the risk of heart disease and cancer by 30%. Long-term use of nicotine medications does not appear to cause medical harm.

Comorbidity

The most common medical conditions from smoking are cardiovascular illnesses, chronic obstructive pulmonary disease, and cancers. Smoking also increases perinatal problems, such as low birth weight and miscarriage. Prevalence of smoking is almost twice as high in individuals with major depressive disorder; although the prevalence of smoking in the United States is higher among individuals with low socioeconomic status, the increased prevalence of smoking among those with depression is independent of socioeconomic status. The most common psychiatric comorbidities associated with smoking are alcohol and other substance, depressive, bipolar, anxiety, personality, and attention-deficit/hyperactivity disorders. In the United States, individuals with a psychiatric disorder are three times more likely than others to have tobacco use disorder. Adults with DSM-5 tobacco use disorder are significantly more likely than other adults to have comorbid psychiatric disorders, including other DSM-5 substance use disorders, major depressive disorder, bipolar I disorder, panic disorder, generalized anxiety disorder, posttraumatic stress disorder, and borderline and antisocial personality disorders.

Tobacco Withdrawal

Diagnostic Criteria	F17.203
A. Daily use of tobacco for at least several weeks. B. Abrupt cessation of tobacco use, or reduction in the amount of tobacco used, followed within 24 hours by four (or more) of the following signs or symptoms: 1. Irritability, frustration, or anger. 2. Anxiety. 3. Difficulty concentrating. 4. Increased appetite. 5. Restlessness. 6. Depressed mood. 7. Insomnia. 650 C. The signs or symptoms in Criterion B cause clinically significant distress or impairment in social, occupational, or other important areas of functioning.	

- D. The signs or symptoms are not attributed to another medical condition and are not better explained by another mental disorder, including intoxication or withdrawal from another substance.

Coding note: The ICD-10-CM code for tobacco withdrawal is **F17.203**. Note that the ICD-10-CM code indicates the comorbid presence of a moderate or severe tobacco use disorder, reflecting the fact that tobacco withdrawal can only occur in the presence of a moderate or severe tobacco use disorder.

Diagnostic Features

Withdrawal symptoms impair the ability to stop tobacco use. The symptoms after abstinence from tobacco are in large part due to nicotine deprivation. Tobacco withdrawal is common among daily tobacco users who stop or reduce their use of tobacco. Symptoms are more intense among individuals who smoke cigarettes and also use smokeless tobacco or electronic cigarettes daily. This symptom intensity is likely attributable to the more rapid onset and higher levels of nicotine with cigarette smoking. Significant withdrawal among those who are nondaily cigarette users or use only nicotine medications is uncommon.

Typically, heart rate decreases by 5–12 bpm in the first few days after stopping smoking, and weight increases an average of 4–7 lb (2–3 kg) over the first year after stopping smoking. Tobacco withdrawal can produce clinically significant mood changes and functional impairment. Because of conditioning effects, withdrawal can be prompted by environmental cues such as seeing others smoking. Gradual reduction of tobacco decreases the severity of withdrawal.

Associated Features

Craving for tobacco or nicotine is very common during abstinence and has a large effect on the ability to remain abstinent. Abstinence can increase impulsivity and anhedonia and can decrease positive affect. Abstinence from tobacco or nicotine also appears to increase craving for sweet or sugary foods and impairs performance on tasks requiring vigilance. Smoking increases the metabolism of many medications used to treat mental disorders; thus, cessation of smoking can increase the blood levels of these medications, and this can produce clinically significant outcomes. This effect appears to be due not to nicotine but rather to other compounds in tobacco.

Prevalence

Approximately 50% of daily smokers who quit for 2 or more days will have four or more symptoms of tobacco withdrawal. The most commonly endorsed signs and symptoms are anxiety, irritability, and difficulty concentrating. The least commonly endorsed symptoms are depression and insomnia.

Development and Course

Tobacco withdrawal usually begins within 24 hours of stopping or cutting down tobacco use, peaks at 2–3 days after abstinence, and usually lasts 2–3 weeks. Tobacco withdrawal symptoms can occur among adolescent tobacco users, even prior to daily tobacco use. Prolonged symptoms beyond 1 month can occur but are uncommon.

Risk and Prognostic Factors

Temperamental. Smokers with depressive disorders, bipolar disorders, anxiety disorders, attention-deficit/hyperactivity disorder, and other substance use disorders have more severe withdrawal.

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Genetic and physiological. Genotype can influence the probability of withdrawal upon abstinence.

Diagnostic Markers

The following biomarkers can be used to measure the extent of tobacco or nicotine use but are only weakly associated with tobacco withdrawal: carbon monoxide in the breath and nicotine and its metabolite cotinine in blood, saliva, or urine.

Functional Consequences of Tobacco Withdrawal

Tobacco withdrawal can cause significant distress and difficulty functioning in a minority of smokers, but this may be uncommon. Withdrawal impairs the ability to stop or control tobacco use. Whether tobacco withdrawal can prompt the development of a new mental disorder or recurrence of a mental disorder is debatable, but if this occurs, it would be in a small minority of tobacco users.

Differential Diagnosis

The symptoms of tobacco withdrawal overlap with those of other substance withdrawal syndromes (e.g., alcohol withdrawal; sedative, hypnotic, or anxiolytic withdrawal; stimulant withdrawal; caffeine withdrawal; opioid withdrawal); caffeine intoxication; anxiety, depressive, bipolar, and sleep disorders; and medication-induced akathisia. Admission to smoke-free inpatient units or voluntary smoking cessation can induce withdrawal symptoms that mimic, intensify, or disguise other disorders or adverse effects of medications used to treat mental disorders (e.g., irritability thought to be due to alcohol withdrawal could be due to tobacco withdrawal). Reduction in symptoms with the use of nicotine confirms the diagnosis.

Comorbidity

Given the typical overlap of tobacco withdrawal with tobacco use disorder, see “Comorbidity” under Tobacco Use Disorder for more details about co-occurring conditions that are likely to be encountered.

Tobacco-Induced Mental Disorders

Tobacco-induced sleep disorder is discussed in the chapter “Sleep-Wake Disorders” (see “Substance/Medication-Induced Sleep Disorder”).

Unspecified Tobacco-Related Disorder

This category applies to presentations in which symptoms characteristic of a tobacco-related disorder that cause clinically significant distress or impairment in social, occupational, or other important areas of functioning predominate but do not meet the full criteria for any specific tobacco-related disorder or any of the disorders in the substance-related and addictive disorders diagnostic class.

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Other (or Unknown) Substance–Related Disorders

Other (or Unknown) Substance Use Disorder

Other (or Unknown) Substance Intoxication

Other (or Unknown) Substance Withdrawal

Other (or Unknown) Substance–Induced Mental Disorders

Unspecified Other (or Unknown) Substance–Related Disorder

Other (or Unknown) Substance Use Disorder

Diagnostic Criteria

- A. A problematic pattern of use of an intoxicating substance not able to be classified within the alcohol; caffeine; cannabis; hallucinogen (phencyclidine and others); inhalant; opioid; sedative, hypnotic, or anxiolytic; stimulant; or tobacco categories and leading to clinically significant impairment or distress, as manifested by at least two of the following, occurring within a 12-month period:
 1. The substance is often taken in larger amounts or over a longer period than was intended.
 2. There is a persistent desire or unsuccessful efforts to cut down or control use of the substance.
 3. A great deal of time is spent in activities necessary to obtain the substance, use the substance, or recover from its effects.
 4. Craving, or a strong desire or urge to use the substance.
 5. Recurrent use of the substance resulting in a failure to fulfill major role obligations at work, school, or home.

6. Continued use of the substance despite having persistent or recurrent social or interpersonal problems caused or exacerbated by the effects of its use.
7. Important social, occupational, or recreational activities are given up or reduced because of use of the substance.
8. Recurrent use of the substance in situations in which it is physically hazardous.
9. Use of the substance is continued despite knowledge of having a persistent or recurrent physical or psychological problem that is likely to have been caused or exacerbated by the substance.
10. Tolerance, as defined by either of the following:
 - a. A need for markedly increased amounts of the substance to achieve intoxication or desired effect.
 - b. A markedly diminished effect with continued use of the same amount of the substance.
11. Withdrawal, as manifested by either of the following:
 - a. The characteristic withdrawal syndrome for other (or unknown) substance (refer to Criteria A and B of the criteria sets for other [or unknown] substance withdrawal).

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- b. The substance (or a closely related substance) is taken to relieve or avoid withdrawal symptoms.

Specify if:

In early remission: After full criteria for other (or unknown) substance use disorder were previously met, none of the criteria for other (or unknown) substance use disorder have been met for at least 3 months but for less than 12 months (with the exception that Criterion A4, "Craving, or a strong desire or urge to use the substance," may be met).

In sustained remission: After full criteria for other (or unknown) substance use disorder were previously met, none of the criteria for other (or unknown) substance use disorder have been met at any time during a period of 12 months or longer (with the exception that Criterion A4, "Craving, or a strong desire or urge to use the substance," may be met).

Specify if:

In a controlled environment: This additional specifier is used if the individual is in an environment where access to the substance is restricted.

Code based on current severity/remission: If an other (or unknown) substance intoxication, other (or unknown) substance withdrawal, or other (or unknown) substance-induced mental disorder is present, do not use the codes below for other (or unknown) substance use disorder. Instead, the comorbid other (or unknown) substance use disorder is indicated in the 4th character of the other (or unknown)

substance-induced disorder code (see the coding note for other [or unknown] substance intoxication, other [or unknown] substance withdrawal, or specific other [or unknown] substance-induced mental disorder). For example, if there is comorbid other (or unknown) substance-induced depressive disorder and other (or unknown) substance use disorder, only the other (or unknown) substance-induced depressive disorder code is given, with the 4th character indicating whether the comorbid other (or unknown) substance use disorder is mild, moderate, or severe: F19.14 for other (or unknown) substance use disorder with other (or unknown) substance-induced depressive disorder or F19.24 for a moderate or severe other (or unknown) substance use disorder with other (or unknown) substance-induced depressive disorder.

Specify current severity/remission:

F19.10 Mild: Presence of 2–3 symptoms.

F19.11 Mild, In early remission

F19.11 Mild, In sustained remission

F19.20 Moderate: Presence of 4–5 symptoms.

F19.21 Moderate, In early remission

F19.21 Moderate, In sustained remission

F19.20 Severe: Presence of 6 or more symptoms.

F19.21 Severe, In early remission

F19.21 Severe, In sustained remission

Specifiers

“In a controlled environment” applies as a further specifier of remission if the individual is both in remission and in a controlled environment (i.e., in early remission in a controlled environment or in sustained remission in a controlled environment). Examples of these environments are closely supervised and substance-free jails, therapeutic communities, and locked hospital units.

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Diagnostic Features

The diagnostic class other (or unknown) substance-related disorders applies to substances that are not included within any of the nine substance classes presented earlier in this chapter (i.e., to alcohol; caffeine; cannabis; hallucinogens [phencyclidine and others]; inhalants; opioids; sedatives, hypnotics, or anxiolytics; stimulants; or tobacco). Such substances include anabolic steroids; nonsteroidal anti-inflammatory drugs; corticosteroids; antiparkinsonian medications; antihistamines; nitrous oxide; amyl-, butyl-, or isobutyl-nitrites; betel nut, which is chewed in many geographic regions to produce mild euphoria and a floating sensation; and kava (from a South Pacific pepper plant), which produces mild euphoria, sedation, incoordination, and weight loss, as well as health effects (e.g., mild hepatitis, lung abnormalities). Note that gaseous substances are included with the inhalant category only if they are hydrocarbon agents; other

gaseous substances (including nitrous oxide mentioned above) are included in the other (or unknown) substance category. Unknown substance-related disorders are associated with unidentified substances, such as intoxications in which the individual cannot identify the ingested drug, or substance use disorders involving either new, black market drugs not yet identified or familiar drugs illegally sold under false names.

Note that substances included within the scope of one of the substance classes should be coded within that respective substance class and are inappropriate to include in the “other substance” category. For example, the following substances are explicitly included in specific substance classes and should not be included in the “other substance” category: synthetic cannabinoids are included in the cannabis category; propofol is included in the sedative, hypnotic, or anxiolytic category; and cathinones (including *khât* plant agents and synthetic chemical derivatives) are included in the stimulant category.

Other (or unknown) substance use disorder is a mental disorder in which repeated use of an other or unknown substance typically continues, despite the individual’s knowing that the substance is causing serious problems for the individual. Those problems are reflected in the diagnostic criteria. When the substance is known but does not fit within any of the other nine substance classes, it should be reflected when recording and coding the name of the disorder (e.g., “nitrous oxide use disorder,” using the applicable code for other [or unknown] substance use disorder).

Associated Features

A diagnosis of other (or unknown) substance use disorder is supported by any of the following: the individual’s reported use of a substance that is not among the nine classes listed in this chapter; recurring episodes of intoxication with negative results in standard drug screens, which may not detect new or rarely used substances; and the presence of symptoms characteristic of an unidentified substance that has newly appeared in the individual’s community.

Because of access to nitrous oxide (“laughing gas”), membership in certain populations may be associated with frequent use of the substance and possibly with a diagnosis of nitrous oxide use disorder. The role of this gas as an anesthetic agent leads to misuse by some medical and dental professionals, and its use as a propellant for commercial products (e.g., whipped cream dispensers) contributes to misuse by food service workers. Nitrous oxide misuse by adolescents and young adults is significant, and some individuals with very frequent use may present with serious medical complications and mental conditions, including myeloneuropathy, spinal cord subacute combined degeneration, peripheral neuropathy, and psychosis.

Use of amyl-, butyl-, and isobutyl (and similar) nitrite gases is prevalent among homosexual men and some adolescents, especially those with conduct disorder.

Substance use disorders generally are associated with elevated risks of suicide, but there is no evidence of unique risk factors for suicide with other (or unknown) substance use disorder.

Prevalence

Based on extremely limited data, the prevalence of most other (or unknown) substance use disorders is likely lower than that of use disorders involving the nine substance classes in this

chapter. For certain gaseous substances, prevalence of use is not rare (lifetime prevalence in the U.S. household population for individuals age 12 and older is estimated at 4.6% for nitrous oxide and 2.5% for nitrites), but how often the patterns of use qualify for a use disorder is unknown.

Development and Course

No single pattern of development or course characterizes the pharmacologically varied other (or unknown) substance use disorders. Often unknown substance use disorders will be reclassified when the unknown substance eventually is identified.

Risk and Prognostic Factors

Risk and prognostic factors for other (or unknown) substance use disorders are thought to be similar to those for most substance use disorders and include the presence of any other substance use disorders, conduct disorder, or antisocial personality disorder in the individual or the individual's family; early onset of substance problems; easy availability of the substance in the individual's environment; childhood maltreatment or trauma; and evidence of limited early self-control and behavioral disinhibition.

Culture-Related Diagnostic Issues

Certain cultures may be associated with other (or unknown) substance use disorders involving specific indigenous substances within the cultural region, such as betel nut.

Diagnostic Markers

Urine, breath, or saliva tests may correctly identify a commonly used substance falsely sold as a novel product. However, routine clinical tests usually cannot identify truly unusual or new substances, which may require testing in specialized laboratories.

Differential Diagnosis

Use of other or unknown substances without meeting criteria for other (or unknown) substance use disorder.

Use of unknown substances is not rare among adolescents, but most use does not meet the diagnostic standard of two or more criteria for other (or unknown) substance use disorder in a 12-month period.

Substance use disorders. Other (or unknown) substance use disorder may co-occur with various substance use disorders that involve any of the nine substance classes presented earlier in this chapter, and the symptoms of the disorders may be similar and overlapping. To disentangle symptom patterns, it is helpful to inquire about which symptoms persisted during periods when some of the substances were not being used.

Other (or unknown) substance intoxication, other (or unknown) substance withdrawal, and other (or unknown) substance-induced mental disorders.

Other (or unknown) substance use disorder is differentiated from other (or unknown) substance intoxication, other (or unknown) substance withdrawal, and other-(or unknown) substance-induced mental disorders (e.g., corticosteroid-induced bipolar and related disorder) in that other (or unknown) substance use disorder describes a problematic pattern of use of the other (or unknown) substance that involves impaired control over the use of the substance, social

impairment attributable to use of the substance, risky use of the substance (e.g., continued use despite medical complications), and pharmacological symptoms (the development of

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tolerance or withdrawal), whereas other (or unknown) substance intoxication, other (or unknown) substance withdrawal, and other (or unknown) substance-induced mental disorders describe psychiatric syndromes that occur in the context of heavy use. Other (or unknown) substance intoxication, other (or unknown) substance withdrawal, and other (or unknown) substance-induced mental disorders may occur in individuals with other (or unknown) substance use disorder. In such cases, a diagnosis of other (or unknown) substance intoxication, other (or unknown) substance withdrawal, or other (or unknown) substance-induced mental disorder should be given in addition to a diagnosis of other (or unknown) substance use disorder, the presence of which is indicated in the diagnostic code.

Comorbidity

Substance use disorders, including other (or unknown) substance use disorder, are commonly comorbid with one another, with conduct disorder in adolescence, and with antisocial personality disorder.

Other (or Unknown) Substance Intoxication

Diagnostic Criteria

- A. The development of a reversible substance-specific syndrome attributable to recent ingestion of (or exposure to) a substance that is not listed elsewhere or is unknown.
- B. Clinically significant problematic behavioral or psychological changes that are attributable to the effect of the substance on the central nervous system (e.g., impaired motor coordination, psychomotor agitation or retardation, euphoria, anxiety, belligerence, mood lability, cognitive impairment, impaired judgment, social withdrawal) and develop during, or shortly after, use of the substance.
- C. The signs or symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including intoxication with another substance.

Specify if:

With perceptual disturbances: This specifier may be noted when hallucinations with intact reality testing or auditory, visual, or tactile illusions occur in the absence of a delirium.

Coding note: The ICD-10-CM code depends on whether there is a comorbid other (or unknown) substance use disorder involving the same substance and whether or not there are perceptual disturbances.

For other (or unknown) substance intoxication, without perceptual disturbances: If a mild other (or unknown) substance use disorder is comorbid, the ICD-10-CM code is **F19.120**, and if a moderate or severe other (or unknown) substance use disorder is comorbid, the ICD-10-CM code is **F19.220**. If there is no comorbid other (or unknown) substance use disorder, then the ICD-10-CM code is **F19.920**.

For other (or unknown) substance intoxication, with perceptual disturbances: If a mild other (or unknown) substance use disorder is comorbid, the ICD-10-CM code is **F19.122**, and if a moderate or severe other (or unknown) substance use disorder is comorbid, the ICD-10-CM code is **F19.222**. If there is no comorbid other (or unknown) substance use disorder, then the ICD-10-CM code is **F19.922**.

Note: For information on Risk and Prognostic Factors, Culture-Related Diagnostic Issues, and Diagnostic Markers, see the corresponding sections in Other (or Unknown) Substance Use Disorder.

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Diagnostic Features

The essential feature of other (or unknown) substance intoxication is the presence of clinically significant behavioral or psychological changes that develop during, or immediately after, use of either a) a substance not included within one of the nine substance classes presented in this chapter (i.e., alcohol; caffeine; cannabis; phencyclidine and other hallucinogens; inhalants; opioids; sedatives, hypnotics, or anxiolytics; stimulants; or tobacco) or b) an unknown substance. If the substance is known, it should be reflected in the name of the disorder upon coding (e.g., “kava intoxication”).

Application of the diagnostic criteria for other (or unknown) substance intoxication is very challenging. Criterion A requires development of a reversible “substance-specific syndrome,” but if the substance is unknown, that syndrome usually will be unknown. To resolve this conflict, clinicians may ask the individual or obtain collateral history as to whether the individual has experienced a similar episode after using substances with the same “street” name or from the same source. Similarly, hospital emergency departments sometimes recognize over a few days numerous presentations of a severe, unfamiliar intoxication syndrome from a newly available, previously unknown substance. Because of the great variety of intoxicating substances, Criterion B can provide only broad examples of signs and symptoms from some intoxications, with no threshold for the number of symptoms required for a diagnosis; clinical judgment guides those decisions. Criterion C requires ruling out other medical conditions, mental disorders, or intoxications.

Prevalence

The prevalence of other (or unknown) substance intoxication is unknown.

Development and Course

Intoxications usually appear and then peak minutes to hours after use of the substance, but the onset and course vary with the substance and the route of administration. Generally, substances used by pulmonary inhalation and intravenous injection have the most rapid onset of action, whereas those ingested by mouth and requiring metabolism to an active product are much slower. (For example, after ingestion of certain mushrooms, the first signs of an eventually fatal intoxication may not appear for a few days.) Intoxication effects usually resolve within hours to a very few days. However, the body may completely eliminate an anesthetic gas such as nitrous oxide just minutes after use ends. At the other extreme, some “hit-and-run” intoxicating substances poison systems, leaving permanent impairments. For example, MPTP (1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine), a contaminating by-product in the synthesis of a certain opioid, kills dopaminergic cells and induces permanent parkinsonism in individuals who had sought opioid intoxication.

Functional Consequences of Other (or Unknown) Substance Intoxication

Impairment from intoxication with any substance may have serious consequences, including dysfunction at work, social indiscretions, problems in interpersonal relationships, failure to fulfill role obligations, traffic accidents, fighting, high-risk behaviors (i.e., having unprotected sex), and substance or medication overdose. The pattern of consequences will vary with the particular substance.

Differential Diagnosis

Use of other or unknown substance, without meeting criteria for other (or unknown) substance intoxication.

The individual used an other or unknown substance(s), but the dose was insufficient to produce symptoms that meet the diagnostic criteria required for the diagnosis.

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Substance intoxication or other substance/medication-induced mental disorders. Familiar substances may be sold in the black market as novel products, and individuals may experience intoxication from those substances. History, toxicology screens, or chemical testing of the substance itself may help to identify it. Other substance intoxication is distinguished from other substance/medication-induced mental disorders (e.g., corticosteroid-induced anxiety disorder) because the symptoms (e.g., anxiety) in these latter disorders are in excess of those (if known) usually associated with the specific substance intoxication, predominate in the clinical presentation, and are severe enough to warrant clinical attention.

Other toxic, metabolic, traumatic, neoplastic, vascular, or infectious disorders that impair brain function and cognition.

Numerous neurological and other medical conditions may produce rapid onset of signs and symptoms mimicking those of intoxications, including the examples in Criterion B. Paradoxically, drug withdrawals also must be ruled out; for example, lethargy may indicate withdrawal from one drug or intoxication with another substance.

Comorbidity

As with all substance-related disorders, conduct disorder in adolescence, antisocial personality disorder, and other substance use disorders tend to co-occur with other (or unknown) substance intoxication.

Other (or Unknown) Substance Withdrawal

Diagnostic Criteria

- A. Cessation of (or reduction in) use of a substance that has been heavy and prolonged.
- B. The development of a substance-specific syndrome shortly after the cessation of (or reduction in) substance use.
- C. The substance-specific syndrome causes clinically significant distress or impairment in social, occupational, or other important areas of functioning.
- D. The symptoms are not attributable to another medical condition and are not better explained by another mental disorder, including withdrawal from another substance.
- E. The substance involved cannot be classified under any of the other substance categories (alcohol; caffeine; cannabis; opioids; sedatives, hypnotics, or anxiolytics; stimulants; or tobacco) or is unknown.

Specify if:

With perceptual disturbances: This specifier may be noted when hallucinations with intact reality testing or auditory, visual, or tactile illusions occur in the absence of a delirium.

Coding note: The ICD-10-CM code depends on whether or not there is a comorbid other (or unknown) substance use disorder and whether or not there are perceptual disturbances.

For other (or unknown) substance withdrawal, without perceptual disturbances: If a mild other (or unknown) substance use disorder is comorbid, the ICD-10-CM code is **F19.130**, and if a moderate or severe other (or unknown) substance use disorder is comorbid, the ICD-10-CM code is **F19.230**. If there is no comorbid other (or unknown) substance use disorder (e.g., in a patient taking an other [or unknown] substance solely under appropriate medical supervision), then the ICD-10-CM code is **F19.930**.

For other (or unknown) substance withdrawal, with perceptual disturbances: If a mild other (or unknown) substance use disorder is comorbid, the ICD-10-CM code is

F19.132, and if a moderate or severe other (or unknown) substance use disorder

is comorbid, the ICD-10-CM code is **F19.232**. If there is no comorbid other (or unknown) substance use disorder (e.g., in a patient taking an other [or unknown] substance solely under appropriate medical supervision), then the ICD-10-CM code is **F19.932**.

Note: For information on Risk and Prognostic Factors and Diagnostic Markers, see the corresponding sections in Other (or Unknown) Substance Use Disorder.

Diagnostic Features

Other (or unknown) substance withdrawal is a clinically significant syndrome that develops during or within a few hours to days after reducing or terminating dosing with a substance (Criteria A and B). Although recent dose reduction or termination usually is clear in the history, other diagnostic procedures are very challenging if the drug is unknown. Criterion B requires development of a “substance-specific syndrome” (i.e., the individual’s signs and symptoms must correspond with the known withdrawal syndrome for the recently stopped drug)—a requirement that rarely can be met with an unknown substance. Consequently, clinical judgment must guide such decisions when this information is limited. Criterion D requires ruling out other medical conditions, mental disorders, or withdrawals from familiar substances. When the substance is known, it should be reflected in the name of the disorder upon coding (e.g., “betel nut withdrawal”).

Prevalence

The prevalence of other (or unknown) substance withdrawal is unknown.

Development and Course

Withdrawal signs commonly appear some hours after use of the substance is terminated, but the onset and course vary greatly, depending on the dose typically used and the rate of elimination of the specific substance from the body. At peak severity, withdrawal symptoms from some substances involve only moderate levels of discomfort, whereas withdrawal from other substances may be fatal. Withdrawal-associated dysphoria often motivates relapse to substance use. Withdrawal symptoms slowly abate over days, weeks, or months, depending on the particular drug and doses to which the individual became tolerant.

Functional Consequences of Other (or Unknown) Substance Withdrawal

Withdrawal from any substance may have serious consequences, including physical signs and symptoms (e.g., malaise, vital sign changes, abdominal distress, headache), intense drug craving, anxiety, depression, agitation, psychotic symptoms, or cognitive impairments. These consequences may lead to problems such as dysfunction at work, problems in interpersonal relationships, failure to fulfill role obligations, traffic accidents, fighting, high-risk behavior (e.g., having unprotected sex), suicide attempts, and substance or medication overdose. The pattern of consequences will vary with the particular substance.

Differential Diagnosis

Dose reduction after extended dosing, but not meeting the criteria for other (or unknown) substance withdrawal.

The individual used other (or unknown) substances, but the dose that was used was insufficient to produce symptoms that meet the criteria required for the withdrawal diagnosis.

Substance withdrawal or other substance/medication-induced mental disorders. Familiar substances may be sold in the black market as novel products, and individuals may experience withdrawal when discontinuing those substances. History, toxicology screens, or

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chemical testing of the substance itself may help to identify it. Other substance withdrawal is distinguished from other substance/medication-induced mental disorders (e.g., venlafaxine-induced anxiety disorder, with onset during withdrawal) because the symptoms (e.g., anxiety) in these latter disorders are in excess of symptoms (if known) usually associated with the specific substance withdrawal, predominate in the clinical presentation, and are severe enough to warrant clinical attention.

Other toxic, metabolic, traumatic, neoplastic, vascular, or infectious disorders that impair brain function and cognition.

Numerous neurological and other medical conditions may produce rapid onset of signs and symptoms mimicking those of withdrawals. Paradoxically, drug intoxications also must be ruled out; for example, lethargy may indicate withdrawal from one drug or intoxication with another substance.

Comorbidity

As with all substance-related disorders, conduct disorder in adolescence, antisocial personality disorder, and other substance use disorders are likely to co-occur with other (or unknown) substance withdrawal.

Other (or Unknown) Substance–Induced Mental Disorders

Because the category of other or unknown substances is inherently ill-defined, the extent and range of these substance-induced mental disorders are uncertain. Nevertheless, other (or unknown) substance–induced mental disorders are possible and are described in other chapters of the manual with disorders with which they share phenomenology (see the substance/medication-induced mental disorders in these chapters): other (or unknown) substance–induced psychotic disorder (“Schizophrenia Spectrum and Other Psychotic Disorders”); other (or unknown) substance–induced bipolar and related disorder (“Bipolar and Related Disorders”); other (or unknown) substance–induced depressive disorder (“Depressive Disorders”); other (or unknown) substance–induced anxiety disorders (“Anxiety Disorders”); other (or unknown) substance–induced obsessive-compulsive disorder (“Obsessive-Compulsive and Related Disorders”); other (or unknown) substance–induced sleep disorder (“Sleep-Wake Disorders”); other (or unknown) substance–induced sexual dysfunction (“Sexual Dysfunctions”); and other (or unknown)

substance/medication–induced major or mild neurocognitive disorder (“Neurocognitive Disorders”). For other (or unknown) substance–induced intoxication delirium, other (or unknown) substance–induced withdrawal delirium, and delirium induced by other (or unknown) substance taken as prescribed, see the criteria and discussion of delirium in the chapter “Neurocognitive Disorders.” These other (or unknown) substance–induced mental disorders are diagnosed instead of other (or unknown) substance intoxication or other (or unknown) substance withdrawal only when the symptoms are sufficiently severe to warrant independent clinical attention.

Unspecified Other (or Unknown) Substance–Related Disorder

F19.99

This category applies to presentations in which symptoms characteristic of an other (or unknown) substance–related disorder that cause clinically significant distress or

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impairment in social, occupational, or other important areas of functioning predominate but do not meet the full criteria for any specific other (or unknown) substance–related disorder or any of the disorders in the substance-related disorders diagnostic class.

Non-Substance-Related Disorders

Gambling Disorder

Diagnostic Criteria

F63.0

- A. Persistent and recurrent problematic gambling behavior leading to clinically significant impairment or distress, as indicated by the individual exhibiting four (or more) of the following in a 12-month period:
 1. Needs to gamble with increasing amounts of money in order to achieve the desired excitement.
 2. Is restless or irritable when attempting to cut down or stop gambling.
 3. Has made repeated unsuccessful efforts to control, cut back, or stop gambling.
 4. Is often preoccupied with gambling (e.g., having persistent thoughts of

reliving past gambling experiences, handicapping or planning the next venture, thinking of ways to get money with which to gamble).

5. Often gambles when feeling distressed (e.g., helpless, guilty, anxious, depressed).
6. After losing money gambling, often returns another day to get even (“chasing” one’s losses).
7. Lies to conceal the extent of involvement with gambling.
8. Has jeopardized or lost a significant relationship, job, or educational or career opportunity because of gambling.
9. Relies on others to provide money to relieve desperate financial situations caused by gambling.

B. The gambling behavior is not better explained by a manic episode.

Specify if:

Episodic: Meeting diagnostic criteria at more than one time point, with symptoms subsiding between periods of gambling disorder for at least several months.

Persistent: Experiencing continuous symptoms, to meet diagnostic criteria for multiple years.

Specify if:

In early remission: After full criteria for gambling disorder were previously met, none of the criteria for gambling disorder have been met for at least 3 months but for less than 12 months.

In sustained remission: After full criteria for gambling disorder were previously met, none of the criteria for gambling disorder have been met during a period of 12 months or longer.

Specify current severity:

Mild: 4–5 criteria met.

Moderate: 6–7 criteria met.

Severe: 8–9 criteria met.

Note: Although some behavioral conditions that do not involve ingestion of substances have similarities to substance-related disorders, only one disorder—gambling disorder—has sufficient data to be included in this section.

Specifiers

Severity is based on the number of criteria endorsed. Individuals with mild gambling disorder may exhibit only 4–5 of the criteria, with the most frequently endorsed criteria usually related to preoccupation with gambling and “chasing” losses. Individuals with moderately severe gambling disorder exhibit more of the criteria (i.e., 6–7). Individuals with the most severe form will exhibit

all or most of the nine criteria (i.e., 8–9). Jeopardizing relationships or career opportunities because of gambling and relying on others to provide money for gambling losses are typically the least often endorsed criteria and most often occur among those with more severe gambling disorder. Furthermore, individuals presenting for treatment of gambling disorder typically have moderate to severe forms of the disorder.

Diagnostic Features

Gambling involves risking something of value in the hopes of obtaining something of greater value. In many cultures, individuals gamble on games and events, and most do so without experiencing problems. However, some individuals develop substantial impairment related to their gambling behaviors. The essential feature of gambling disorder is persistent and recurrent maladaptive gambling behavior that disrupts personal, family, and/or vocational pursuits (Criterion A). Gambling disorder is defined as a cluster of four or more of the symptoms listed in Criterion A occurring at any time in the same 12-month period.

A pattern of “chasing one’s losses” may develop, with an urgent need to continue gambling (often with placing larger bets or taking greater risks) to undo a loss or series of losses. The individual may abandon a gambling strategy and try to win back losses all at once. Although many gamblers may “chase” for short periods of time, it is the frequent, and often long-term, “chase” that is characteristic of gambling disorder (Criterion A6). Individuals may lie to family members, therapists, or others to conceal the extent of involvement with gambling; these instances of deceit may also include, but are not limited to, covering up illegal behaviors such as forgery, fraud, theft, or embezzlement to obtain money with which to gamble (Criterion A7). Individuals may also engage in “bailout” behavior, turning to family or others for help with a desperate financial situation that was caused by gambling (Criterion A9).

In some cases, symptoms meeting diagnostic criteria for gambling disorder may occur as a direct physiological consequence of taking dopaminergic medications, such as those used to treat Parkinson’s disease. When such symptoms are induced by a medication, these cases would be diagnosed as gambling disorder.

Associated Features

Distortions in thinking (e.g., denial, superstitions, a sense of power and control over the outcome of chance events, overconfidence) may be present in individuals with gambling disorder. Many individuals with gambling disorder believe that money is both the cause of and the solution to their problems. Some individuals with gambling disorder are impulsive, competitive, energetic, restless, and easily bored; they may be overly concerned with the approval of others and may be generous to the point of extravagance when winning. Other individuals with gambling disorder are depressed and lonely, and they may gamble when feeling helpless, guilty, or depressed.

Prevalence

The past-year prevalence rate of gambling disorder is about 0.2%–0.3% in the general U.S. population, with a range of 0.1%–0.7% observed across international studies. In the

general U.S. population, the lifetime prevalence rate is about 0.4%–1.0%. For women, the lifetime prevalence rate of gambling disorder is about 0.2%, and for men it is about 0.6%. The 12-month prevalence of DSM-5 gambling disorder varies among ethnoracial groups in the United States: it is 0.52% in African Americans, 0.25% in Latinx, and 0.23% in non-Latinx Whites.

Development and Course

The onset of gambling disorder can occur during adolescence or young adulthood, but in other individuals it manifests during middle or even older adulthood. Generally, gambling disorder develops over the course of years, although the progression appears to be more rapid in women than in men. National data from the United States and Canada show that most individuals who develop a gambling disorder evidence a pattern of gambling that gradually increases in both frequency and amount of wagering. Certainly, milder forms can develop into more severe cases. Most individuals with gambling disorder report that one or two types of gambling are most problematic for them, although some individuals participate in many forms of gambling. Individuals are likely to engage in certain types of gambling (e.g., buying scratch tickets daily) more frequently than others (e.g., playing slot machines or blackjack at the casino weekly). Frequency of gambling can be related more to the type of gambling than to the severity of the overall gambling disorder. For example, purchasing a single scratch ticket each day may not be problematic, while less frequent casino, sports, or card gambling may be part of a gambling disorder. Similarly, amounts of money spent wagering are not in themselves indicative of gambling disorder. Some individuals can wager thousands of dollars per month and not have a problem with gambling, while others may wager much smaller amounts but experience substantial gambling-related difficulties.

Gambling patterns may be regular or episodic, and gambling disorder can be persistent or in remission. Gambling can increase during periods of stress or depression and during periods of substance use or abstinence. There may be periods of heavy gambling and severe problems, times of total abstinence, and periods of nonproblematic gambling. Gambling disorder is sometimes associated with spontaneous, long-term remissions. Nevertheless, some individuals underestimate their vulnerability to develop gambling disorder or to relapse following remission. When in a period of remission, they may incorrectly assume that they will have no problem regulating gambling and that they can engage in some forms of gambling nonproblematically, only to experience a relapse of gambling disorder.

Early expression of gambling disorder is more common among young men (ages 18–21 years) than among young women. Individuals who begin gambling in youth often do so with family members or friends. Development of early-life gambling disorder appears to be associated with impulsivity and substance abuse. Internet gambling has been linked to risky and problematic gambling among youth and may be conducted in a more isolative (i.e., nonpeer) fashion. Some video gaming characteristics (e.g., loot boxes or loot crates containing prizes determined by chance that may be of higher or lower value or desirability) overlap with gambling behavior and may influence the course of gambling disorder. Many high school and college students who develop gambling disorder grow out of the disorder over time, although it remains a lifelong problem for some. Mid- and later-life onset of gambling disorder is more common among women than among men.

There are age and gender variations in the type of gambling activities and the prevalence rates of gambling disorder. Gambling disorder in the United States is more common among younger and middle-age individuals than among older adults. Among U.S. young adults (ages 18–21 years), the disorder is more prevalent in young men than in young women. Younger individuals prefer different forms of gambling (e.g., sports betting), whereas older adults are more likely to develop problems with slot machine and bingo gambling. Although the proportions of individuals who seek treatment for gambling disorder are low across all age groups in the United States, younger individuals are especially unlikely to present for treatment.

Risk and Prognostic Factors

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Temperamental. Gambling that begins in childhood or early adolescence is associated with increased rates of gambling disorder. Gambling disorder also appears to aggregate with antisocial personality disorder, depressive and bipolar disorders, and other substance use disorders, particularly alcohol use disorder.

Genetic and physiological. Gambling disorder can aggregate in families, and this effect appears to relate to both environmental and genetic factors. Gambling problems are more frequent in monozygotic than in dizygotic twins. Gambling disorder is also more prevalent among first-degree relatives of individuals with moderate to severe alcohol use disorder than among the general population.

Course modifiers. Many individuals, including adolescents and young adults, are likely to resolve their problems with gambling disorder over time, although a strong predictor of future gambling problems is previous gambling problems. Psychopathology, including attention-deficit/hyperactivity and anxiety disorders, has been found to be associated with increased risk of onset of gambling disorder among those who gamble and with persistence of gambling disorder symptoms over time.

Culture-Related Diagnostic Issues

Types of gambling activities vary across cultural contexts and ethnoracial groups (e.g., pai gow, cockfights, blackjack, horse racing). Some Indigenous populations in Canada, New Zealand, and the United States have high prevalence rates of gambling problems, possibly related to limited economic opportunities, the expectation that gambling may help advance social goals, and the location of casinos on some U.S. tribal lands. U.S.-born individuals have higher rates of gambling problems than first-generation immigrants to the United States. Endorsement of specific disorder criteria may vary across ethnoracial groups. For example, among individuals with gambling problems, Asian Americans may be less likely than other groups to endorse being preoccupied with gambling (Criterion A4), while African Americans and Latinx may be more likely to endorse repeated unsuccessful efforts to control gambling (Criterion A3).

Sex- and Gender-Related Diagnostic Issues

Men develop gambling disorder at higher rates than women, although this gender gap may be narrowing. Data from treatment-seeking populations have suggested that women may develop

gambling problems more rapidly after the onset of gambling (so-called telescoping), although general population data suggest that men progress more rapidly to disordered gambling than women do. Although women seek treatment sooner than men do, rates of treatment seeking in U.S. national surveys are low (< 10%) among individuals with gambling disorder regardless of gender.

Women may gamble as a maladaptive approach to negative affect, whereas men may gamble more for the thrill of it. Compared with men, women may also experience more shame related to gambling. Men tend to wager on different forms of gambling than women, with cards, sports, and horse race gambling more prevalent among men, and slot machine and bingo gambling more common among women. Women with gambling disorder are more likely than men with gambling disorder to have depressive, bipolar, and anxiety disorders.

Association With Suicidal Thoughts or Behavior

In a U.S. study, up to half of individuals in treatment for gambling disorder in Connecticut reported suicidal thoughts, and about 17% reported attempted suicide. A nationwide register study in Sweden showed that compared with individuals without gambling disorder, individuals ages 20–74 years with gambling disorder have a 15-fold increased suicide mortality rate.

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Functional Consequences of Gambling Disorder

Areas of psychosocial, health, and mental health functioning may be adversely affected by gambling disorder. Specifically, individuals with gambling disorder may, because of their involvement with gambling, jeopardize or lose important relationships with family members or friends. Such problems may occur from repeatedly lying to others to cover up the extent of gambling or from requesting money that is used for gambling or to pay off gambling debts. Employment or educational activities may likewise be adversely impacted by gambling disorder; absenteeism or poor work or school performance can occur with gambling disorder, as individuals may gamble during work or school hours or be preoccupied with gambling or its adverse consequences when they should be working or studying. Individuals with gambling disorder in a U.S. national sample had poor general health and utilized medical services at high rates.

Differential Diagnosis

Nondisordered gambling. Gambling disorder must be distinguished from professional and social gambling. In professional gambling, risks are limited and discipline is central. Social gambling typically occurs with friends or colleagues and lasts for a limited period of time, with acceptable losses. Some persons can experience problems associated with gambling (e.g., short-term chasing behavior and loss of control) that do not meet the full criteria for gambling disorder.

Manic episode. Loss of judgment and excessive gambling may occur during a manic episode. An additional diagnosis of gambling disorder should be given only if the gambling behavior is not better explained by manic episodes (e.g., a history of maladaptive gambling behavior at times other than during a manic episode). Alternatively, an individual with gambling disorder may,

during a period of gambling, exhibit behavior that resembles a manic episode, but once the individual is away from the gambling, these manic-like features dissipate.

Personality disorders. Problems with gambling may occur in individuals with antisocial personality disorder and other personality disorders. If the criteria are met for both disorders, both can be diagnosed.

Gambling symptoms due to dopaminergic medications. Some individuals taking dopaminergic medications (e.g., for Parkinson's disease) may experience urges to gamble that might be distressing or impairing enough to meet criteria for gambling disorder. In such cases, a diagnosis of gambling disorder would be warranted.

Comorbidity

Gambling disorder is associated with poor general health. In addition, some specific medical conditions, such as tachycardia and angina, are more common among individuals with gambling disorder than in the general population, even when other substance use disorders, including tobacco use disorder, are controlled for. In U.S. national surveys, individuals with gambling disorder have high rates of comorbidity with other mental disorders, such as substance use disorders, depressive disorders, anxiety disorders, and personality disorders. In some individuals, other mental disorders may precede gambling disorder and be either absent or present during the manifestation of gambling disorder. Gambling disorder may also occur prior to the onset of other mental disorders, especially bipolar and related disorders, anxiety disorders, and substance use disorders. In a U.S. national survey, in approximately three-quarters of cases of individuals with gambling disorder and another mental disorder, other psychopathology preceded the gambling disorder.

Neurocognitive Disorders

The neurocognitive disorders (NCDs) begin with delirium, followed by the syndromes of major NCD, mild NCD, and their etiological subtypes. The major or mild NCD subtypes are NCD due to Alzheimer's disease; vascular NCD; NCD with Lewy bodies; NCD due to Parkinson's disease; frontotemporal NCD; NCD due to traumatic brain injury; NCD due to HIV infection; substance/medication-induced NCD; NCD due to Huntington's disease; NCD due to prion disease; NCD due to another medical condition; NCD due to multiple etiologies; and unspecified NCD. The NCD category encompasses the group of disorders in which the primary clinical deficit is in cognitive function, and that are acquired rather than developmental. Although cognitive deficits are present in many if not all mental disorders (e.g., schizophrenia, bipolar disorders), only disorders whose core features are cognitive are included in the NCD category. The NCDs are those in which impaired cognition has not been present since birth or very early life, and thus represents a decline from a previously attained level of functioning.

The NCDs are unique among DSM-5 categories in that these are syndromes for which the underlying pathology, and frequently the etiology as well, can potentially be determined. The various underlying disease entities have all been the subject of extensive research, clinical experience, and expert consensus on diagnostic criteria. The DSM-5 criteria for these disorders have been developed in close consultation with the expert groups for each of the disease entities and align as closely as possible with the current consensus criteria for each of them. The potential utility of biomarkers is also discussed in relation to diagnosis. Dementia is subsumed under the newly named entity *major neurocognitive disorder*, although the term *dementia* is not precluded from use in the etiological subtypes in which that term is standard. Furthermore, DSM-5 recognizes a less severe level of cognitive impairment, *mild neurocognitive disorder*, which can also be a focus of care. Diagnostic criteria are provided for both these syndromic entities, followed by diagnostic criteria for the different etiological subtypes. Several of the NCDs frequently coexist with one another, and their relationships may be multiply characterized under different chapter subheadings, including "Differential Diagnosis" (e.g., NCD due to Alzheimer's disease vs. vascular NCD), "Risk and Prognostic Factors" (e.g., vascular pathology increasing the clinical expression of Alzheimer's disease), or "Comorbidity" (e.g., mixed Alzheimer's disease–vascular pathology).

The term *dementia* is retained in DSM-5 for continuity and may be used in settings where physicians and patients are accustomed to this term. Although dementia is the customary term for disorders like the degenerative dementias that usually affect older adults, the term *neurocognitive disorder* is widely used and often preferred for conditions affecting younger individuals, such as impairment secondary to traumatic brain injury or HIV infection. Furthermore, the major NCD definition is somewhat broader than the term *dementia*, in that a diagnosis of major NCD can be made if there is a significant cognitive decline in only one cognitive domain, whereas a diagnosis of dementia in ICD-10 and ICD-11 (and formerly in

DSM-IV) requires multiple cognitive deficits. Thus, cases that would qualify in ICD-10 and ICD-11 (and formerly DSM-IV) for a diagnosis of amnesic disorder (memory impairment in the absence of other cognitive deficits) are diagnosed as major NCD in DSM-5.

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Neurocognitive Domains

The criteria for the various NCDs are based on defined cognitive domains. [Table 1](#) provides for each of the key domains a working definition, examples of symptoms or observations regarding impairments in everyday activities, and examples of assessments. The domains thus defined, along with guidelines for clinical thresholds, form the basis on which the NCDs, their levels, and their subtypes may be diagnosed.

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TABLE 1 Neurocognitive domains

Cognitive domain	Examples of symptoms or observations	Examples of assessments
Complex attention (sustained attention, divided attention, selective attention, processing speed)	<i>Major:</i> Has increased difficulty in environments with multiple stimuli (TV, radio, conversation); is easily distracted by competing events in the environment. Is unable to attend unless input is restricted and simplified. Has difficulty holding new information in mind, such as recalling phone numbers or addresses just given, or reporting what was just said. Is unable to perform mental calculations. All thinking takes longer than usual, and components to be processed must be simplified to one or a few. <i>Mild:</i> Normal tasks take longer than previously. Begins to find errors in routine tasks; finds work needs more double-checking than previously. Thinking is easier when not competing with other things (radio, TV, other conversations, cell phone, driving).	<i>Sustained attention:</i> Maintenance of attention over time (e.g., pressing a button every time a tone is heard, and over a period of time). <i>Selective attention:</i> Maintenance of attention despite competing stimuli or distractors: hearing numbers and letters read and asked to count only letters. <i>Divided attention:</i> Attending to two tasks within the same time period: rapidly tapping while learning a story being read. Processing speed can be quantified on any task by timing it (e.g., time to put together a design of blocks; time to match symbols with numbers; speed in responding, such as counting speed or serial 3 speed).
Executive function (planning, decision-making, working memory, responding to feedback/error correction, overriding habits/inhibition, mental flexibility)	<i>Major:</i> Abandons complex projects. Needs to focus on one task at a time. Needs to rely on others to plan instrumental activities of daily living or make decisions. <i>Mild:</i> Increased effort required to complete multistage projects. Has increased difficulty multitasking or difficulty resuming a task interrupted by a visitor or phone call. May complain of increased fatigue from the extra effort required to organize, plan, and make decisions. May report that large social gatherings are more taxing or less enjoyable because of increased effort required to follow shifting conversations.	<i>Planning:</i> Ability to find the exit to a maze; interpret a sequential picture or object arrangement. <i>Decision-making:</i> Performance of tasks that assess process of deciding in the face of competing alternatives (e.g., simulated gambling). <i>Working memory:</i> Ability to hold information for a brief period and to manipulate it (e.g., adding up a list of numbers or repeating a series of numbers or words backward). <i>Feedback/error utilization:</i> Ability to benefit from feedback to infer the rules for solving a problem. <i>Overriding habits/inhibition:</i> Ability to choose a more complex and effortful solution to be correct (e.g., looking away from the direction indicated by an arrow; naming the color of a word's font rather than naming the word).

		word is often faster than naming the word). <i>Mental/cognitive flexibility:</i> Ability to shift between two concepts, tasks, or response rules (e.g., from number to letter, from verbal to key-press response, from adding numbers to ordering numbers, from ordering objects by size to ordering by color).
670 Learning and memory (immediate memory, recent memory [including free recall, cued recall, and recognition memory], very-long-term memory [semantic; autobiographical], implicit learning)	<i>Major:</i> Repeats self in conversation, often within the same conversation. Cannot keep track of short list of items when shopping or of plans for the day. Requires frequent reminders to orient to task at hand. <i>Mild:</i> Has difficulty recalling recent events, and relies increasingly on list making or calendar. Needs occasional reminders or re-reading to keep track of characters in a movie or novel. Occasionally may repeat self over a few weeks to the same person. Loses track of whether bills have already been paid. Note: Except in severe forms of major neurocognitive disorder, semantic, autobiographical, and implicit learning are relatively preserved, compared with recent memory.	<i>Immediate memory span:</i> Ability to repeat a list of words or digits. Note: Immediate memory sometimes subsumed under “working memory” (see “Executive Function”). <i>Recent memory:</i> Assesses the process of encoding new information (e.g., word lists, a short story, or diagrams). The aspects of recent memory that can be tested include 1) free recall (the individual is asked to recall as many words, diagrams, or elements of a story as possible); 2) cued recall (examiner aids recall by providing semantic cues such as “List all the food items on the list” or “Name all of the children from the story”); and 3) recognition memory (examiner asks about specific items—e.g., “Was ‘apple’ on the list?” or “Did you see this diagram or figure?”). Other aspects of memory that can be assessed include semantic memory (memory for facts), autobiographical memory (memory for personal events or people), and implicit (procedural) learning (unconscious learning of skills).
Language (expressive language [including naming, word finding, fluency, and grammar, and syntax] and receptive language)	<i>Major:</i> Has significant difficulties with expressive or receptive language. Often uses general-use phrases such as “that thing” and “you know what I mean,” and prefers general pronouns rather than names. With severe impairment, may not even recall names of closer friends and family. Idiosyncratic word usage, grammatical errors, and spontaneity of output and economy of utterances occur. Stereotypy of speech occurs; echolalia and automatic speech typically precede mutism. <i>Mild:</i> Has noticeable word-finding difficulty. May substitute general for specific terms. May avoid use of specific names of acquaintances. Grammatical errors involve subtle omission or incorrect use of articles, prepositions, auxiliary verbs, etc.	<i>Expressive language:</i> Confrontational naming (identification of objects or pictures); fluency (e.g., name as many items as possible in a semantic [e.g., animals] or phonemic [e.g., words starting with “f”] category in 1 minute). <i>Grammar and syntax</i> (e.g., omission or incorrect use of articles, prepositions, auxiliary verbs): Errors observed during naming and fluency tests are compared with norms to assess frequency of errors and compare with normal slips of the tongue. <i>Receptive language:</i> Comprehension (word definition and object-pointing tasks involving animate and inanimate stimuli); performance of actions/activities according to verbal command.
671 Perceptual-motor (includes abilities subsumed under the terms <i>visual perception</i>, <i>visuoconstructional</i>, <i>perceptual-motor</i>, <i>praxis</i>, and <i>gnosis</i>)	<i>Major:</i> Has significant difficulties with previously familiar activities (using tools, driving motor vehicle), navigating in familiar environments; is often more confused at dusk, when shadows and lowering levels of light change perceptions. <i>Mild:</i> May need to rely more on maps or others for directions. Uses notes and follows others to get to a new place. May find self lost or turned around when not concentrating on task. Is less precise in parking. Needs to expend greater effort for spatial tasks such as carpentry, assembly, sewing, or knitting.	<i>Visual perception:</i> Line bisection tasks can be used to detect basic visual defect or attentional neglect. Motor-free perceptual tasks (including facial recognition) require the identification and/or matching of figures—best when tasks cannot be verbally mediated (e.g., figures are not objects); some require the decision of whether a figure can be “real” or not based on dimensionality. <i>Visuoconstructional:</i> Assembly of items requiring hand-eye coordination, such as drawing, copying, and block assembly.

	assembly, sewing, or knitting.	<i>Perceptual-motor</i>: Integrating perception with purposeful movement (e.g., inserting blocks into a form board without visual cues; rapidly inserting pegs into a slotted board). <i>Praxis</i>: Integrity of learned movements, such as ability to imitate gestures (wave goodbye) or pantomime use of objects to command (“Show me how you would use a hammer”). <i>Gnosis</i>: Perceptual integrity of awareness and recognition, such as recognition of faces and colors.
Social cognition (recognition of emotions, theory of mind)	<i>Major</i>: Behavior clearly out of acceptable social range; shows insensitivity to social standards of modesty in dress or of political, religious, or sexual topics of conversation. Focuses excessively on a topic despite group’s disinterest or direct feedback. Behavioral intention without regard to family or friends. Makes decisions without regard to safety (e.g., inappropriate clothing for weather or social setting). Typically, has little insight into these changes. <i>Mild</i>: Has subtle changes in behavior or attitude, often described as a change in personality, such as less ability to recognize social cues or read facial expressions, decreased empathy, increased extraversion or introversion, decreased inhibition, or subtle or episodic apathy or restlessness.	<i>Recognition of emotions</i>: Identification of emotion in images of faces representing a variety of both positive and negative emotions. <i>Theory of mind</i>: Ability to consider another person’s mental state (thoughts, desires, intentions) or experience—story cards with questions to elicit information about the mental state of the individuals portrayed, such as “Where will the girl look for the lost bag?” or “Why is the boy sad?”

Delirium

Diagnostic Criteria

- A. A disturbance in attention (i.e., reduced ability to direct, focus, sustain, and shift attention) accompanied by reduced awareness of the environment.
- B. The disturbance develops over a short period of time (usually hours to a few days), represents a change from baseline attention and awareness, and tends to fluctuate in severity during the course of a day.
- C. An additional disturbance in cognition (e.g., memory deficit, disorientation, language, visuospatial ability, or perception).
- D. The disturbances in Criteria A and C are not better explained by another preexisting, established, or evolving neurocognitive disorder and do not occur in the context of a severely reduced level of arousal, such as coma.
- E. There is evidence from the history, physical examination, or laboratory findings that the disturbance is a direct physiological consequence of another medical condition, substance intoxication or withdrawal (i.e., due to a drug of abuse or to a medication), or exposure to a toxin, or is due to multiple etiologies.

Specify if:

Acute: Lasting a few hours or days.

Persistent: Lasting weeks or months.

Specify if:

Hyperactive: The individual has a hyperactive level of psychomotor activity that may be accompanied by mood lability, agitation, and/or refusal to cooperate with medical care.

Hypoactive: The individual has a hypoactive level of psychomotor activity that may be accompanied by sluggishness and lethargy that approaches stupor.

Mixed level of activity: The individual has a normal level of psychomotor activity even though attention and awareness are disturbed. Also includes individuals whose activity level rapidly fluctuates.

Specify whether:

Substance intoxication delirium: This diagnosis should be made instead of substance intoxication when the symptoms in Criteria A and C predominate in the clinical picture and when they are sufficiently severe to warrant clinical attention.

Coding note: The ICD-10-CM codes for the [specific substance] intoxication delirium are indicated in the table below. Note that the ICD-10-CM code depends on whether or not there is a comorbid substance use disorder present for the same class of substance. If a mild substance use disorder is comorbid with the substance intoxication delirium, the 4th position character is “1,” and the clinician should record “mild [substance] use disorder” before the substance intoxication delirium (e.g., “mild cocaine use disorder with cocaine intoxication delirium”). If a moderate or severe substance use disorder is comorbid with the substance intoxication delirium, the 4th position character is “2,” and the clinician should record “moderate [substance] use disorder” or “severe [substance] use disorder,” depending on the severity of the comorbid substance use disorder. If there is no comorbid substance use disorder (e.g., after a one-time heavy use of the substance), then the 4th position character is “9,” and the clinician should record only the substance intoxication delirium.

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ICD-10-CM			
Substance intoxication delirium	With moderate or severe use disorder		
	With mild use disorder	With moderate or severe use disorder	Without use disorder
Alcohol	F10.121	F10.221	F10.921
Cannabis	F12.121	F12.221	F12.921
Phencyclidine	F16.121	F16.221	F16.921
Other hallucinogen	F16.121	F16.221	F16.921

Inhalant	F18.121	F18.221	F18.921
Opioid	F11.121	F11.221	F11.921
Sedative, hypnotic, or anxiolytic	F13.121	F13.221	F13.921
Amphetamine-type substance (or other stimulant)	F15.121	F15.221	F15.921
Cocaine	F14.121	F14.221	F14.921
Other (or unknown) substance	F19.121	F19.221	F19.921

Substance withdrawal delirium: This diagnosis should be made instead of substance withdrawal when the symptoms in Criteria A and C predominate in the clinical picture and when they are sufficiently severe to warrant clinical attention.

Coding note: The ICD-10-CM codes for the [specific substance] withdrawal delirium are indicated in the table below. Note that the ICD-10-CM code depends on whether or not there is a comorbid substance use disorder present for the same class of substance. If a mild substance use disorder is comorbid with the substance withdrawal delirium, the 4th position character is “1,” and the clinician should record “mild [substance] use disorder” before the substance withdrawal delirium (e.g., “mild alcohol use disorder with alcohol withdrawal delirium”). If a moderate or severe substance use disorder is comorbid with the substance withdrawal delirium, the 4th position character is “2,” and the clinician should record “moderate [substance] use disorder” or “severe [substance] use disorder,” depending on the severity of the comorbid substance use disorder. If there is no comorbid substance use disorder (e.g., after regular use of an anxiolytic substance taken as prescribed), then the 4th position character is “9,” and the clinician should record only the substance withdrawal delirium.

Substance withdrawal delirium	ICD-10-CM		
	With mild use disorder	With moderate or severe use disorder	Without use disorder
Alcohol	F10.131	F10.231	F10.931
Opioid	F11.188	F11.288	F11.988
Sedative, hypnotic, or anxiolytic	F13.131	F13.231	F13.931
Other (or unknown) substance	F19.131	F19.231	F19.931

Medication-induced delirium: This diagnosis applies when the symptoms in Criteria A and C arise as a side effect of a medication taken as prescribed.

Code [specific medication]–induced delirium: **F11.921** opioid taken as prescribed (or **F11.988** if during withdrawal from opioid taken as prescribed); **F12.921** pharmaceutical cannabis receptor agonist taken as prescribed; **F13.921** sedative, hypnotic, or anxiolytic taken as prescribed (or **F13.931** if